Artificial Intelligence (BCS701)

Unit -1 Introduction to AI & AI Agents

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Unit	Topic
I	**Introduction to Artificial Intelligence & Intelligent Agents:** Definition and scope of AI, History and applications of AI, Characteristics of Intelligent Agents, Types of agents and environments, Agent architecture, Problem Solving Approach to Typical AI problems, Problem-solving agents. Example problems and approaches.

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❖ What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the **ability of a computer or a machine to think and act like humans** – i.e., to learn, reason, and make decisions.

Matlab, agar ek system **human intelligence ke tasks** perform kar sake jaise:

- Problem solving
- Learning
- Decision making
- Understanding language
- Recognizing images and voices

toh use **Artificial Intelligence** kehte hain.

AI is a branch of computer science that aims to create machines that can perform tasks which normally require **human intelligence**.

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◆ Goals of Artificial Intelligence(IMP)

The **main goals of AI** are:

1. Automation of Tasks

2. **Problem Solving & Reasoning**

3. **Learning from Data (Machine Learning)**

4. **Understanding Natural Language**

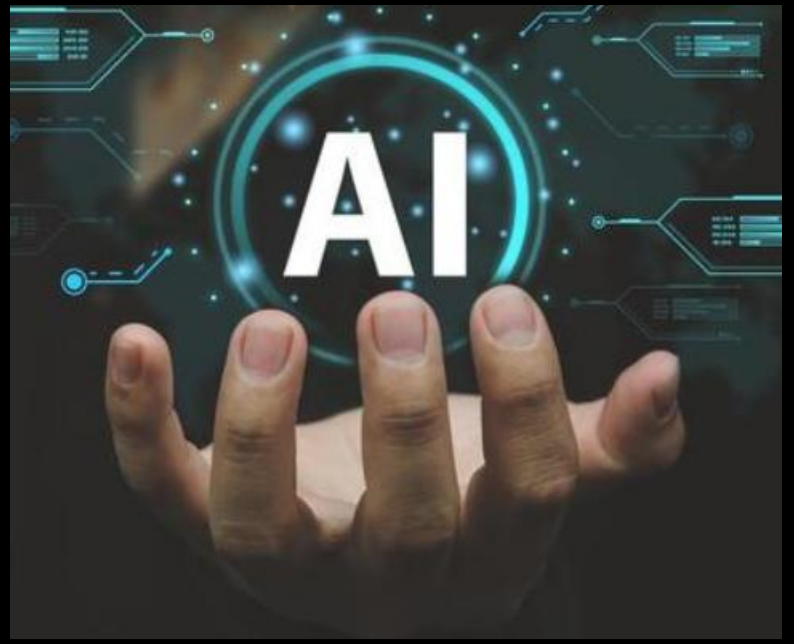
5. **Perception**

To give machines the ability to see, hear, and interpret the world.

Example: Self-driving cars recognizing road signs.

6. Robotics

7. **General Intelligence**



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📒 Branches of Artificial Intelligence(Application) (V.IMP)

Artificial Intelligence is a vast field. It has several branches that deal with different aspects of making machines intelligent. The **major branches of AI** are:

1. Machine Learning (ML)

Focus: Enabling machines to learn from data and improve performance without being explicitly programmed.

Types:

1. **Supervised Learning** → learns from labeled data.
2. **Unsupervised Learning** → finds patterns in unlabeled data.
3. **Reinforcement Learning** → learns through rewards and punishments.

Example: Spam email filtering, Netflix recommendations.

2. Natural Language Processing (NLP)

Focus: Helping machines understand, interpret, and respond to human languages.

Tasks: Language translation, sentiment analysis, chatbots.

Example: Google Translate, ChatGPT, Siri.

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3. Robotics

Focus: Designing intelligent robots that can interact with the physical environment.

Tasks: Navigation, object manipulation, human–robot interaction.

Example: Industrial robots, autonomous drones, humanoid robots like Sophia.

4. Computer Vision

Focus: Enabling machines to interpret and understand visual information from the world.

Tasks: Image recognition, object detection, facial recognition.

Example: Face unlock on smartphones, self-driving car vision systems.

7. Neural Networks & Deep Learning

Focus: Mimicking the structure of the human brain (neurons) to recognize complex patterns.

Deep Learning: Multi-layered neural networks for high-level learning.

Example: Speech recognition, image classification, driverless cars.

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📒 Categories of Artificial Intelligence(V.IMP)

AI is generally classified into **four categories** based on its capability and functionality.

1. Reactive Machines

These are the most basic type of AI systems. They can only **react to present situations** and do not store past experiences (no memory).

Key Features:

- No learning from history.
- Works only on current input.
- Limited capability.

Example:

IBM's Deep Blue (the chess-playing computer that defeated Garry Kasparov).
It can evaluate chess moves but cannot learn from previous games.

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2. Limited Memory AI

These AI systems can use past data or experiences for a short time to make better decisions.

Key Features:

- Can learn from historical data.
- Improves over time with training.
- Most modern AI applications fall in this category.

Example:

- **Self-driving cars** (they observe nearby vehicles, speed, lane positions, and use past data to make decisions).
- **Virtual assistants like Siri or Alexa** (they improve responses based on past interactions).

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3. Theory of Mind AI

This is a more advanced type of AI (still under research) that can understand human emotions, beliefs, and intentions.

Key Features:

Can interact socially.

Understands human feelings and adapts behavior accordingly.

Example:

- Not fully developed yet.
- Early research is seen in **emotional AI chatbots** and **robots that respond to human emotions**.

4. Self-Aware AI

The most advanced type of AI (currently only theoretical). These systems will have **consciousness, self-awareness, and independent decision-making abilities like humans**.

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Key Features:

- Understand themselves.
- Possess emotions, desires, and independent intelligence.
- Potential to surpass human intelligence.

Example:

- No real-world examples yet.
- Depicted in **movies** like *Terminator (Skynet)*, *Ex Machina*, *Iron Man's Jarvis*.

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📒 Future and Scope of Artificial Intelligence(IMP)

Artificial Intelligence is not just a subject—it is shaping the **future of technology, industries, and human life**. Its scope is expanding day by day.

◆ 1. Healthcare

AI will revolutionize diagnosis and treatment.

Future doctors will use AI for **predicting diseases, drug discovery, robotic surgeries, and personalized medicine**.

Example: AI predicting cancer in early stages, robotic surgery assistants.

◆ 2. Education

AI-powered tutors will provide **personalized learning experiences**.

Automatic grading and smart learning platforms will reduce workload for teachers.

Example: AI platforms that adapt lessons according to student's learning speed.

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◆ 3. Transportation

AI will power self-driving cars, drones, and traffic management systems.
This will reduce accidents and improve efficiency.
Example: Tesla's self-driving technology.

◆ 4. Business and Finance

AI will optimize decision-making, fraud detection, customer service (chatbots), and stock market prediction.
Future companies will rely on AI to remain competitive.
Example: AI detecting fraudulent transactions in banks.

◆ 5. Robotics and Automation

AI-powered robots will take over repetitive and dangerous tasks.
Future industries will depend on automation for efficiency.
Example: Industrial robots in manufacturing, warehouse automation by Amazon.

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6. Cybersecurity

AI will strengthen security by **detecting cyberattacks in real-time**.
Future: AI will protect sensitive data from hackers.
Example: AI-based anomaly detection systems.

7. Daily Life Applications

Smart assistants, AI-based home appliances, and recommendation systems will become more advanced.
Example: Alexa, Google Assistant, AI-powered home automation.

Overall :

The future and scope of AI are very broad. It is expected to transform fields like healthcare, education, transportation, business, cybersecurity, and daily life applications. AI will create new job opportunities and may lead to the development of general intelligence, but it also requires careful handling of ethical issues.

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📒 History of Artificial Intelligence

AI has developed over decades. Here's a **timeline view**:

◆ 1940s – Early Foundations

First computers were built (ENIAC, 1943–45).

Concept of storing information and solving mathematical problems began.

◆ 1950 – Alan Turing's Test

Alan Turing published the famous paper *"Computing Machinery and Intelligence"*.

He introduced the **Turing Test** to check if a machine can "think" like a human.

◆ 1956 – Birth of AI as a Field

John McCarthy coined the term **"Artificial Intelligence"** at the Dartmouth Conference.

This is officially considered the **birth of AI**.

◆ 1960s–70s – Early AI Programs

Development of **Expert Systems** that could solve domain-specific problems.

Example: ELIZA (early chatbot), DENDRAL (chemistry expert system).

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◆ 1980s – Machine Learning Boom

Focus shifted to **neural networks and machine learning**.

Expert systems widely used in business.

◆ 1997 – AI Defeats Human in Chess

IBM's Deep Blue defeated world chess champion Garry Kasparov.

◆ 2000s – Rise of Data & Internet

AI used in **recommendation systems, search engines, and spam filters**.

More data + better computers = stronger AI systems.

◆ 2010s – Deep Learning Era

Major breakthroughs in **image recognition, natural language processing, and self-driving cars**.

Virtual assistants like Siri, Alexa, Google Assistant became popular.

◆ Present & Future

AI is now used in healthcare, robotics, finance, education, etc.

Focus on **Explainable AI, Ethical AI, and General Intelligence**.

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📒 Agent and Agent Program

◆ What is an Agent?

An **Agent** is anything that **perceives the environment through sensors and acts on that environment using actuators**.

1. **Sensors** → input devices (e.g., camera, microphone, radar).
2. **Actuators** → output devices (e.g., motors, speakers, robotic arms).

✓ Example:

A **self-driving car**:

- Sensors → cameras, GPS, radar.
- Actuators → steering, brakes, accelerator.
- It perceives traffic and acts by steering/braking.

◆ What is an Agent Program?

An **Agent Program** is the **software** (set of instructions or code) that decides what action the agent should take based on the input it receives.

It implements the **perception-to-action mapping**.

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✅ **Example:**

In a self-driving car, the **agent program** processes camera images (perception) and decides: "turn left, slow down, or stop" (action).

📒 **Intelligent Agent Program(Characterisitics)**

An **Intelligent Agent** is an agent that **acts rationally** → meaning it chooses the **best possible action** to achieve its goals, given what it perceives and knows.

To ensure an agent program is an **Intelligent Agent Program**, it must have:

1. Autonomy

It should operate without human intervention and improve with experience.

Example: A chatbot learning from past conversations.

2. Perception of Environment

It must sense and understand its environment accurately.

Example: A self-driving car detecting traffic signals.

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3. Rational Decision-Making

It should always choose the action that maximizes performance (best possible outcome).

Example: GPS suggesting the fastest route.

4. Adaptability & Learning

It should learn from past actions/data and improve over time.

Example: YouTube's recommendation system getting better with your watching history.

5. Goal-Oriented Behavior

It must act to achieve specific goals effectively.

Example: A vacuum cleaning robot's goal = clean the room.

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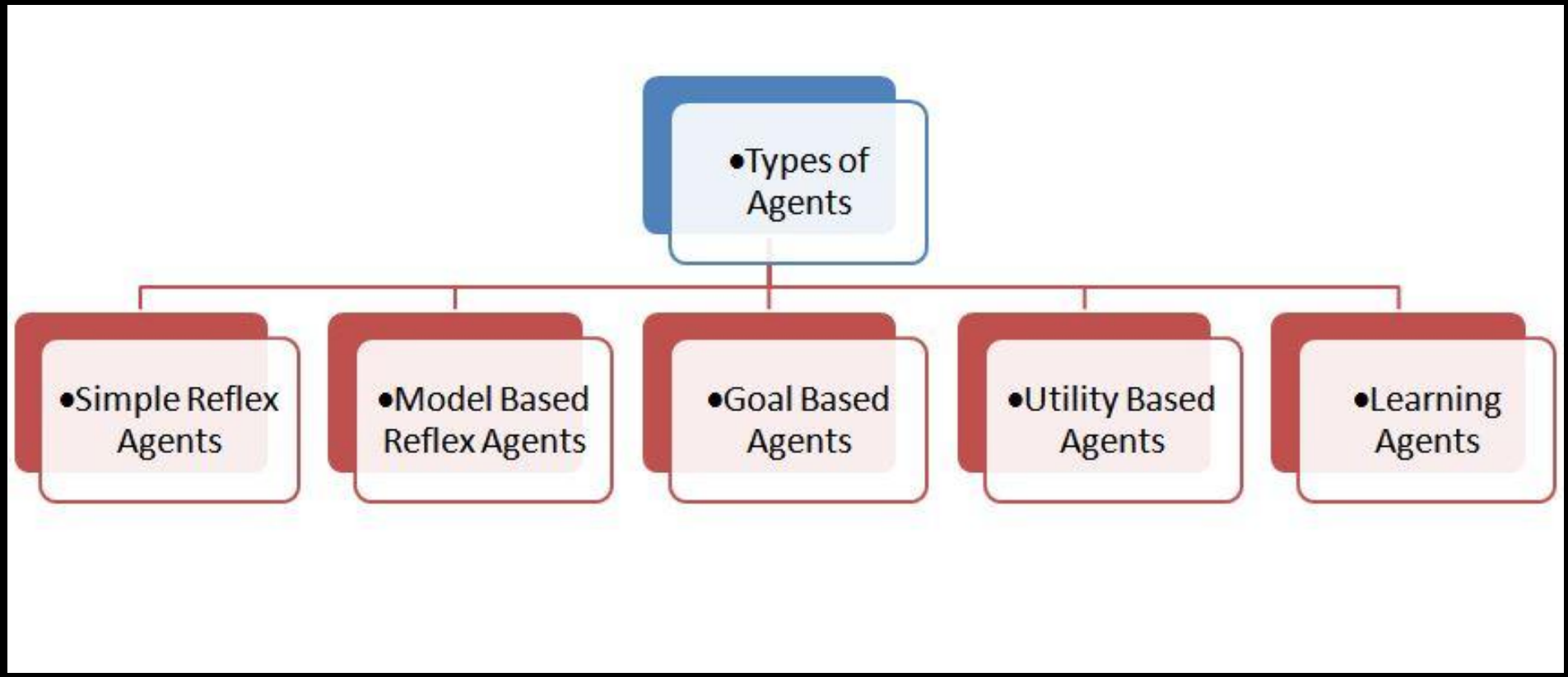
📒 Types of Intelligent Agents(Architecture)(V.V.VIMP)

There are **five main types** of intelligent agents (based on their design and capability):



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1. Simple Reflex Agent

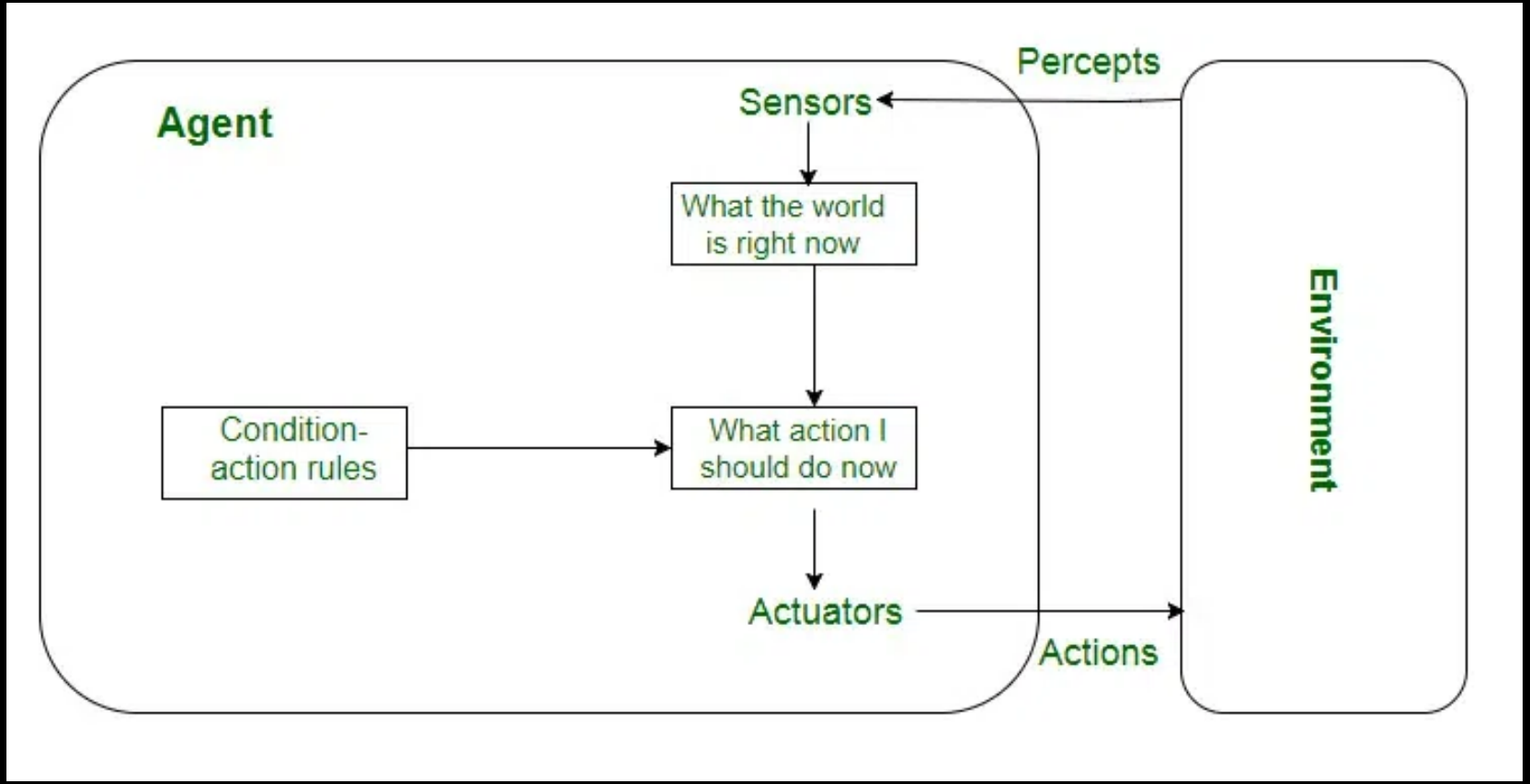
A **Simple Reflex Agent** is the **most basic type of intelligent agent**.

It chooses actions based only on the **current percept (situation)**, ignoring the history of past percepts.

It uses the rule:

☞ **Condition → Action**

(*If something happens, then do something immediately.*)



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◆ How it Works (From Diagram) :

1. Sensors → Collect information about the environment (percepts).
Example: Thermometer sensing temperature.

2. Condition–Action Rules → Agent checks the condition and matches it with a rule.
Example: IF temperature < 20°C → THEN turn on heater.

3. Action Selection → Based only on "what the world is right now."
It does not remember past states.

4. Actuators → Perform the selected action to affect the environment.
Example: Heater turns ON.

Example 2: Traffic Light Controlled Car

Condition: If light = Red.
Action: Stop car.
Condition: If light = Green.
Action: Move car.

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2. Model-Based Reflex Agent

A **Model-Based Reflex Agent** is an improvement over the **Simple Reflex Agent**.
Unlike the simple one (which only looks at the current percept), this agent:

- Keeps track of the **internal state of the world** (memory).
- Uses a **model** to understand how the world evolves and what effects its actions will have.

☞ It can handle **partially observable environments** (when sensors cannot give complete information at once).

◆ How it Works (from the diagram)

1. **Sensors** → Perceive the current environment (percepts).
2. **State** → Maintains an **internal state (memory)** of the world, updated over time.

It stores information about what happened earlier.
Example: If the agent cannot see the whole map, it remembers previously visited places.

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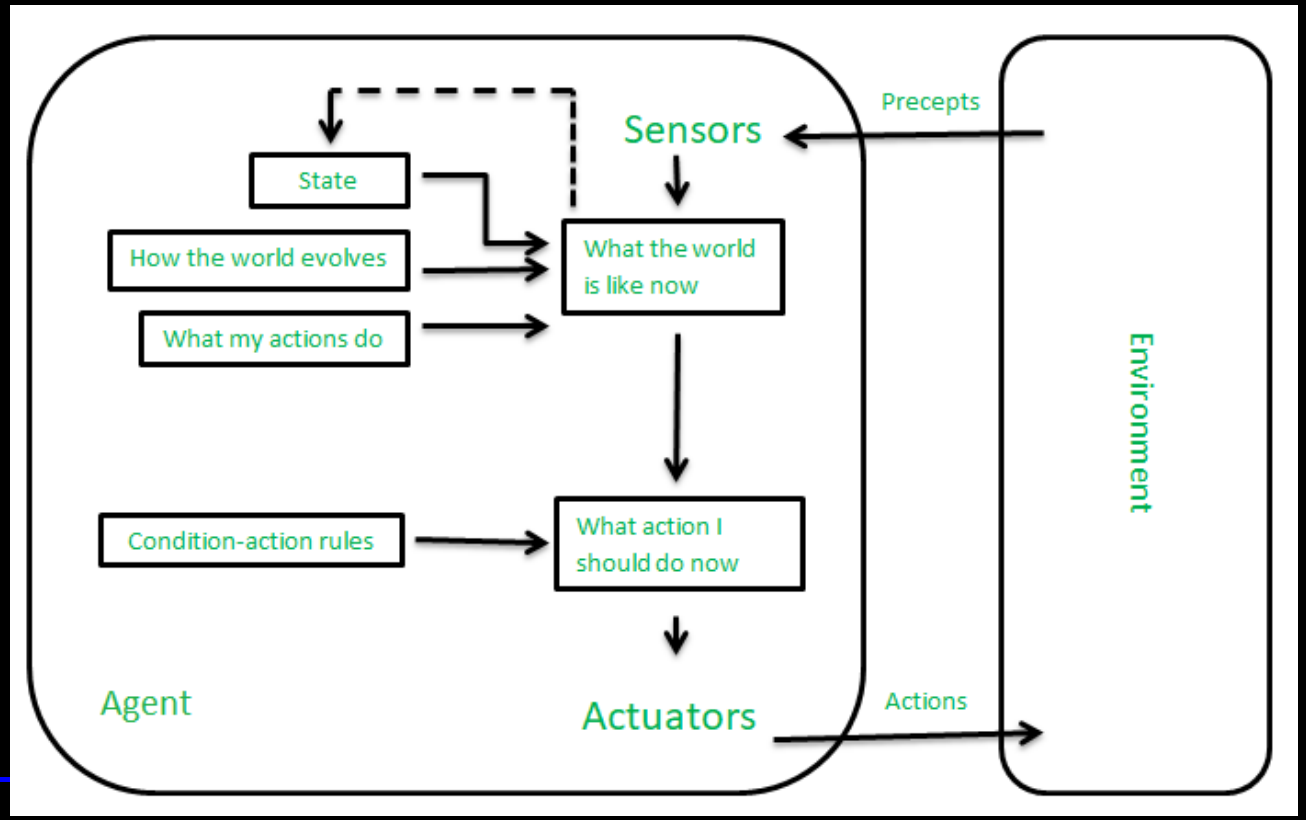
3. Model of the World

- **How the world evolves:** Predicts how the environment changes with time.
- **What my actions do:** Predicts the effect of its own actions.

4. Condition–Action Rules → Uses both the **current percept + internal state** to choose the right action.

5. Actuators → Execute the selected action in the environment.



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◆ Example 1: Self-Driving Car 🚗

I. **Sensors**: Cameras, radar, GPS.
II. **Internal State**: Remembers where obstacles and roads were detected earlier.
III. **Model**: Knows how traffic moves and how its steering/brakes affect the car.
IV. **Action**: Safely drive by considering both current view + memory of past surroundings.

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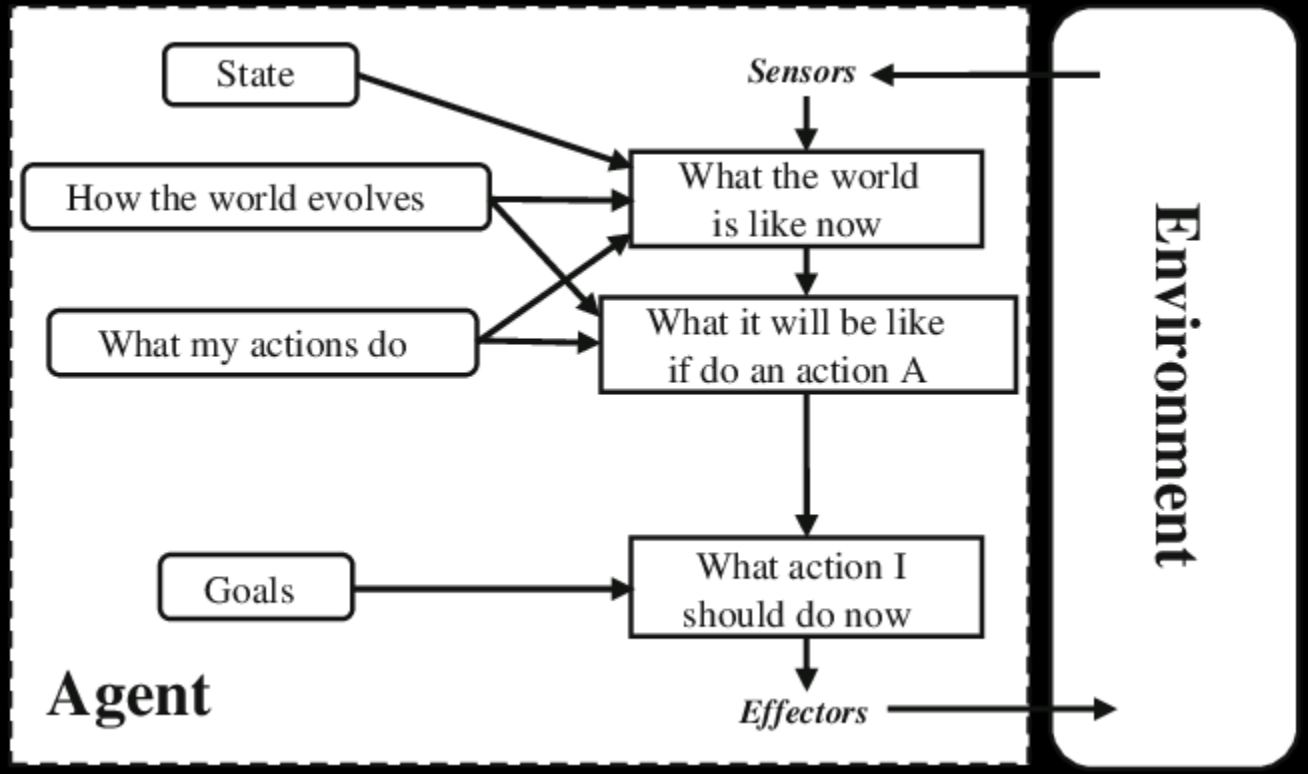
3. Goal-Based Agent

A **Goal-Based Agent** is an intelligent agent that doesn't just react or use memory…

☞ It also looks at a **goal (target)** and chooses actions that help achieve that goal.

So, it answers:

- "What is the world like now?"
- "What will happen if I take this action?"
- "Does this help me reach my goal?"



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◆ How it Works (from the diagram)

1. **Sensors** – The agent sees the current situation (world state).
2. **Model** – It predicts:
   - How the world changes over time.
   - What effect its actions will have.
3. **Goals** – The agent has some goal or target to achieve.
4. **Decision** – The agent compares possible actions and chooses the one that **leads closer to the goal**.
5. **Effectors/Actuators** – Perform the action in the environment.

◆ Example 1: Chess Playing Agent ♟

Goal: Win the game.

It checks the current board (state).

Predicts: "If I move my queen, what will happen next?"

Chooses the move that helps achieve the goal (checkmate opponent).

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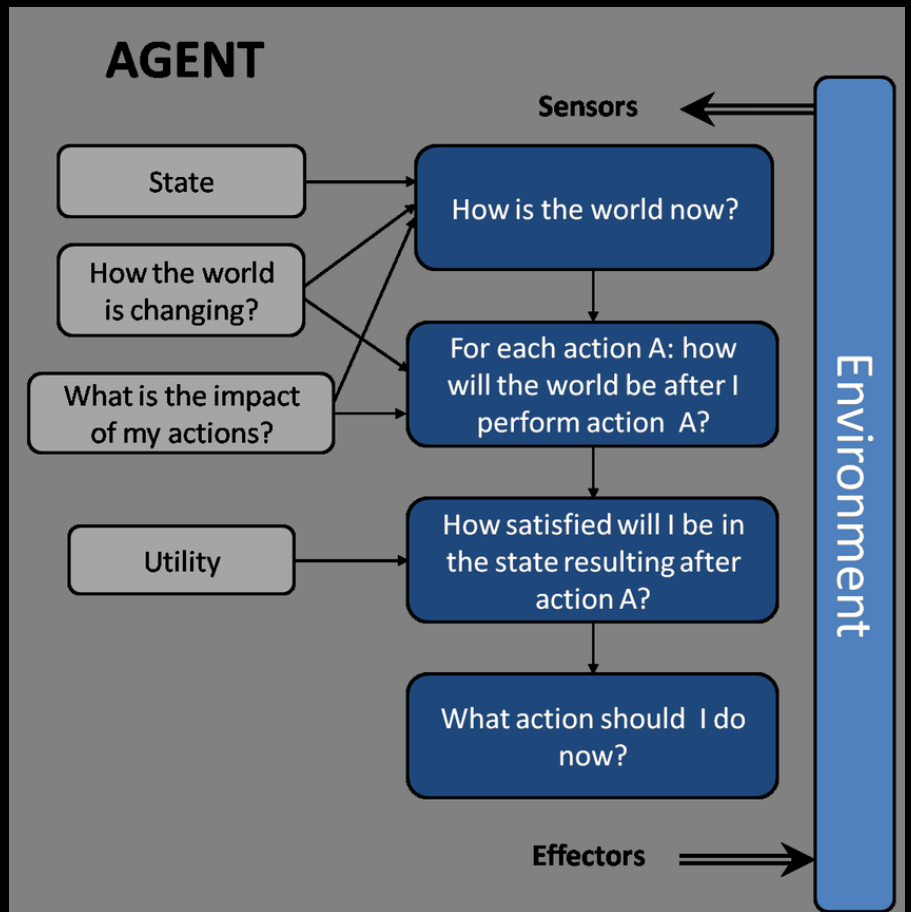
4. Utility-Based Agent

A **Utility-Based Agent** is the most advanced type of intelligent agent.
It does not just aim for a **goal**, but also measures **how good or bad** a situation is using something called **utility** (happiness or satisfaction level).

☞ In short:

1. **Goal-based agent** → "Does this action achieve my goal?"
2. **Utility-based agent** → "Which action gives me the *best outcome* or *maximum happiness*?"



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◆ How it Works (from diagram)

1. **Sensors** – Observe the current world (state).
2. **Model** – Predict what will happen if the agent takes action A.

- "How the world evolves"
- "What my actions do"

3. **Utility Function** – Calculates **how happy** or **satisfied** the agent will be in that situation.

Example: "Is this result better than the other option?"

4. **Decision** – Choose the action that **maximizes utility (happiness, benefit, or safety)**.
5. **Actuators** – Perform the chosen action.

◆ Example 1: Online Shopping Recommender 🛒

Goal: Recommend a product.

Utility: Recommend the product that the user is most likely to buy and enjoy.

Chooses the item with the **highest rating + best match** for the user.

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5. Learning Agent

A **Learning Agent** is an intelligent agent that can **learn from its past experiences**, **improve its performance over time**, and **adapt** to changes in the environment.

👉 Unlike other agents, it **does not stay fixed** → It becomes smarter with experience.

🔸 How it Works (Step by Step)

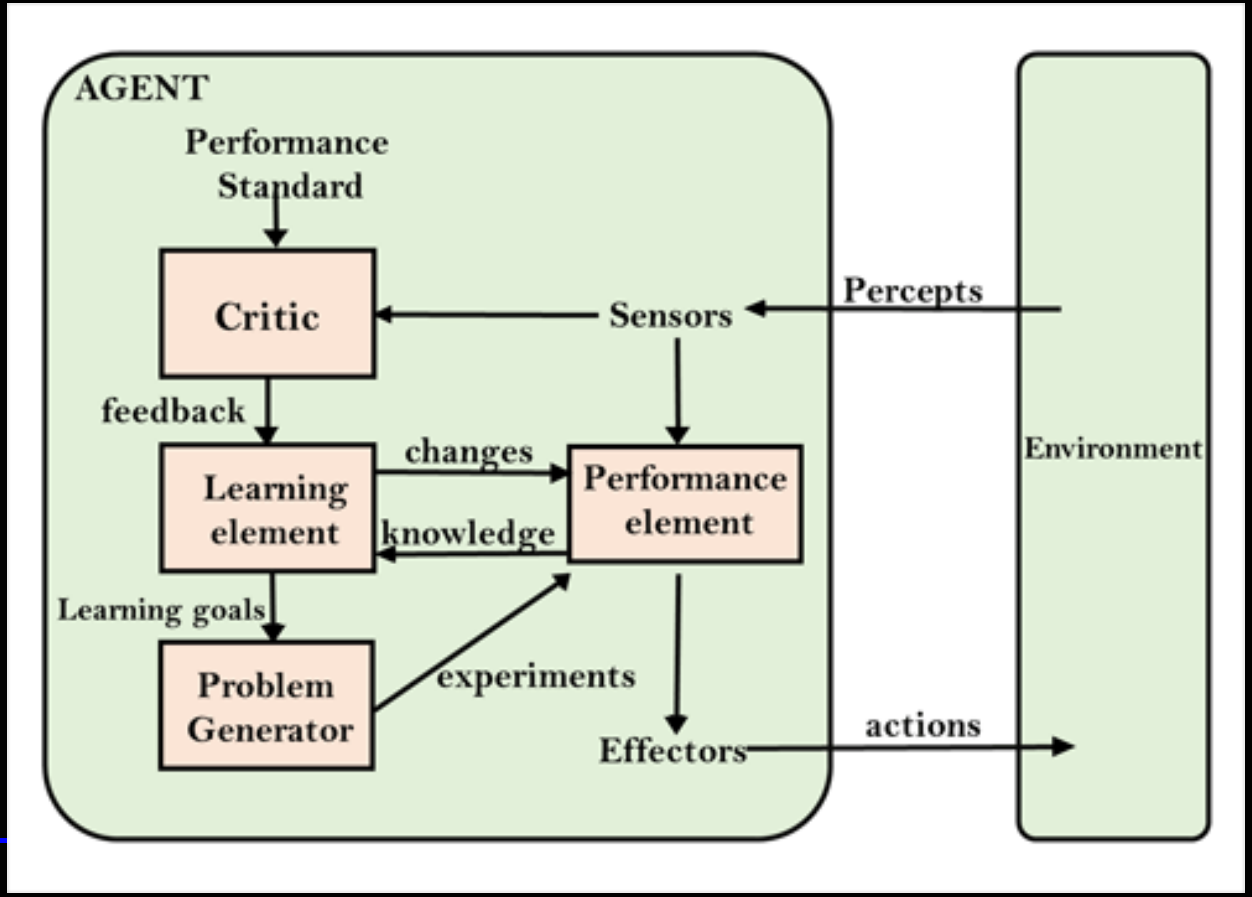
1. Agent observes the environment (percepts via sensors).

2. Performance element takes an action through actuators.

3. Critic checks if the action was good or bad.

4. Learning element updates knowledge to improve next time.

5. Problem generator sometimes suggests trying new actions to learn faster.



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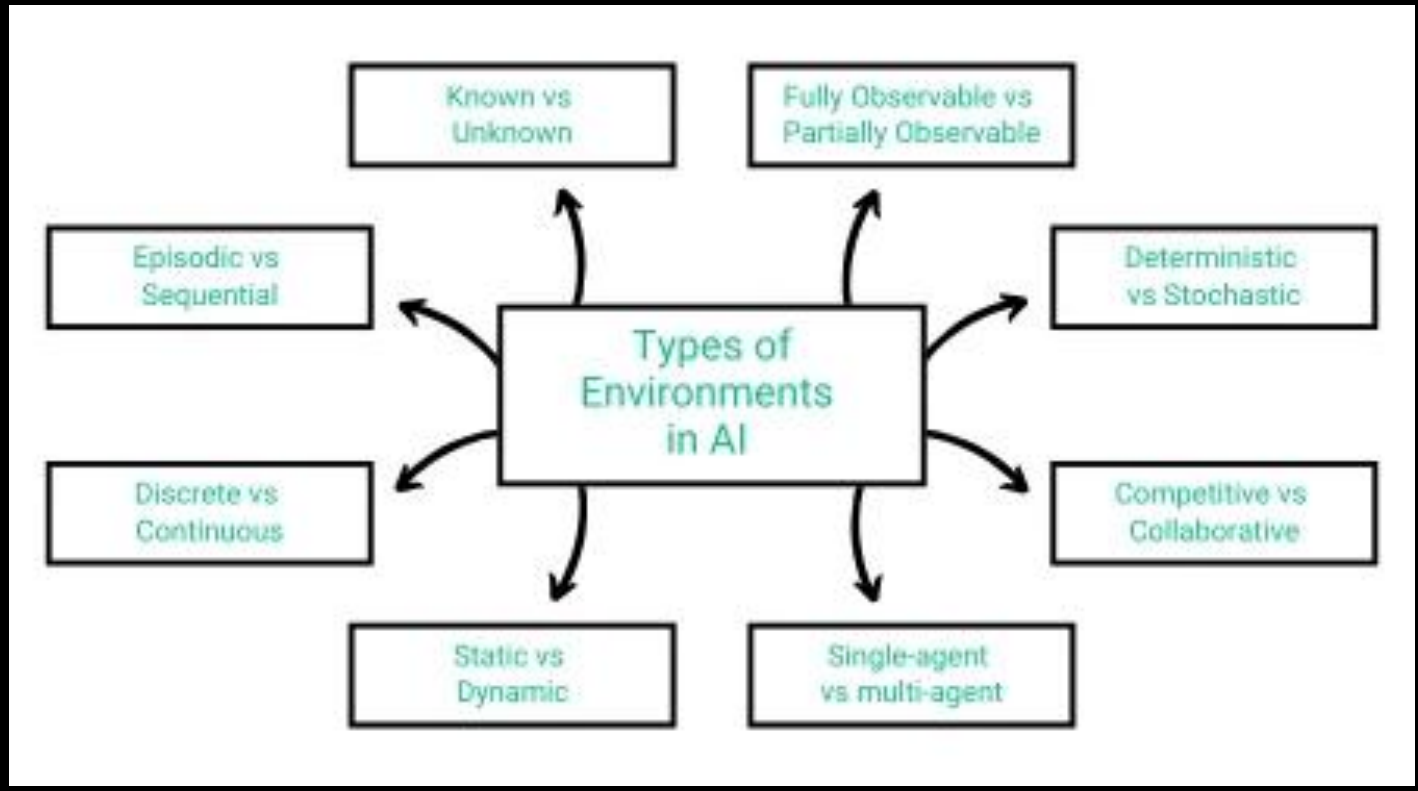
◆ Example 1: Chess-Playing Agent ♟

At first, the agent may lose many games.
The **critic** tells it where mistakes happened.
The **learning element** improves strategies.
With experience, it starts winning more games.

❖ Type of Environment in AI : (V.V.VIMP)

The environment is the world in which the AI agent operates. Different problems need different environments.



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(a) Fully Observable vs. Partially Observable

- **Fully Observable:** The agent can see the complete environment at all times.
  ☞ Example: Chess game – the board is fully visible.
- **Partially Observable:** The agent can only see part of the environment.
  ☞ Example: Driving a car in fog – you can't see everything.

(b) Deterministic vs. Stochastic

- **Deterministic:** The next state of the environment is completely predictable if the current state and action are known.
  ☞ Example: Solving a math puzzle.
- **Stochastic:** The next state is uncertain; randomness is involved.
  ☞ Example: Weather prediction.

(c) Episodic vs. Sequential

- **Episodic:** Each task is independent of previous tasks.
  ☞ Example: Image classification – each photo is separate.
- **Sequential:** Current action affects future actions.
  ☞ Example: Playing chess – one move affects the next.

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(d) Static vs. Dynamic

- **Static:** Environment does not change while the agent is thinking.
  - 👉 Example: Crossword puzzle.
- **Dynamic:** Environment keeps changing.
  - 👉 Example: Self-driving car in traffic.

(e) Discrete vs. Continuous

- **Discrete:** Finite number of possible states/actions.
  - 👉 Example: Tic-tac-toe game.
- **Continuous:** Infinite states/actions possible.
  - 👉 Example: Driving a car – infinite speed/steering angles.

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(f). Competitive vs Collaborative Environment

-> Competitive Environment

Agents work *against each other* (like a game or business competitors).
Example: Chess → if one agent wins, the other loses.
Goal: maximize own performance, minimize opponent's.

-> Collaborative Environment

Agents *work together* to achieve a common goal.
Example: Robots working together to move a heavy object.
Goal: maximize group performance.

(g). Single Agent vs Multi-Agent Environment

-> Single Agent

Only *one agent* operates in the environment.
Example: Vacuum cleaner robot cleaning a room.
Simpler, as no need to consider other agents.

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-> Multi-Agent

Multiple agents exist and interact (can be cooperative or competitive).

Example: Self-driving cars on the road (each car = agent).

More complex due to coordination, competition, or communication.

(h). Known vs Unknown Environment

-> Known Environment

The agent has *complete knowledge* of the environment and rules.

Example: Chess → rules and possible moves are fully known.

Easier for planning.

-> Unknown Environment

The agent does *not fully know* the rules or structure.

Must *learn and explore*.

Example: A new video game the AI has never seen before.

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❖ Problem Solving Approach of AI Problem : (V.V.VIMP)

When AI solves a problem, it needs some important components (parts). These are like the building blocks of AI.

🔑 AI Components required to solve a problem:

1. Learning

AI must learn from data or experience.

Example: If you want AI to recognize cats, it must first learn from images of cats.

Learning can be:

- Supervised Learning (learn with labeled examples)
- Unsupervised Learning (find patterns without labels)
- Reinforcement Learning (learn by trial and error with rewards)

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2. Knowledge Representation

AI needs to store knowledge in some format so it can use it later.

Just like we keep knowledge in our brain, AI keeps it in:

- Rules
- Semantic networks
- Ontologies
- Frames

Example: A medical AI system stores knowledge about diseases, symptoms, and treatments.

3. Reasoning

AI must be able to think logically and make decisions using the knowledge it has.

Example: If AI knows “All humans are mortal” and “Socrates is a human”, it should conclude “Socrates is mortal.”

Reasoning helps AI to solve puzzles, make predictions, and plan actions.

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4. Problem Solving

AI uses search algorithms and techniques to find solutions.

Problem solving has two approaches:

- Special-purpose (for one specific problem)
- General-purpose (for many types of problems)

Example: GPS uses AI to solve the problem of finding the best route.

5. Perception

AI must be able to see, hear, or sense the world.

This includes:

- Computer vision (understanding images/videos)
- Speech recognition (understanding spoken words)
- Sensors (detecting environment)

Example: Self-driving cars use cameras (vision) and LiDAR (sensing).

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6. Language Understanding (NLP – Natural Language Processing)

AI must understand and communicate in human language.

Example: ChatGPT understanding your question and replying in English.

Tasks: speech recognition, translation, question answering, chatbots.

7. Planning

AI should be able to set goals and plan steps to achieve them.

Example: A robot planning how to move from one room to another while avoiding obstacles.

8. Action (Execution)

After planning, AI must act on the environment.

Example: A robot moves its arm, a car turns left, or an AI system executes a trade in the stock market.

⚡ Simple Flow:

👉 Learning → Knowledge → Reasoning → Problem Solving → Perception → NLP → Planning → Action

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❖ Problem Solving agent (Steps) : (V.V.IMP)

A Problem-Solving Agent is an AI agent that finds the best sequence of actions to reach a goal.

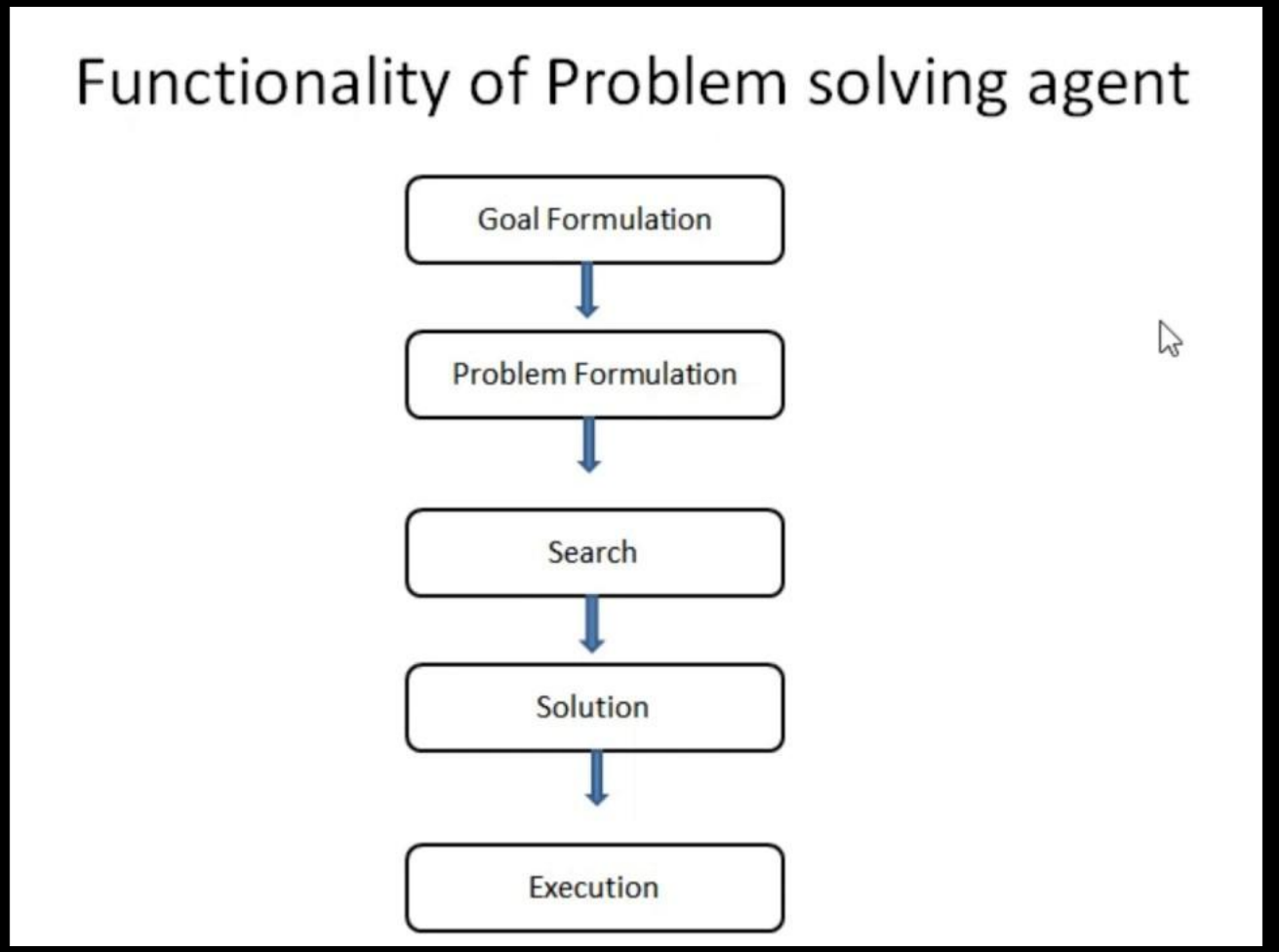
It works step by step.

❖ Steps Involved in Problem-Solving Agent :

1.Define the Goal 🏁

First, the agent should know **what it wants to achieve**.

Example: "I want to reach my school."



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2. Define the Problem 🔍

Describe the problem in AI terms:

- **Initial state** → Where are we now?
- **Actions** → What actions are possible?
- **Transition model** → What happens if we take an action?
- **Goal test** → How do we know we have reached the goal?
- **Path cost** → What is the cost of the solution (time, distance, money)?

3. Search for a Solution 🔑

The agent looks at **different possible paths** and chooses one.

Uses **search algorithms** (like BFS, DFS, A* etc.)

4. Choose the Best Action ✔

From the solution path, the agent selects the **first step/action** to take.

5. Execute the Action 🚶

Perform the action in the real environment.

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6. Observe the New State 👀

Check **where we are now** after taking action.

7. Repeat Until Goal is Reached 🔄

- If not at goal → again search → act → observe → repeat.
- If goal reached → **done!** 🎉

Example 🚗

- **Suppose the agent is a taxi driver AI:**
- **Goal: "Drop passenger at destination."**
- **Problem: Define start location, roads, traffic rules, etc.**
- **Search: Find shortest route using Google Maps.**
- **Choose action: "Turn left."**
- **Execute: The taxi turns left.**
- **Observe: Now taxi is on another road.**
- **Repeat: Continue until destination reached.**

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Thank You....

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Unit -2 Problem Solving & Search Strategies

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II	**Problem Solving & Search Strategies:** Uninformed Search Strategies: BFS, DFS, Iterative Deepening, Informed Search Strategies: Greedy Best-First Search, A* Search, Heuristics and Optimization, Hill Climbing, Simulated Annealing, Constraint Satisfaction Problems, Game Playing: Min-max, Alpha-Beta Pruning, Stochastic & Partially Observable Games.

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❖ Two Categories of Problem :

1. Computational Problems

These are problems where we want the **computer to calculate something**.

The input is given, and the computer must produce the correct output by **processing** it.

Example:

- Sorting numbers (arranging 5, 2, 9 into 2, 5, 9).
- Finding the shortest path in Google Maps.
- Checking if a number is prime.

👉 In short: **Problems that require calculations, logic, or algorithm.**

2. Non-Computational / Non-Algorithmic Problems

These are problems that **cannot be solved completely by a computer** using an algorithm.

Sometimes they involve **human judgment, creativity, or emotions**, which computers can't fully handle.

Example:

Understanding feelings (like love, happiness, sadness).

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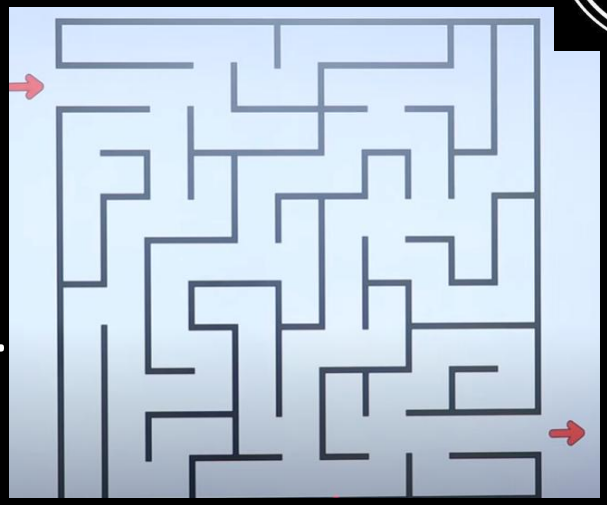


❖ Uninformed Search Strategies :

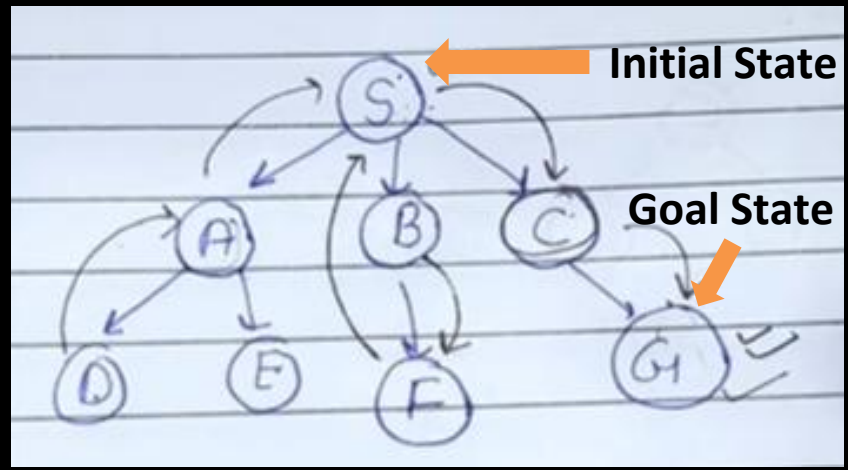
Uninformed search means:

- We **do not have extra information** about the goal or path.
- We only know the **start state, goal state, and possible moves**.
- That's why it is also called **Blind Search**.
- It explores the search space **without any guidance**.

Uninformed search algorithms provide basic search strategies for exploring problem spaces where no additional knowledge is available beyond the problem definition.

These algorithms are important for solving a wide range of problems in AI, such as pathfinding, puzzle solving, and state-space search.

While these algorithms may not always be the most efficient, they provide a baseline for understanding and solving complex problems in AI.



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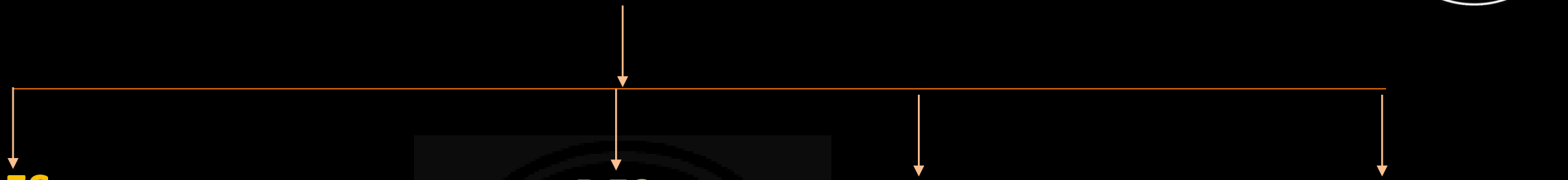
Type of Uninformed Search Strategies (V.VIMP)



1. Breadth-First Search (BFS)(Complete)

- BFS explores the search tree **level by level**.
- It starts from the root node (start state), visits all children first, then moves to the next level.
- It uses a **queue (FIFO)** data structure.

✓ **Properties:**

- Complete (will always find a solution if one exists).
- Optimal (finds the shortest path if cost is same).
- Time complexity: O(b^d) (where b = branching factor, d = depth).

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✓ Algorithm (BFS):

1.Put the start node in a queue.

2.Repeat until the queue is empty:

- Remove the first node.
- If it is the goal → return solution.
- Otherwise, expand the node and add all children to the queue.

2. Depth-First Search (DFS)(Incomplete)

- DFS explores as **deep as possible** along one branch before backtracking.
- It uses a **stack (LIFO)** or recursion.

✓ Properties:

- Not complete (may go infinite in deep tree).
- Not optimal (may not give shortest path).
- Time complexity: O(b^m) (where m = maximum depth).

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✓ Algorithm (DFS):

1.Put the start node in a stack.

2.Repeat until stack is empty:

- Pop the top node.
- If it is the goal → return solution.
- Otherwise, expand the node and push children onto the stack.

DFS is called **incomplete** because:

In infinite search trees, it may not terminate.

It can get stuck in loops.

It cannot guarantee finding a solution even if one exists.

✓ But if the search space is finite (limited), DFS **can** be complete.

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◆ Depth Limited Search (DLS)(Imcomplete)

Depth Limited Search is a modified form of **Depth First Search (DFS)** where we **put a limit on the depth/level** of the search tree.

It means: the algorithm will not go beyond a certain depth (say L).

- **Why it is needed?**
  - DFS may go **too deep** (sometimes infinite in an infinite tree/graph).
  - To avoid this, we put a **depth bound** → so that search is **controlled** and will **stop** after reaching that depth.

◆ Algorithm (Step by Step)

1. Start from the root node.
2. Explore nodes in **depth-first manner**.
3. Keep a counter for the **current depth**.
4. If current depth > L, **stop exploring further** (backtrack).
5. If goal is found before reaching depth L → return success.
6. If no solution found till depth L → return failure.

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❖ Iterative Deepening Depth First Search (IDDFS / ID-DFS)

- It is a **search strategy** that combines the **space efficiency** of DFS and the **completeness** of BFS.
- In simple words:
- It works like **DFS** but avoids the problem of going too deep (infinite path).
- It works like **BFS** by ensuring that all levels are explored.
- So it's a **mixture of DFS + BFS**.

◆ How it works?

- First, run **DFS** with depth limit = 0.
  (Only check the root node)
- If the solution is not found, increase the depth limit to 1 and run DFS again.
  (Check root + children)
- If still not found, increase depth limit to 2 and run DFS again.
  (Check root + children + grandchildren)
- Continue this process until the goal is found.

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- 👉 This way, it keeps increasing the depth like **layers in BFS**, but searches like **DFS**.
- **Time Complexity:** O(b^d) (same as BFS, where b = branching factor, d = depth).

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◆ What is Heuristic in AI?

☞ **Heuristic = Smart Guess / Shortcut Rule** that helps us make decisions faster without checking everything.

Heuristic is a **rule of thumb** or **educated guess** used to solve problems faster when an exact solution is difficult or time-consuming.

It does not always give the perfect/optimal solution but provides a good enough solution in less time.

For example:

If you want to go **from Lucknow to Delhi**, instead of checking *every road in India*, you take a **smart guess** like "Go towards Kanpur Expressway because it's usually the fastest route."

That smart guess = **Heuristic**.

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◆ Components of Heuristic Search

- **Initial State** – Starting point (example: Lucknow).
- **Goal State** – Destination (example: Delhi).
- **Operators/Actions** – The steps/moves possible (roads, flights, trains).
- **Path Cost (g(n))** – Actual distance/time already covered.
- **Heuristic Function (h(n))** – Estimated cost to reach the goal (like Google Maps telling *remaining distance*).
- **Evaluation Function (f(n))** – Formula used in A* search:
- f(n)=g(n)+h(n)f(n) (total cost = actual cost till now + estimated cost left).

◆ Advantages of Heuristic Search

✓ **Faster than blind search** (BFS, DFS).
✓ **More practical** for real-world problems (route planning, games, maps).
✓ **Can find optimal solution** if heuristic is good (like A*).
✓ **Saves time and memory** by reducing unnecessary paths.

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◆ Disadvantages of Heuristic Search

✕ **Not always correct** – heuristic is just a guess, may mislead.
✕ **Depends on heuristic quality** – if h(n) is poor, search becomes slow.
✕ **May be costly to design heuristics** (for complex problems).
✕ **Sometimes non-optimal** solution if heuristic is not accurate.

◆ Real-life Example

Imagine you want to travel from **Lucknow → Delhi**:
Blind Search = checking all possible roads, wasting fuel & time.
Heuristic Search = Using Google Maps estimate (traffic + distance) and directly choosing the fastest path.
That Google Maps estimate = **heuristic function (h(n))**.

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Types of Heuristic Search Techniques (V.V.IMP)

- Greedy Best First Search
- A* Search
- Hill Climbing
- Simulated Annealing

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What are Informed Search Strategies?

- Informed search strategies are search techniques that use **extra knowledge (heuristics)** about the problem to find the solution more efficiently.
- Unlike uninformed search (which knows nothing except the problem definition), informed search uses a **heuristic function (h(n))** that **estimates the cost or distance from the current node to the goal**.
- Because of this extra information, informed search can **find solutions faster and with fewer steps**.

➢ **Heuristic Function (h(n)):**

A rule of thumb or estimate that tells how close a node is to the goal.

Example: In a map, “straight-line distance” from your current city to the destination.

➢ **Evaluation Function (f(n)):**

Combines actual cost and heuristic.

Formula:

f(n) = g(n) + h(n)

g(n) = cost from start to current node.

h(n) = estimated cost from current node to goal.

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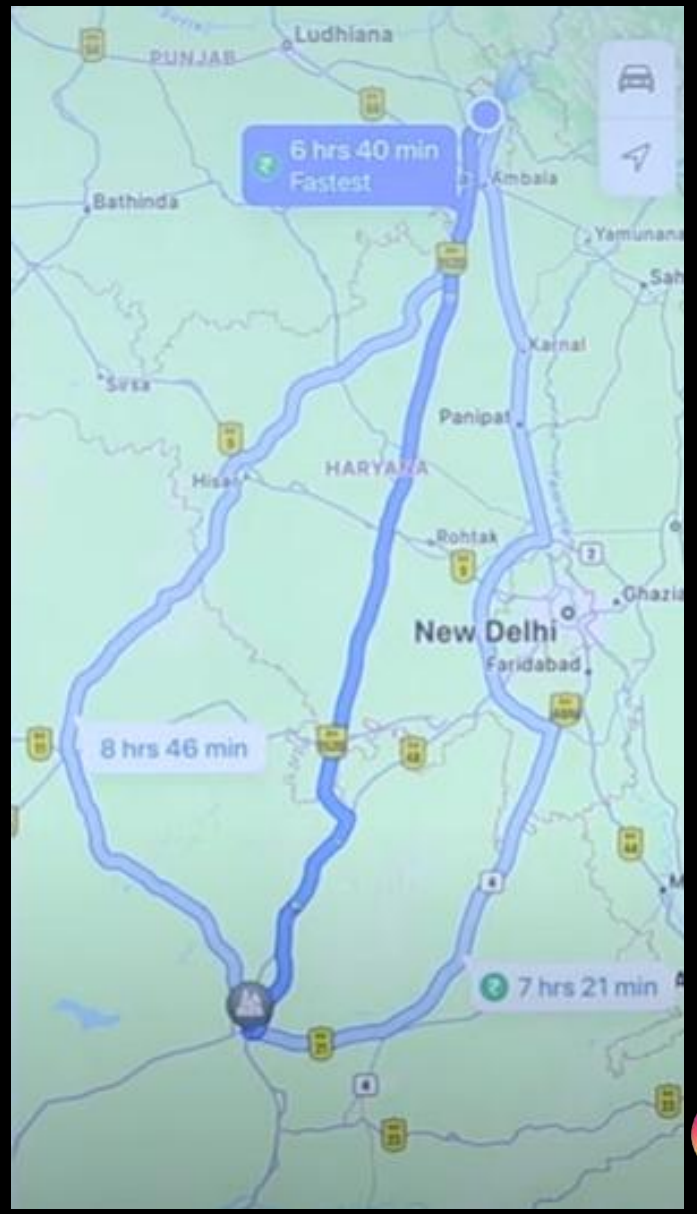
Informed search algorithms in AI are search methods that use extra knowledge, called heuristics, to prioritize which paths to explore. By estimating how close each possible step is to the goal, these algorithms can find solutions more quickly and efficiently than uninformed or “blind,” search. They are widely used in AI for tasks like pathfinding and puzzle solving because they help navigate large, complex search spaces.

Let's see some popular informed search algorithm,



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❖ Difference between Uninformed Search & Informed Search :

Uninformed Search	Informed Search
1) Search without any extra information.	1) Search with additional information (heuristics).
2) No knowledge about the goal direction.	2) Uses knowledge/heuristics to find steps to solution.
3) Time consuming.	3) Provides quicker solution.
4) More complexity (Time & Space).	4) Less complexity (Time & Space).
5) Examples: DFS, BFS etc.	5) Examples: A*, Heuristic DFS, Best-First Search.

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❖ Type of Informed Search Strategies:

i. Greedy Best First Search
ii. A* Search

✓ What is Greedy Best-First Search?(V.IMP)

- It is an **Informed Search Algorithm**.
- "Greedy" means it always tries to go towards the **goal as quickly as possible**.
- It **uses only heuristic function (h(n))** → which estimates "How close the current node is to the goal".
- It **doesn't care about the cost so far (g(n))**, only about the guess of the future cost.
- GBFS always **chooses the path that looks closest to the goal** according to the heuristic function.

◆ Formula Used:

Greedy Best-First Search selects the node with the **lowest heuristic value**:

$$f(n)=h(n)$$

Where **h(n)** = estimated cost from current node → goal

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How Greedy Best-First Search Works

1. Start from the Initial Node

- The search begins from the starting point (let's say node **A**).

2. Evaluate Paths using Heuristic Function

- The algorithm checks all possible paths from the current node.
- Each possible path to the next node is given a **heuristic value (h(n))**.
- This value tells us how close the node is to the goal.
- The smaller the heuristic value, the nearer it is assumed to be to the goal.

3. Choose the Best Path

- At every step, we look at all available paths.
- The path with the **smallest heuristic value** is chosen for expansion.

4. Repeat the Process

- Continue moving along the chosen path.
- Keep checking and selecting the path with the lowest heuristic value.
- This is repeated until the **goal node is reached**.

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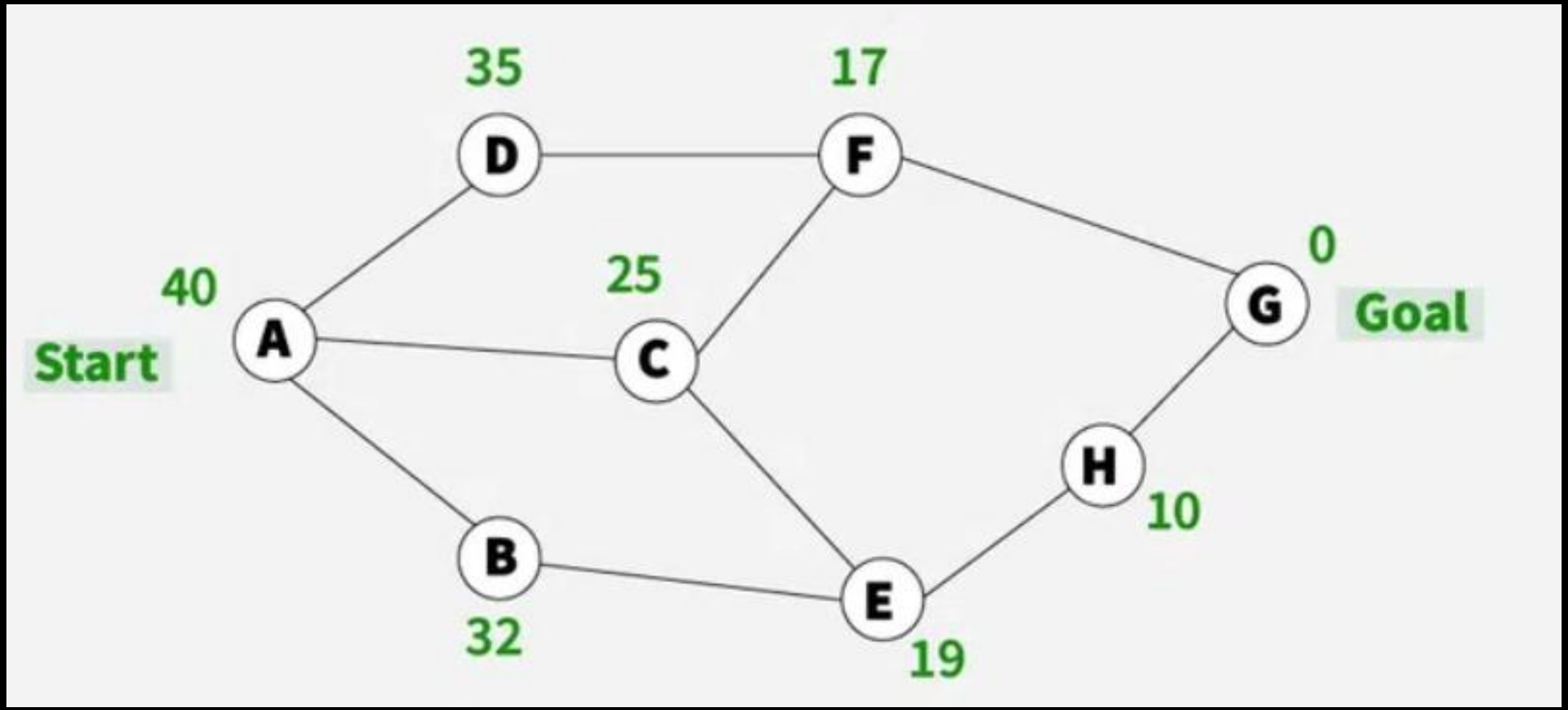

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Step 1: Start at Node A

Possible paths from A:

A → B (32)

A → C (25)

A → D (35)

👉 **We choose C** because it has the **smallest heuristic value (25).**

Step 2: From Node C

Possible paths from C:

C → F (17)

C → E (19)

👉 **We choose F** because 17 < 19 (smallest heuristic value).

Step 3: From Node F

Possible path from F:

F → G (0)

👉 **We choose G** because the heuristic value = **0** (goal reached).

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✓ Final Path Selected:

A → C → F → G

✓ Advantages of Greedy Best First Search

1. **Faster than uninformed searches**

Since it uses a heuristic function, it generally expands fewer nodes compared to BFS/DFS.

2. **Memory efficient (compared to A*)**

Stores only the nodes necessary for the heuristic-based expansion.

3. **Simple to implement**

Easy to code since it just selects the node with the lowest heuristic value.

4. **Good for finding a solution quickly**

Often provides a fast, though not necessarily optimal, path.

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✕ Disadvantages of Greedy Best First Search

1. **Not optimal**

It may not find the shortest or best path, only a "fast-looking" one.

2. **Incomplete**

In some cases (like infinite paths or loops), it may never find a solution.

3. **Depends heavily on heuristic**

If the heuristic is poor, it can perform worse than even BFS.

4. **Can get stuck in local minima**

Chooses the path that looks good immediately, but may ignore the global best solution.

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❖ A* Search Algorithm :

- **A*** is an **informed search algorithm** used in Artificial Intelligence.
- It is like a **smart GPS navigation system** 🚗 → it finds the **best and shortest path** from start to goal.
- It uses both:
  1. **Cost to reach the current node** (past knowledge) → g(n)
  2. **Estimated cost from current node to goal** (future guess using heuristic) → h(n)

☞ The total formula is:

$$f(n)=g(n)+h(n)$$

- **f(n):** Total estimated cost of the path through node n.
- **g(n):** Actual cost from the start node to n.
- **h(n):** Estimated cost from n to the goal (heuristic).

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❖ Step by Step Working (Algorithm):

- Start node se shuru karo.
- Har neighbor ke liye g(n), h(n), aur f(n) nikalo.
- Jis node ka f(n) **sabse chhota hoga**, usko choose karo agla step.
- Ye process repeat karo jab tak goal mil na jaaye.
- Result = **best shortest path**.

Example – In Video

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❖ Optimization :

Optimization is the process of **finding the best solution** from all possible solutions, by **maximizing or minimizing** a particular objective (such as cost, time, distance, accuracy, or profit).

In short:

☞ **Optimization = Best choice among many choices**

Optimization means selecting the **best path, action, or parameter** for solving a problem efficiently.

Example: In pathfinding (like A*), optimization means **choosing the shortest path**.

Types of Optimization

1. **Linear Optimization** – when the problem is linear.
2. **Non-linear Optimization** – when equations are non-linear.
3. **Combinatorial Optimization** – finding the best combination (e.g., Traveling Salesman Problem).

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◆ What is Hill Climbing?(V.VIMP)

- Hill Climbing is a **heuristic search algorithm** used in **Artificial Intelligence** to find the best solution by **step by step improving** the current solution.
- Hill climbing is a heuristic search algorithm that belongs to the family of local search methods. It is designed to solve problems where the goal is to find an optimal (or near-optimal) solution by iteratively moving from the current state to a better neighboring state, according to a heuristic or evaluation function.

☞ Think of it like **climbing a hill**:

- You are standing at some point on the hill.
- You look around for a higher place (better solution).
- If you find a higher point, you move there.
- You keep moving upward until you cannot go higher.
- That’s where you stop.
- This final point is your **solution**.

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❖ Hill Climbing Algorithms

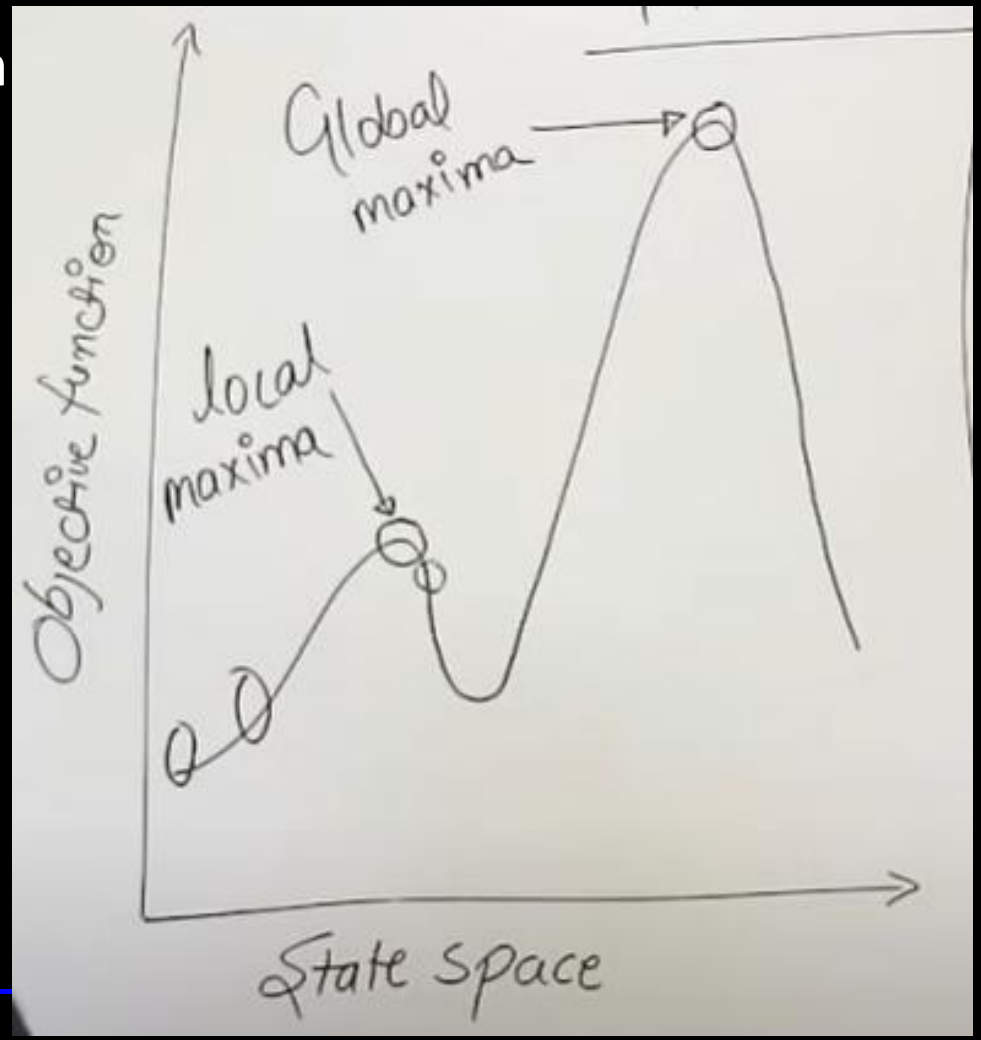
Hill climbing follows these steps:

a) **Initial State:** Start with an arbitrary or random solution (initial state).

b) **Neighboring States:** Identify neighboring states of the current solution by making small adjustments (mutations or tweaks).

c) **Move to Neighbor:** If one of the neighboring states offers a better solution (according to some evaluation function), move to this new state.

d) **Termination:** Repeat this process until no neighboring state is better than the current one. At this point, we have reached a local maximum or minimum.



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◆ Real-Life Example :

- Imagine you are in **Lucknow** and want to reach the **top of Bara Imambara (hill-like viewpoint)**.
- You start at the ground (initial solution).
- You look around: if you see stairs going up (better option), you take them.
- At every step, you again check where to go next: left or right stairs? Choose the higher one.
- You keep climbing until you reach a point where no stair goes higher.
- That is your **final position (solution)**.
- ⚠ But sometimes you may stop at a **small peak (local maximum)**, not the tallest one. This is one disadvantage.

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```
Hill_Climbing(Problem)
{
    Current ← Make_Node(Initial_State[Problem])

    loop
    {
        Neighbor ← a highest-valued successor of Current

        if Value[Neighbor] ≤ Value[Current]
            return State[Current]

        Current ← Neighbor
    }
}
```

- Start from **initial state**.
- Look at all **neighbors** (next possible moves).
- Choose the **best one (highest value)**.
- If no better neighbor exists → **stop** (you are at peak).
- Otherwise → move to the neighbor and repeat.

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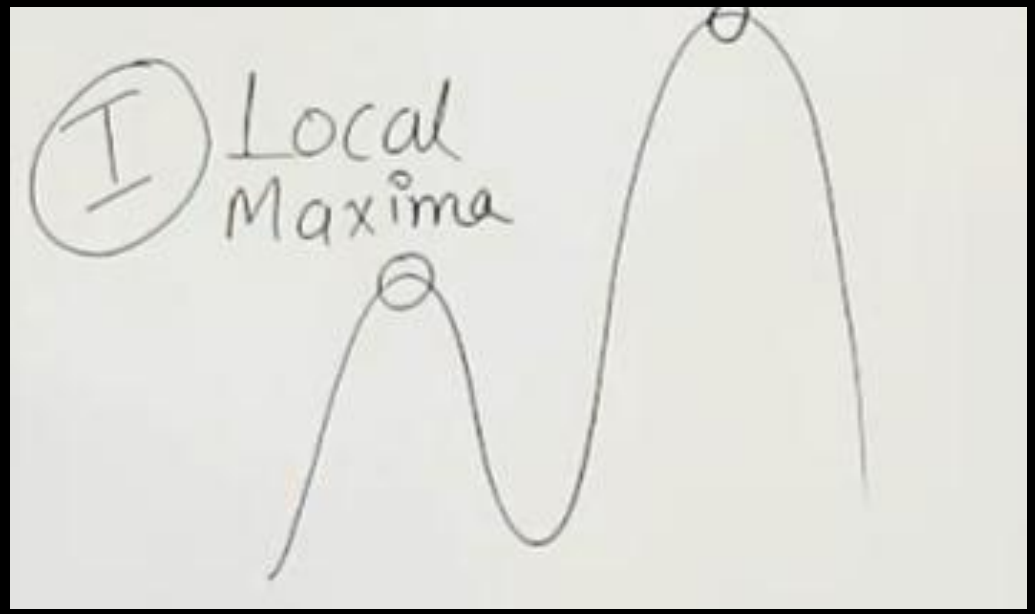
🚧 Limitations of Hill Climbing:

1. Local Maximum Problem ⛰️ :

Sometimes you reach a **small peak** (local maximum) but not the **tallest peak** (global maximum).

The algorithm thinks "I can’t go higher now" and stops, even though a bigger mountain exists somewhere else.



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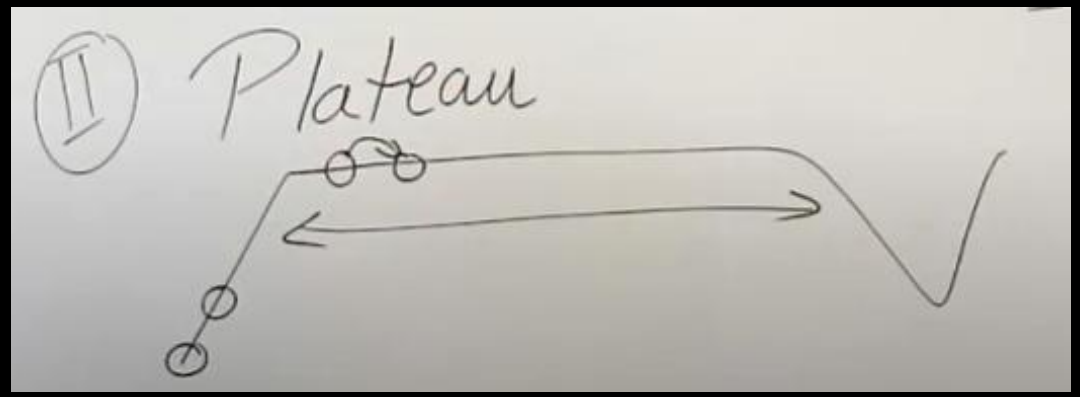
2. Plateau Problem :

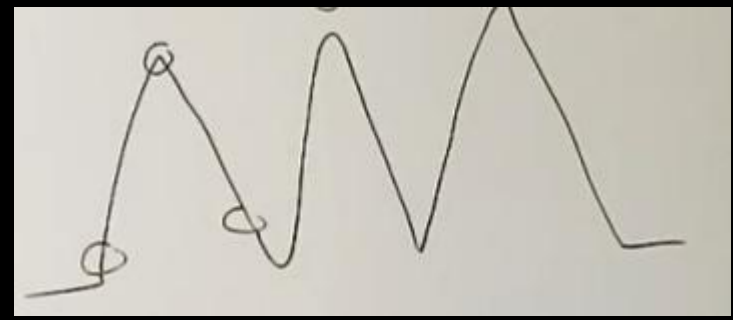
A plateau means a **flat area** (like a flat roof).

Here, no move makes you go higher or lower → the algorithm gets **stuck doing nothing**.



3. Ridge Problem ▲

- Imagine a **narrow path on a mountain ridge** where you can only go up if you move in a very specific direction.
- Hill climbing usually checks only simple directions (up, down, left, right).
- It may miss the right diagonal direction and get stuck.



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🌍 What is Simulated Annealing?

Simulated Annealing is a **search algorithm** used to solve optimization problems (finding the best solution out of many).

It is inspired by the **annealing process in metallurgy** → where metal is heated and then slowly cooled to remove defects and make it stronger.

In computer science, instead of metal, we have a **solution** that we try to improve step by step.

नाम क्यों "Simulated Annealing" है?

Annealing असल में **लोहे को गरम करके ठंडा करने की प्रक्रिया** होती है।

जब लोहा बहुत गरम होता है, तब उसके atom इधर-उधर हिलते-डुलते रहते हैं → इस वजह से उनकी energy ज़्यादा होती है।

धीरे-धीरे ठंडा करने पर atom अपनी जगह settle हो जाते हैं और सबसे **strong (best)** structure बना लेते हैं।

वैसे ही कंप्यूटर में हम किसी problem का **best solution (optimal solution)** ढूँढने के लिए यही idea copy करते हैं → इसलिए नाम है **Simulated (मतलब नकली) Annealing**।

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Stimulate annealing allows downward steps .

❖ Simulated Annealing Method :

1.Think of a hiker (you) climbing hills and mountains.

- The goal = reach the **highest mountain (best solution)**.

2.Problem:

- Sometimes you may get stuck on a **small hill** (local solution) and think it's the top.
- But far away, there is a **bigger mountain (global solution)**.

3.What SA does:

- In the beginning (high temperature):
  You walk **randomly** → sometimes up, sometimes down.

Why? To explore different hills, so you don't get stuck too early.

- Later (temperature cools):
  You become **careful** → mostly go **upwards** (towards better solutions).

Means, as time passes, randomness reduces and you focus on the best path.

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4.Main Idea:

- **"Sometimes go down (accept worse solution) so you can later find a higher hill."**
- Slowly reduce randomness until you only move towards the **best peak**.

❖ Simulated Annealing Algorithm

1.Start with an initial solution S.

2.Set an initial **temperature T** (a large value).

3.Repeat until stopping condition (like T becomes very small or max iterations reached):

a. Pick a **new neighbor solution S_new** (slightly change the current solution).

b. Calculate the **change in cost**:

$$\Delta E = Cost(Snew) - Cost(S)$$

c. **If ΔE < 0** (new solution is better):

→ Accept S_new as the current solution.

d. **If ΔE > 0** (new solution is worse):

→ Accept it **with a probability**:

$$P = e^{-\Delta E/T}$$

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- (This means sometimes we accept worse solutions).
- e. **Decrease temperature T** slowly (cooling schedule).
- **Return the best solution** found.

मान लो तुम exam दे रहे हो और तुम्हें कोई question solve करना है:
पहले तुम अलग-अलग तरीके try करते हो (कुछ सही, कुछ गलत)। → ये है **high temperature (exploration phase)**।
बाद में तुमने देख लिया कौन सा तरीका काम कर रहा है, तो उसी को refine करते हो। → ये है **low temperature (exploitation phase)**।
आखिर में तुम्हें सही answer मिल जाता है।

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🌟 What is a CSP?(V.VIMP)

A **Constraint Satisfaction Problem** is a problem where:

1. You have a set of **variables**.
2. Each variable has a **domain** (possible values it can take).
3. There are **constraints** (rules/conditions) that restrict what combinations of values are allowed.

👉 The goal: **Assign values to variables** such that **all constraints are satisfied**.

👉 A **CSP is a type of problem where you need to find values for some things (variables), but you must follow certain rules (constraints).**

The goal is to assign values in such a way that **all rules are satisfied**.

✅ Real-life Examples

- **Scheduling:** Assigning classes to rooms → No two classes in the same room at the same time.
- **Resource Allocation:** Distributing resources → A machine cannot be used by two people at once.
- **Sudoku / Crossword / Map Coloring** → Classic CSP problems.

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👉 CSP ek aisa problem hai jisme humein “cheezon ko arrange/fill karna” hota hai lekin kuch rules follow karke.
“Cheezein” = Variables (jo fill karni hai)
“Kya bharna hai” = Domain (possible values)
“Kaise bharna hai” = Constraints (rules)

⚙ Components of CSP

1. Variables:

The unknowns we need to decide.
Example: In Sudoku → each empty cell is a variable.

2. Domain:

The possible values each variable can take.
Example: In Sudoku → numbers 1 to 9.

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3. Constraints (Rules):

Restrictions that must be followed.

Example:

- In Sudoku → No two same numbers in a row/column/box.
- In Map Coloring → Two neighboring states can't have the same color.

A CSP can be written as:

- **Variables:** V1, V2, …, Vn
- **Domains:** D1, D2, …, Dn
- **Constraints:** C1, C2, …, Cm

🎯 Why CSP is important in AI?

Because many AI problems can be seen as CSPs → **just find the right values under rules.**

It makes solving complex problems (like planning, scheduling, puzzles, games) much easier.

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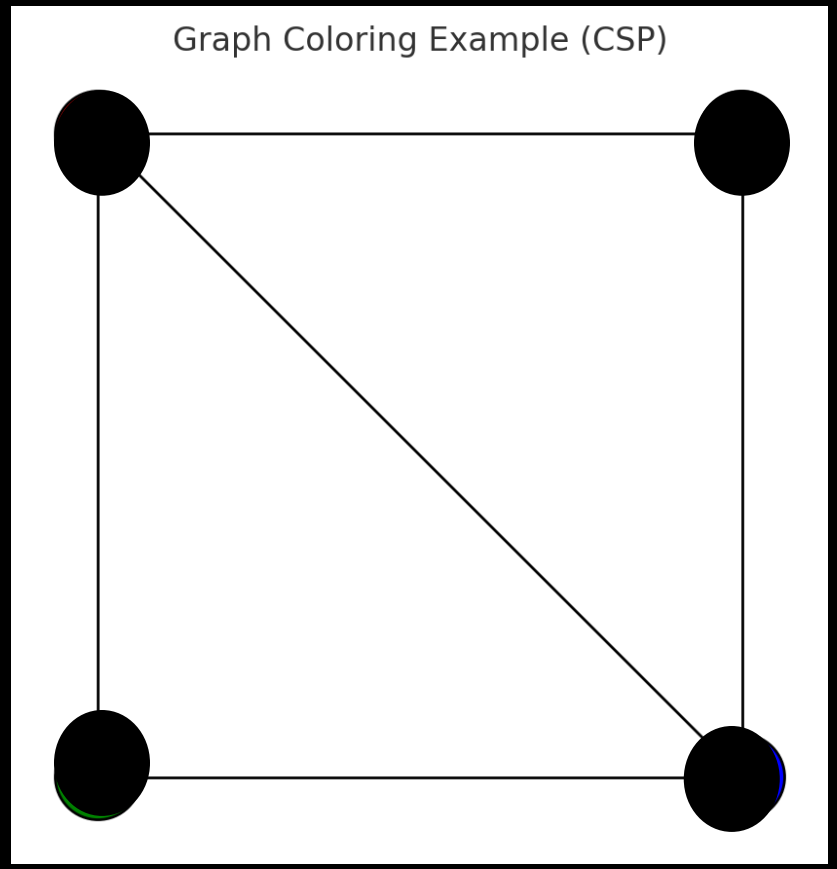
Example :

This is a **graph coloring** problem.

- **Nodes:** 1, 2, 3, 4
- **Goal:** Give each node a color from {Red, Green, Blue}.
- **Rule (Constraints):** Any two nodes that are connected (adjacent) must have **different** colors.



- **Variables (V):** {1, 2, 3, 4} → these are the graph's nodes.
- **Domain (D):** {Red, Green, Blue} → the possible colors for each node.
- **Constraints (C):**
- 1 ≠ 2
- 1 ≠ 3
- 1 ≠ 4
- 2 ≠ 4
- 3 ≠ 4

☞ These pairs are connected by edges in the graph, so their colors must differ.

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Backtracking Table :

Step	Variable	Domain	Chosen color	Result
1	Node 1	{R, G, B}	**R** (choose Red)	✓
2	Node 2	{R, G, B}	**G** (since 1 ≠ 2)	✓
3	Node 3	{R, G, B}	**G** (allowed; 1 ≠ 3 holds)	✓
4	Node 4	{R, G, B}	**B** (since 2 ≠ 4 and 3 ≠ 4)	✓ Solution found

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❖ *Solve the following CSP problem of crypt arithmetic.(V.V.IMP)*

Problem:

SEND
+MORE

MONEY

Ans :

9567 + 1085 = 10652).

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❖ What is Game Playing in AI?

AI **plays games** (like chess, tic-tac-toe, or any decision-making game) by **choosing moves** that lead to winning.

The AI does not guess randomly — it **looks ahead**, evaluates possible moves, and picks the best one.

- AI game playing means making a computer **choose the best move in a game** by **thinking ahea**
- Works for **two-player, turn-based, zero-sum games** (like chess, tic-tac-toe, checkers).
- **Zero-sum:** one player's gain is another player's loss.
- AI models games using **a game tree**:
- **Nodes = positions** in the game.
- **Edges = moves**.
- AI tries to **maximize its own score** while **minimizing the opponent's score**.

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❖ Types of Games

1. Deterministic vs. Stochastic

Deterministic: No luck involved — same moves → same result (e.g., chess).

Stochastic: Luck involved — dice rolls, cards, etc. (e.g., ludo, poker).

2. Fully vs. Partially Observable

Fully observable: Both players see everything (e.g., chess).

Partially observable: Some information is hidden (e.g., poker).

❖ Minimax Algorithm (for Two-player Games)(IMP)

Imagine you and your opponent take turns.

✓ Minimax logic:

- *You (MAX)* try to **maximize your score**.
- *Opponent (MIN)* tries to **minimize your score** (or maximize theirs).
- The computer **builds a game tree**:
- Nodes = game positions.
- Branches = possible moves.

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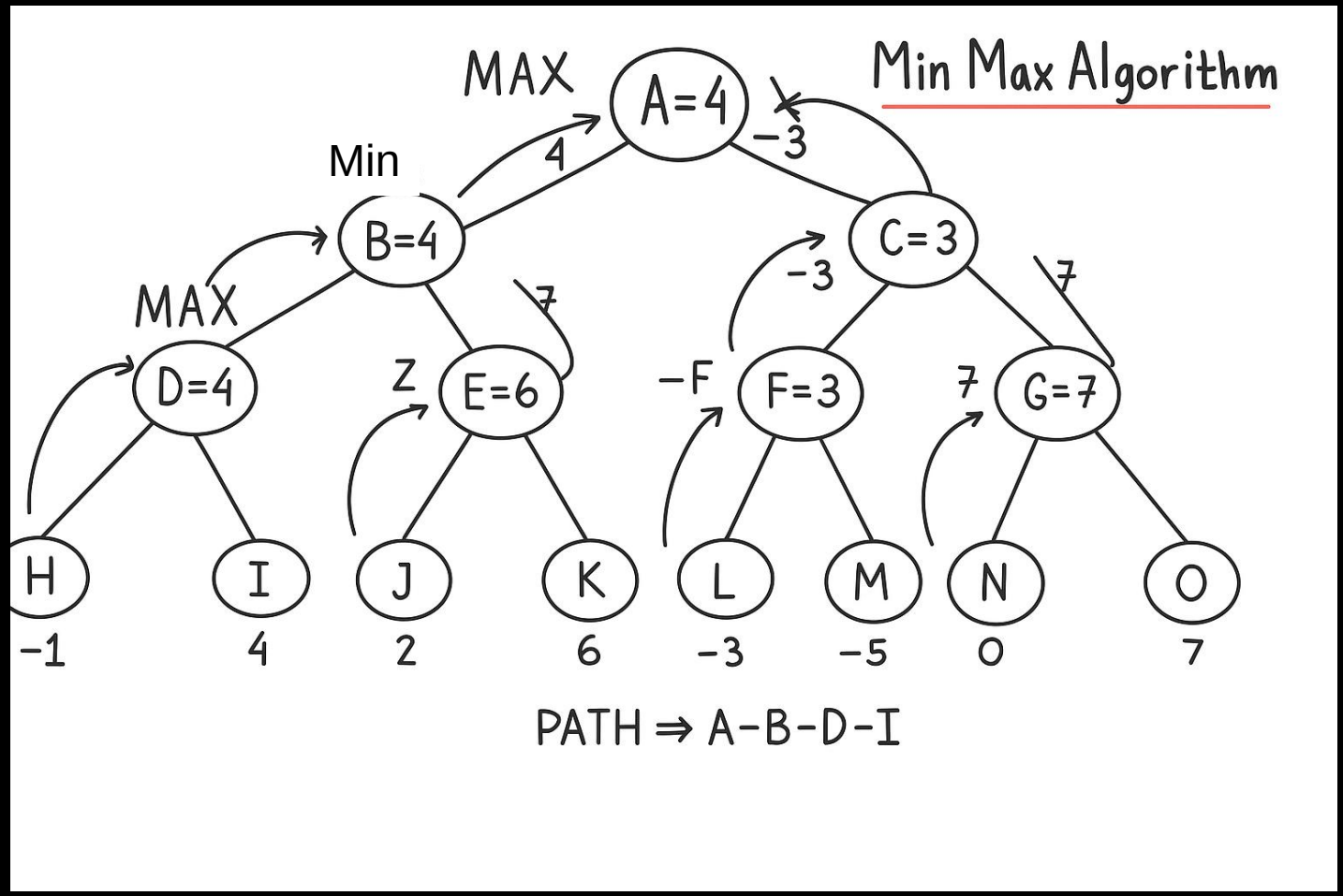
- It evaluates the outcome at the bottom (win = +1, draw = 0, lose = –1)
- Then works **backward** to choose the best move now.



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❖ How it Works (Algorithm) :

Step 1: Make the Game Tree

Draw a tree showing all possible moves in the game.

Each circle (node) = one game situation.

Each arrow (edge) = one move to reach the next situation.

Step 2: Give Values to Last Nodes (Terminal States)

- At the very end of the tree (where no more moves are possible), write down a value:
  - Positive value → Good for the maximizing player (like winning).
  - Negative value → Bad for the maximizing player (like losing).
  - Zero → Draw.

Step 3: Pass Values Up the Tree (Backtracking)

- Start from the bottom and go up:
  - **If it's MAX player's turn:** choose the **highest** value among the child nodes.
  - **If it's MIN player's turn:** choose the **lowest** value among the child nodes.
- Write this chosen value in the parent node. Do this until you reach the root.

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Step 4: Pick the Best Move

- At the very top (root node), the MAX player will choose the move that has the **maximum value**.
- This is the **optimal move**.

❖ Alpha-Beta Pruning : (IMP)

Alpha-Beta Pruning is an **improvement of Mini-Max Algorithm**.
It **does not change the final result**.
It **skips checking some branches** of the game tree that will never affect the final decision.
This saves **time and effort**.

Key Idea

- **Alpha (α):** Best value **MAX player** can guarantee **so far** (initially $-\infty$).
- **Beta (β):** Best value **MIN player** can guarantee **so far** (initially $+\infty$).
- **If ever $\alpha \geq \beta$ at a node → stop exploring that branch** (pruning happens).

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❖ Steps of Alpha-Beta Pruning

Step 1: Build the Game Tree (like Mini-Max)

Draw all possible moves from the current game state.

Step 2: Evaluate Terminal Nodes

Just like Mini-Max, assign utility values (win, lose, draw) to leaf nodes.

Step 3: Traverse the Tree with α and β

- Start from root with $\alpha = -\infty$ and $\beta = +\infty$.
- **For MAX nodes:**
  - Update $\alpha = \max(\alpha, \text{value of child})$.
  - If $\alpha \geq \beta \rightarrow$ **prune remaining branches** (no need to check them).
- **For MIN nodes:**
  - Update $\beta = \min(\beta, \text{value of child})$.
  - If $\alpha \geq \beta \rightarrow$ **prune remaining branches**.

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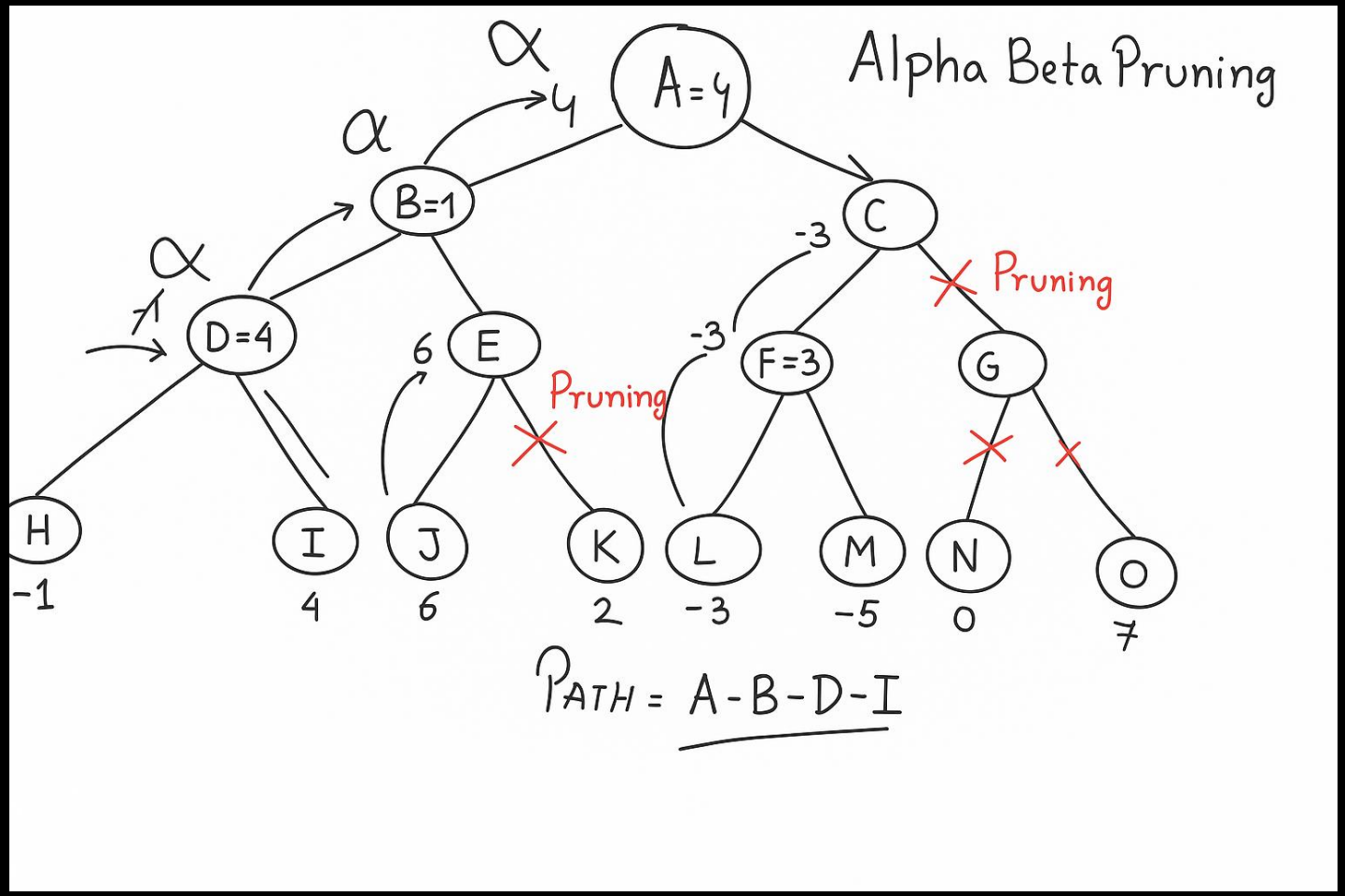
Step 4: Choose Optimal Move

At the root node, MAX chooses the move with the highest value, **but fewer nodes are visited** due to pruning.



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1. Stochastic Game

- **What it is:** A game where **outcomes depend on both players' actions AND some random events**.
- **Key idea:** Even if you play perfectly, **luck or probability can affect the result**.
- **Example:**
  - In a board game like *Snakes and Ladders*, you decide how to move, but a dice roll decides how many steps you take.
  - In card games, even if you choose the best strategy, the card you draw is random.

In short: *Player choices + random chance → next state.*

2. Partially Observable Game

What it is: A game where **players do not have full information about the current state of the game**.

Key idea: You have to **guess or predict hidden information**.

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Example:

In *Poker*, you can see your own cards but **not your opponent's cards**.

In *Battleship*, you don't know where the enemy ships are until you make moves and gather clues.

In short: *Players cannot see everything → must reason with limited information.*

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Thank You....

Download Notes :
https://rzp.io/rzp/lxROMpy

Artificial Intelligence (BCS701)

Unit – 3 Knowledge Representation & Reasoning

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III	**Knowledge Representation & Reasoning:** Propositional and First Order Logic, Syntax, Semantics, and Inference, Knowledge-based agents: Wumpus world, Logic Programming using Prolog, Forward and Backward Chaining, Resolution, Ontological Engineering and Reasoning.

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Download Notes : https://rzp.io/rzp/VsXqbmB

Artificial Intelligence (BCS701)

📌 What is Propositional Logic?

Propositional Logic is the **simplest form of logic** used in AI.
It deals with **statements (propositions)** that can be either **True (T)** or **False (F)**.
It does **not** deal with details inside the statement, only with the **whole statement's truth value**.

👉 Example:

"It is raining." → This is a proposition.
"2 + 2 = 4." → This is also a proposition (True).
"The moon is made of cheese." → Proposition (False).

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🔑 Basic Concepts in Propositional Logic

1. Proposition

A simple statement that is either True or False.

Example: "The sky is blue" ✓

2. Logical Connectives (Operators)

We can combine propositions using special operators:

3. Negation (¬ or NOT): Reverses truth.

If P = "It is raining" (True) → ¬P = "It is NOT raining" (False).

4. Conjunction (∧ or AND): Both must be True.

P: "It is raining" (T), Q: "I have an umbrella" (T) → P ∧ Q = T.

If any one is False → result is False.

5. Disjunction (∨ or OR): At least one is True.

P: "I will go to park", Q: "I will play football".

If either happens → True.

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6. Implication (→ or IF...THEN):

If P → Q = "If P happens, then Q must happen".

Example: "If it rains, then the ground gets wet."

7. Biconditional (↔ or IFF): Both are same (either both True or both False).

Example: "You will pass ↔ You study hard."

Truth Table Examples :

1. AND (∧) -

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

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2. OR (V) :

P	Q	P V Q
T	T	T
T	F	T
F	T	T
F	F	F

3. NOT (¬) :

P	¬P
T	F
F	T

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⚡ Properties of Propositional Logic

1. **Truth Values:** Each proposition is either True (1) or False (0).
2. **Compositionality:** Complex statements can be built using operators.
3. **Precedence:** NOT > AND > OR > → > ↔ (like BODMAS in maths).

Example: ¬P ∨ Q means "NOT P OR Q".

4. **Equivalence:** Some statements mean the same.

Example: P → Q is same as ¬P ∨ Q.

🌟 Properties of Logical Operators :

These properties help us **simplify logical statements** without changing their meaning.

1. Commutativity (Order doesn't matter)

If we swap the order, the result stays the same.

Example:

I. **P ∧ Q = Q ∧ P** (P AND Q is same as Q AND P)

II. **P ∨ Q = Q ∨ P** (P OR Q is same as Q OR P)

👉 **Like in maths:** 2 + 3 = 3 + 2.

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2. Associativity (Grouping doesn't matter)

When we have many ANDs or ORs, it doesn't matter how we group them.

Example:

I. $(P \land Q) \land R = P \land (Q \land R)$

II. $(P \lor Q) \lor R = P \lor (Q \lor R)$

☞ **Like in maths:** $(2 + 3) + 4 = 2 + (3 + 4)$.

3. Distributivity (Mixing AND & OR works like × and + in maths)

Example:

I. $P \land (Q \lor R) = (P \land Q) \lor (P \land R)$

II. $P \lor (Q \land R) = (P \lor Q) \land (P \lor R)$

☞ Just like multiplication distributes over addition.

4. Identity (Adding "True" or "False" doesn't change value)

I. $P \land \textbf{true} = P$ (AND with true → same as P)

II. $P \lor \textbf{false} = P$ (OR with false → same as P)

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5. Domination (Certain values always win)

I. **P ∨ true = true** (OR with true → always true)

II. **P ∧ false = false** (AND with false → always false)

6. Double Negation (Two NOTs cancel each other)

I. **¬(¬P) = P**

☞ Saying "It is NOT NOT raining" = "It is raining".

7. Idempotence (Repeating doesn’t matter)

I. **P ∧ P = P**

II. **P ∨ P = P**

☞ Writing the same thing twice doesn’t change the meaning.

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❖ Applications of Propositional Logic in AI

1. Knowledge Representation

- Store facts in True/False form.

Example: "The patient has fever" = True.

2. Automated Reasoning (AI thinking with logic rules)

- **Modus Ponens**: If $P \rightarrow Q$ and P is true → Q must be true.

Example: If "It rains → ground gets wet", and "It rains", then "ground gets wet".

- **Modus Tollens**: If $P \rightarrow Q$ and $\neg Q$ is true → $\neg P$ is true.

Example: If "It rains → ground gets wet", but "ground is NOT wet", then "It didn't rain".

3. Problem Solving & Planning

AI uses logic to plan actions.

Example: STRIPS planner uses logic to represent "preconditions" and "effects" of actions.

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4. Decision Making

AI can evaluate choices using truth tables.

Example: "If traffic is heavy → don't go. If traffic is clear → go."

5. Natural Language Processing (NLP)

Sentences can be converted into logic for reasoning.

Example: "All birds fly" → logic rule for AI to understand.

⚠ Limitations of Propositional Logic :

1. Not Expressive Enough

Cannot say "All humans are mortal". (Needs **Predicate Logic**).

2. Scalability Problem

Truth tables grow too large when there are many propositions.

3. Binary Only

Only True/False. Cannot handle probabilities like "70% chance of rain".

4. No Quantifiers

Cannot use "For all" (∀) or "There exists" (∃).

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5. No Uncertainty Handling

Cannot deal with uncertain or incomplete knowledge.

6. No Context

Just cares about True/False, not the meaning behind statements.

❖ First-Order Logic in Artificial Intelligence

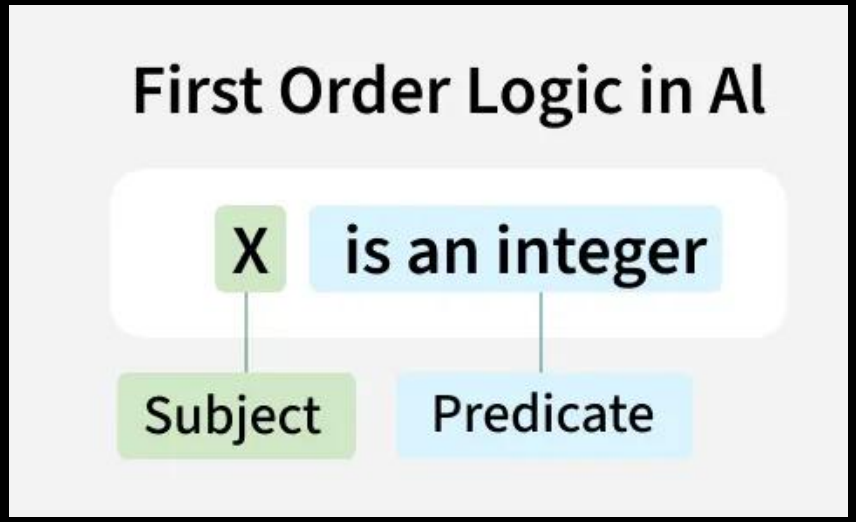
👉 **Propositional logic** only says whether a whole sentence is **true or false**.

👉 **FOL** goes deeper: it talks about **objects, their properties, and relationships** between them.

That's why FOL is also called **Predicate Logic**.

It helps AI and computers reason about the **real world**.

➢ First-Order Logic (FOL) also known as predicate logic helps us represent knowledge, reason through problems and understand language



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Key Components of FOL :

1. Constants

These are names of specific objects or things.

Example:

- Alice (a person)
- 2 (a number)
- NewYork (a city)

Think: Constants = proper nouns.

2. Variables

Placeholders for *any* object (not fixed).

Example:

- x, y, z (can stand for any person, number, city, etc.)

Think: Variables = "someone" or "something".

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3. Predicates

Show **properties** of objects or **relations** between them.

Example:

- Human(x) → "x is a human".
- Likes(Alice, Bob) → "Alice likes Bob".

Think: Predicates = describing words / verbs.

4. Functions

Map one object to another object.

Example:

- MotherOf(x) → gives the mother of person x.
- Age(Alice) → Alice's age.

Think: Functions = special arrows that return one output.

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5. Quantifiers

These tell us about **how many objects** we are talking about.

a) Universal Quantifier (∀) = "for all"

Example:

∀x (Person(x) → Mortal(x))

→ "All people are mortal."

Means: *no exceptions*.

b) Existential Quantifier (∃) = "there exists"

Example:

∃x (Person(x) ∧ Likes(x, IceCream))

→ "There is at least one person who likes ice cream."

Means: *at least one*.

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6. Logical Connectives (same as in propositional logic)

- **AND (∧):** Both must be true → P ∧ Q.
- **OR (∨):** At least one is true → P ∨ Q.
- **NOT (¬):** Negation → ¬P.
- **IMPLIES (→):** If first is true, then second must be true. → P → Q.
- **IFF (↔):** True if both sides are same → P ↔ Q.

📌 Applications of First-Order Logic (FOL) in AI

1. **Knowledge Representation** → Store facts & relationships (e.g., symptoms → diseases).
2. **Automated Theorem Proving** → Prove math/logical statements, verify software.
3. **NLP** → Convert language into logic (e.g., "All cats are animals" → ∀x (Cat(x) → Animal(x))).
4. **Expert Systems** → Decision-making rules (e.g., if fired → compensation).
5. **Semantic Web** → Smarter search by linking data (e.g., author ↔ book).

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⚠ Challenges of FOL :

- ✓ **Complexity** → Needs lots of resources for big knowledge.
- ✓ **Undecidability** → Some problems can't be solved by algorithms.
- ✓ **No Uncertainty** → Only True/False, can't handle "maybe" situations.

Feature	**Propositional Logic (PL)**	**First-Order Logic (FOL)**
Basic Unit	Proposition (a simple statement, e.g., *It is raining*)	Predicate (describes objects & relations, e.g., *Raining(NewYork)*)
Expressiveness	Limited (only True/False facts)	More powerful (can describe relations, properties, quantifiers)
Quantifiers	✗ Not allowed	✓ Has quantifiers: ∀ (for all), ∃ (there exists)
Objects/Entities	✗ Cannot talk about objects	✓ Can talk about objects (Alice, Car, x, y)
Example	"It is hot" OR "It is cold"	"For all x, if x is human → x is mortal"
Use in AI	Simple reasoning (truth tables, circuits)	Complex reasoning (knowledge representation, NLP, expert systems)

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📌 Syntax of First-Order Logic (FOL)

The **syntax** tells us *how to write valid statements* (formulas) in FOL using symbols and rules.
These valid statements are called **Well-Formed Formulas (WFFs)**.

1. Terms in FOL (represent objects/entities)

Terms = things we talk about.

- **Constants** → Specific names (e.g., John, Apple).
- **Variables** → General placeholders (e.g., x, y).
- **Functions** → Give new terms from other terms (e.g., Age(John), Parent(x)).

✅ **Example:** Father(John) means "the father of John."

2. Predicates in FOL (describe properties/relations)

Predicates = statements about objects.

- **One-argument predicate:** property of one object.
  Example: Human(x) → "x is a human".
- **Two-argument predicate:** relation between two objects.

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Example: Parent(x, y) → "x is parent of y".

- ✓ Example: Loves(John, Mary) means "John loves Mary.“

3. Quantifiers in FOL (talk about many objects)

Quantifiers specify scope of statements.

i. **Universal (∀):** "For all"
   Example: ∀x Human(x) → Mortal(x)
   Meaning: "All humans are mortal."

ii. **Existential (∃):** "There exists at least one"
   Example: ∃x Loves(x, IceCream)
   Meaning: "Someone loves ice cream."

4. Connectives in FOL (combine statements)

I. **Conjunction (∧)** → "and"
   Example: P(x) ∧ Q(x) → "x is red and round."

II. **Disjunction (∨)** → "or"
   Example: Cat(x) ∨ Dog(x) → "x is a cat or a dog."

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iii. Implication (→) → "if...then"

Example: Mammal(x) → ProducesMilk(x) → "If x is a mammal, then x produces milk."

iv. Negation (¬) → "not"

Example: ¬Smart(x) → "x is not smart."

5. Well-Formed Formulas (WFFs)

WFFs = valid sentences in FOL made by combining **terms, predicates, quantifiers, and connectives**.

❖ Features of WFFs:

- Must follow syntax rules.
- Quantifiers have proper scope.
- Can be simple (atomic) or complex (nested).

✓ Examples:

- Simple: Human(John)
- Complex: ∀x (Human(x) → Mortal(x))

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6. Importance of WFFs in AI

- ✓ **Knowledge Representation:** Encode facts and rules about the world.
- ✓ **Automated Reasoning:** Used for inference and decision-making.
- ✓ **NLP:** Map natural language into logical form.
- ✓ **Problem-Solving/Planning:** Define states, goals, and rules.

📌 Semantics of First-Order Logic (FOL)

☞ **Semantics** in FOL means: *How to give meaning to formulas* (deciding whether a statement is **true or false** in a given situation or model).

It explains **what the symbols stand for** and **when a formula is true**.

1. Key Elements of Semantics

- **Domain of discourse** → the set of all objects we are talking about (e.g., all people, all numbers).
- **Variables** → placeholders for objects in the domain (x, y).
- **Constants** → fixed objects (e.g., John, 5).

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- **Predicates** → properties or relations (e.g., Human(x) is true if x is a human).
- **Functions** → map one object to another (e.g., Mother(x) gives x's mother).
- **Quantifiers** →
  - ∀ → "for all"
  - ∃ → "there exists at least one"

2. Interpretation (Model)

An **interpretation** tells us how to map symbols to real meaning. It includes:

1.Domain → set of objects.

2.Interpretation of constants → assigns each constant to one object in the domain.

3.Interpretation of predicates → tells us when a predicate is true for objects.

4.Interpretation of functions → gives output objects for input objects.

✓ Example:

- Domain = {John, Mary}
- Constant a = John
- Predicate Loves(x, y) = {(John, Mary)} → meaning John loves Mary.

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3. Truth in FOL

- A formula is **true** or **false** depending on the interpretation.
- **Atomic Formula (like P(x))**: true if predicate holds for that object.
- **Complex Formulas**: truth is checked using connectives (∧, ∨, →, ¬).

4. Quantifiers

I. **Universal (∀x P(x))** → true if P(x) is true for *all* objects in domain.
   Example: ∀x Human(x) → Mortal(x) → "All humans are mortal."

II. **Existential (∃x P(x))** → true if P(x) is true for *at least one* object.
   Example: ∃x Loves(x, IceCream) → "Someone loves ice cream."

5. Satisfaction

- A formula is **satisfied** if it is true under a given interpretation (model).
- Notation: M ⊨ ϕ → Formula ϕ is true in interpretation M.

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6. Validity

- A formula is **valid** if it is true in *every interpretation*.
- Notation: $\models \phi$.

Example: $\forall x\,(P(x) \rightarrow P(x))$ is always valid.

7. Relation between Satisfaction & Validity

I. **Satisfaction** → true in *some model*.

II. **Validity** → true in *all models*.

A formula is valid if its **negation** is never satisfied.

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❖ What is Inference?

In FOL, inference is the process of **deriving new true statements from existing ones** using logical rules.

Example:

- Fact: ∀x (Human(x) → Mortal(x)) (All humans are mortal).
- Fact: Human(Socrates).
- Inference: Mortal(Socrates).

❖ Inference Rules in FOL

These are formal rules used by AI to reason:

1.Modus Ponens (If-Then Rule):

If P → Q and P are true → then Q must be true.

Example:

- If it rains, the ground gets wet.
- It rains.
- → Ground is wet.

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2. Universal Instantiation (UI):

From ∀x P(x), we can take any object "a" and say P(a).

Example:

∀x Human(x) → Mortal(x)

Take "Socrates": Human(Socrates) → Mortal(Socrates)

3. Existential Instantiation (EI):

From ∃x P(x), we can say P(c) for some new constant c.

Example:

- ∃x Loves(x, IceCream) → "Let c be that person" → Loves(c, IceCream).

4. Resolution Rule:

Used in automated theorem proving → combines clauses to remove contradictions.

Example (simplified):

- P ∨ Q
- ¬P
- → Q

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❖ What are Knowledge-Based Agents?

A **knowledge-based agent** is an AI agent that uses **stored knowledge + reasoning** to make smart decisions.

Unlike simple reflex agents (that just react), these agents can:

- **Store facts** about the world.
- **Reason (think)** using those facts.
- **Update** their knowledge after new observations.
- **Act intelligently** based on reasoning.

👉 Example:

An **automated taxi agent** knows the map of a city.

- Knowledge: "Road A connects to Road B".
- Goal: "Go from Station A → Station B".
- It reasons: "I should take Road A → Road B to reach the goal".

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❖ Why use a Knowledge Base (KB)?

- A **Knowledge Base (KB)** is a collection of facts and rules about the world.
- The agent uses **inference** (reasoning) to get new facts from old ones.

☞ Example:

- Fact: All humans are mortal.
- Fact: Socrates is a human.
- Inference: Socrates is mortal.

❑ Two inference methods used:

1.Forward chaining → Start with facts → apply rules → get new conclusions.

2.Backward chaining → Start with a goal → check what facts/rules support it.

❖ Levels of Knowledge-Based Agents

1. Knowledge Level

What the agent knows + its goals.

Example: Taxi agent knows "A → B" route.

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2. Logical Level

How knowledge is **represented** (in logic statements).

Example: "Route(A, B)" is encoded in logic.

3. Implementation Level

Actual execution of logic in the real world.

Example: Taxi agent physically drives from A → B.

❖ Main Components of a Knowledge-Based Agent

1.Knowledge Base (KB):

1. Stores **facts and rules** about the world.
2. Example: "If raining → road is slippery".

2.Inference Engine (IE):

1. The "reasoning brain" → applies logic to KB to get new conclusions.
2. Example: "It is raining → Therefore, road is slippery".

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❖ Features of Knowledge-Based Agents

- Store and update knowledge.
- Think/reason using inference.
- Explain decisions to humans.
- Act intelligently (not just react).

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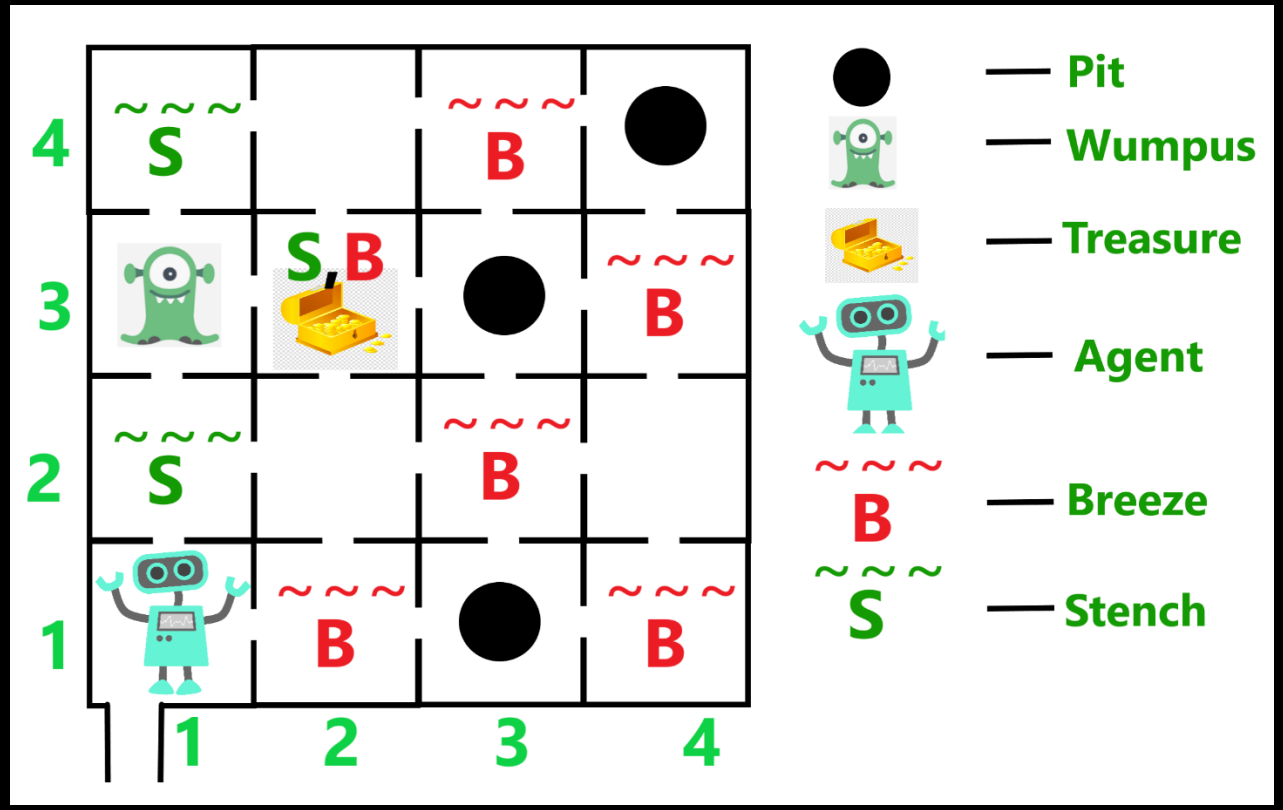
❖ What is Wumpus World?

Wumpus World is a **grid-based game environment**.
The agent (our AI player) lives in a **cave of rooms**, organized as a grid (usually 4×4).
Some rooms are safe, while others are dangerous (with pits or a monster called **Wumpus**).
The goal of the agent:

- Find the **gold** 💰.
- Avoid dangers (Wumpus & pits).
- Safely **come back out** of the cave.



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❖ Elements in Wumpus World

1. Wumpus 🐉

A monster that kills the agent if they enter its room.

2. Pits 🕳️

Bottomless holes; if agent falls, game over.

3. Gold 💰

The treasure that the agent must find.

4. Agent

Our AI player that moves around the grid.

5. Walls

Boundaries of the cave.

❖ Actions the Agent Can Take

- **Move Forward**
- **Turn Left / Turn Right**
- **Grab** (pick up the gold)
- **Shoot** (arrow to kill Wumpus)
- **Climb** (exit the cave if at start square)

❖ Rules of the Game

- Agent starts at location (1,1).
- Agent has **one arrow**.
- Agent must use **reasoning** (knowledge-based logic) to:
  - Avoid unsafe rooms.
  - Locate gold safely.
- Winning = Grab gold + climb out alive.

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❖ Step-by-Step Working :

◆ Step 1: Start (1,1)

Agent starts at (1,1).

Percepts: *Nothing dangerous here → safe room*

◆ Step 2: Move to (2,1)

Percepts: Stench (S).

Meaning: The Wumpus is **nearby** (either in (1,1), (2,2), or (3,1)).

But agent already knows (1,1) is safe → so danger is in (2,2) or (3,1).

◆ Step 3: Move to (1,2)

Percepts: Breeze (B).

Meaning: A pit is **nearby** (in (2,2) or (1,3)).

Now agent can combine knowledge:

- (2,2) is suspected of both pit (breeze) and Wumpus (stench).
- So (2,2) is very dangerous → avoid it.

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◆ **Step 4: Move towards (2,2)? → No! Too risky.**

Agent avoids this square because both *breeze + stench* indicate danger.

◆ **Step 5: Shoots Arrow towards (1,3)**

Because stench was detected, agent guesses Wumpus is in (1,3).

Fires → hears **scream**.

✓ **Wumpus is killed.**

◆ **Step 6: Moves to (2,3)**

Percepts: Glitter (Gold).

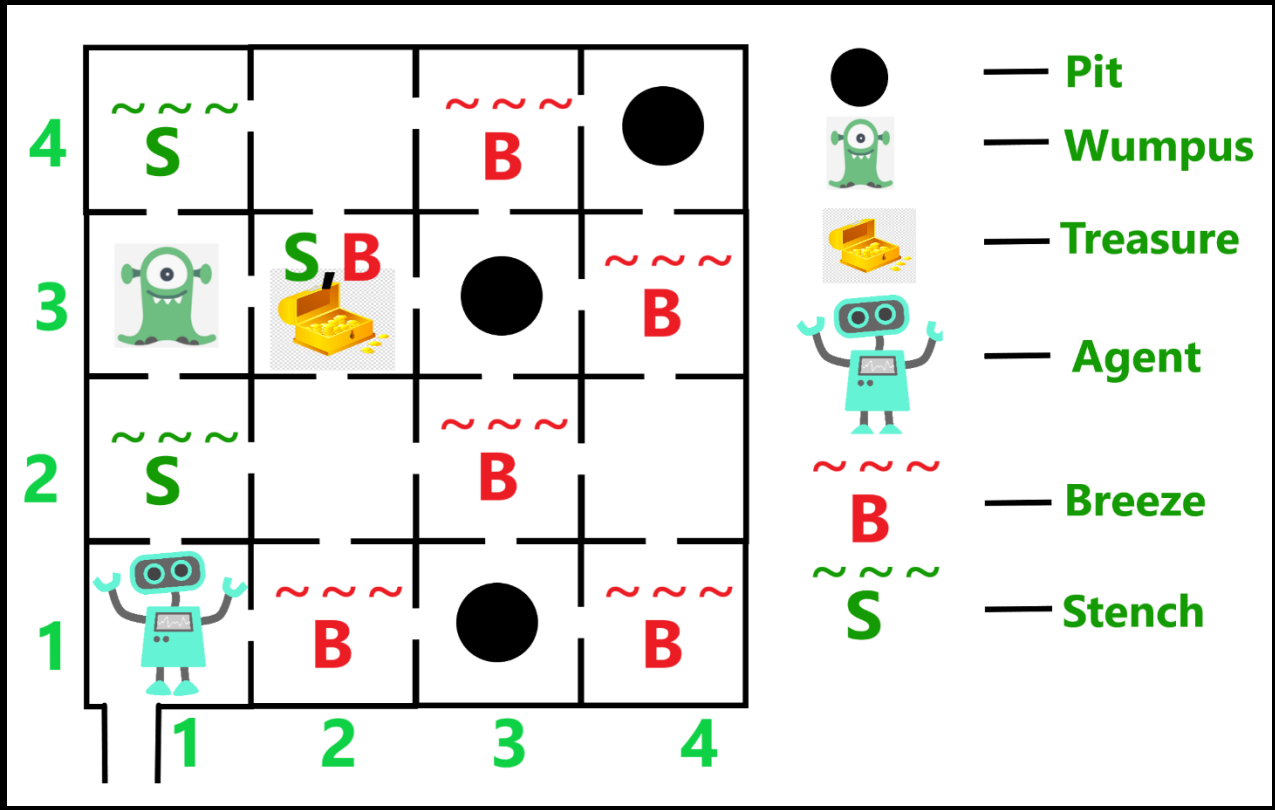
Agent **Grabs Treasure** 💰.

◆ **Step 7: Returns to (1,1)**

Agent traces back safely avoiding pits.

◆ **Step 8: Climbs out (Exit)**

✓ **Mission Successful: Agent escapes with Gold alive!**



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🏹 PEAS Description of Wumpus World :

PEAS = **Performance Measure, Environment, Actuators, Sensors**

It describes how an **agent** (the explorer) interacts with the **Wumpus World**.

1. Performance Measures (How success/failure is judged)

✓ +1000 → Agent gets the gold and returns safely.

✓ –1000 → Agent dies (falls into a pit or eaten by Wumpus).

✓ –1 → For every move (to minimize unnecessary moves).

✓ –10 → If agent uses arrow (penalty for wasting resources).

👉 **Goal:** Maximize total score.

2. Environment (Where the agent operates)

- A cave with **4x4 grid = 16 rooms**.

Hazards:

- **Wumpus** (dangerous monster).
- **Pits** (fall → death).

Clues:

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- **Stench** near Wumpus.
- **Breeze** near Pit.
- **Glitter** in Gold's room.

Agent starts at **(1,1), facing right**.
Gold, Wumpus, and pits can be anywhere except (1,1).

3. Actuators (What the agent can do)

- **Move Forward** → Goes to next square.
- **Turn Right/Left** → Rotates 90°.
- **Shoot Arrow** → Can kill Wumpus.
- **Grab** → Picks up gold.
- **Release** → Drops gold.

4. Sensors (What the agent can sense)

- **Breeze** → Pit nearby.
- **Stench** → Wumpus nearby.
- **Glitter** → Gold in current room.
- **Scream** → Wumpus has been killed.
- **Bump** → Agent hit a wall.

🔄 How the Agent Operates with PEAS

1.Perception → Collects sensory input (breeze, stench, glitter, bump, scream).
2.Inference → Uses logic to guess location of hazards/gold.
3.Planning → Decides next safe step (avoid suspected danger, choose path).
4.Action → Executes move (walk, turn, shoot, grab).
5.Repeats this cycle until it finds gold and exits safely.

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❖ Logic Programming using Prolog :

1. What is Logic Programming?

- **Logic Programming** is a way of programming based on **logic and reasoning**, not step-by-step instructions.
- Instead of telling the computer **how** to do something, we tell it **what is true** and let it figure out the answers.
- It is based on **mathematical logic** (mainly **First-Order Logic**).

👉 Example:

Instead of writing code to calculate if someone is a grandfather, we just define:

- X is a parent of Y
- Y is a parent of Z
- If X is parent of Y and Y is parent of Z, then X is a grandfather of Z.

The system then **reasons** and finds answers.

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❖ What is Prolog?

Prolog = **PROgramming in LOGic**

It is the most popular **logic programming language**.

Works on the principle of **facts**, **rules**, and **queries**.

❑ Key Concepts in Prolog

(a) Facts

Facts tell what is true in the system.

```
parent(john, mary).
parent(mary, ann).
```

☞ **Means:** John is a parent of Mary, Mary is a parent of Ann.

(b) Rules

Rules describe **relationships** using logic.

```
grandparent(X, Z) :- parent(X, Y), parent(Y, Z).
```

☞ **Means:** If X is a parent of Y and Y is a parent of Z, then X is a grandparent of Z.

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(c) Queries

We ask Prolog questions, and it tries to **prove them true** using facts and rules.

?- grandparent(john, ann).

👉 **Output:** **Yes** (because John → Mary → Ann).

❑ How Prolog Works (Execution Style)

Prolog uses a reasoning method called **backtracking**:

1.Start with a query.

2.Match query with facts/rules.

3.If match found → success.

4.If not → try another possibility (backtrack).

👉 **Example:** If we ask ?- parent(john, ann).

- Prolog checks facts.
- Doesn’t find parent(john, ann). → says **No**.

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❖ Applications of Prolog

1. **Artificial Intelligence**
   Expert systems (medical diagnosis, legal reasoning).
   Natural Language Processing (chatbots, grammar checking).
2. **Problem Solving**
   Puzzles, games (like Sudoku, Wumpus world).
3. **Knowledge Representation**
   Representing facts and rules about the world.
4. **Robotics**
   Robots use reasoning to make decisions.

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❖ Forward Chaining (Data-Driven) :

Forward chaining is a data-driven inference technique. It starts with the available data and applies rules to infer new data until a goal is reached. This method is commonly used in situations where the initial data set is extensive, and the goal is to derive conclusions from it.

☞ Think of it like **starting from facts and moving forward step by step** until you reach the goal.

❖ How it works :

1. Start with what you already know (**facts**).
2. Check which **rules** can be applied with these facts.
3. When a rule matches, **create a new fact**.
4. Keep repeating until:

- You reach your **goal**, or
- No more new facts can be created.

✓ Example:

Fact: Patient has **fever**.

Rule: If **fever + rash → measles**.

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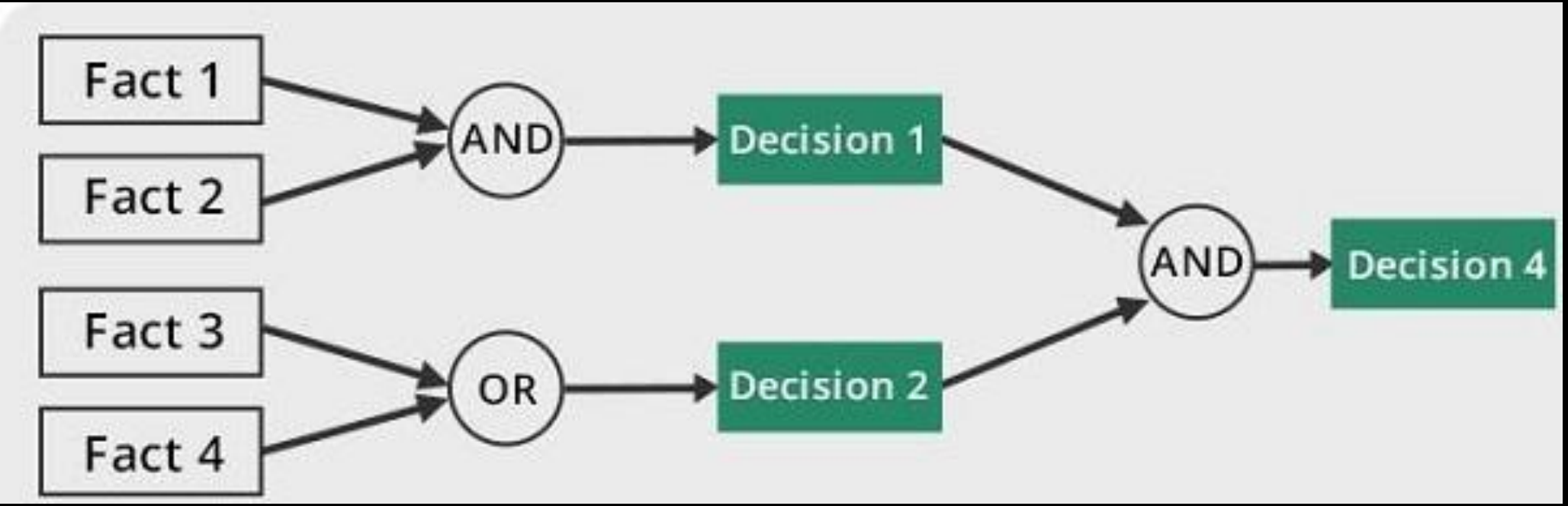
If we later find "rash", system concludes → **measles**.

☞ **Forward chaining = Facts → Rules → New Facts → Goal**



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❖ Backward Chaining (Goal-Driven)

Backward chaining is a goal-driven inference technique. It starts with the goal and works backward to determine which facts must be true to achieve that goal. This method is ideal for situations where the goal is clearly defined, and the path to reach it needs to be established

☞ Think of it like **starting from the goal and moving backward to check if it can be proved**.

❖ How it works :

1. Start with a **goal** (what you want to prove).
2. Look for rules that could conclude this goal.
3. Check if the conditions of those rules are true.
4. If not, break them into **sub-goals** and check again.
5. Continue backward until you reach actual facts or fail.

✓ Example:

Goal: Why is the **network down**?

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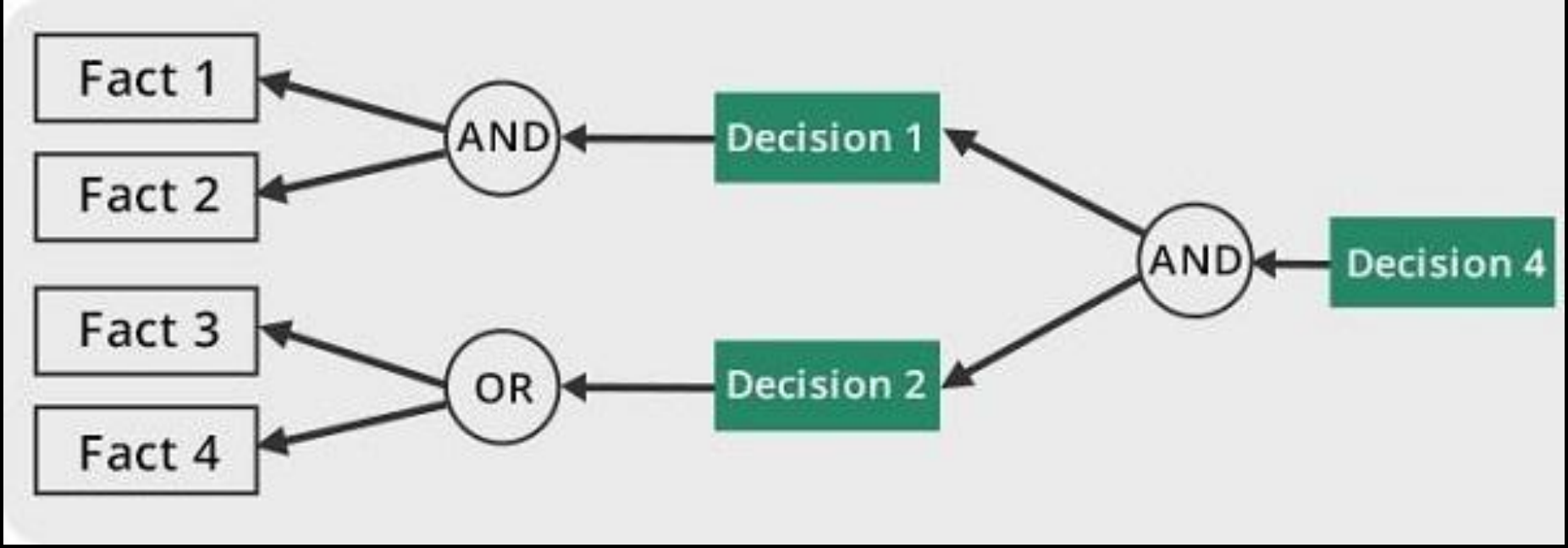
Rule: If router is not working → network down.
Check if router is working.

- If router is faulty → Goal proved.
- If router is fine → Try other rules.

☞ Backward chaining = **Goal → Rules → Check Facts (or Sub-Goals)**



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Feature	Forward Chaining	Backward Chaining
Driven by	Data (facts)	Goal
Starts from	Facts	Goal
Best for	Situations with lots of data, want to explore all possible outcomes	Specific problems where the goal is already known
Example use	Medical diagnosis (finding all possible diseases)	Troubleshooting (finding why something failed)

- **Forward Chaining → Start with facts → Move forward.**
- **Backward Chaining → Start with goal → Move backward.**

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🌟 What is Resolution Algorithm?

Resolution is a method in **Artificial Intelligence** used for **logical reasoning**.

It helps us check whether a given statement (goal) can be proven true using known facts (knowledge base).

👉 In simple words:

It's like solving a puzzle by combining small logical pieces until we either:

I. **Find a contradiction** → means the statement is proved.

II. **No contradiction** → means the statement can't be proved.

🛠How Resolution Algorithm Works (Easy Steps)

Step 1: Convert to CNF (Conjunctive Normal Form)

- CNF means: statements written as **AND of ORs**.
- **Example:**
  - Statement: "If it is raining, then the ground is wet."
  - In logic: Raining → Wet
  - CNF form: ¬Raining ∨ Wet

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Step 2: Apply Resolution Rule

Resolution rule:

If we have

- (A ∨ B)
- (¬A ∨ C)

  👉 We can combine them into (B ∨ C) by removing A and ¬A.

✅ This helps simplify big problems step by step.

Step 3: Keep Repeating

Keep applying resolution on pairs of clauses.

Two possible results:

I. **Empty Clause (Contradiction)** → means the statement is **proved** true.

II. **No Empty Clause** → means the statement is **not provable** with given facts.

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❖ Example (Simple)

Knowledge Base (KB):

1.¬Rain ∨ Wet (If it rains, ground is wet)

2.Rain

Goal: Prove Wet.

👉 Steps:

1.Add negation of goal: ¬Wet.

2.Now KB = { ¬Rain ∨ Wet, Rain, ¬Wet }

3.Resolve (¬Rain ∨ Wet) with (¬Wet) → gives ¬Rain.

4.Resolve (¬Rain) with (Rain) → gives **empty clause**.

✅ Empty clause means **proved** → The ground is wet.

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🌟 What are Ontologies in AI?

☞ Ontology is like a **map of knowledge** that tells us:

- **What kinds of things exist** in a domain (movies, actors, scripts, etc.),
- **What properties they have** (runtime, age, rating, etc.),
- **How they are related** (actor acts in a movie, director directs a movie, etc.).

So, ontology = **categories + properties + relationships** that help computers understand meaning, not just words.

- Ontology means creating a structured map of knowledge where we identify **things, their groups, their properties, and their relationships**.

Think of ontologies as smart organizing systems for knowledge. Just as a library uses categories to organize books (fiction, non-fiction, science, history), ontologies create structured ways to organize information, enabling computers and people to understand it more effectively. Instead of just throwing information into random buckets, ontologies define how different pieces of information connect. They're like creating a family tree, but for ideas and concepts.

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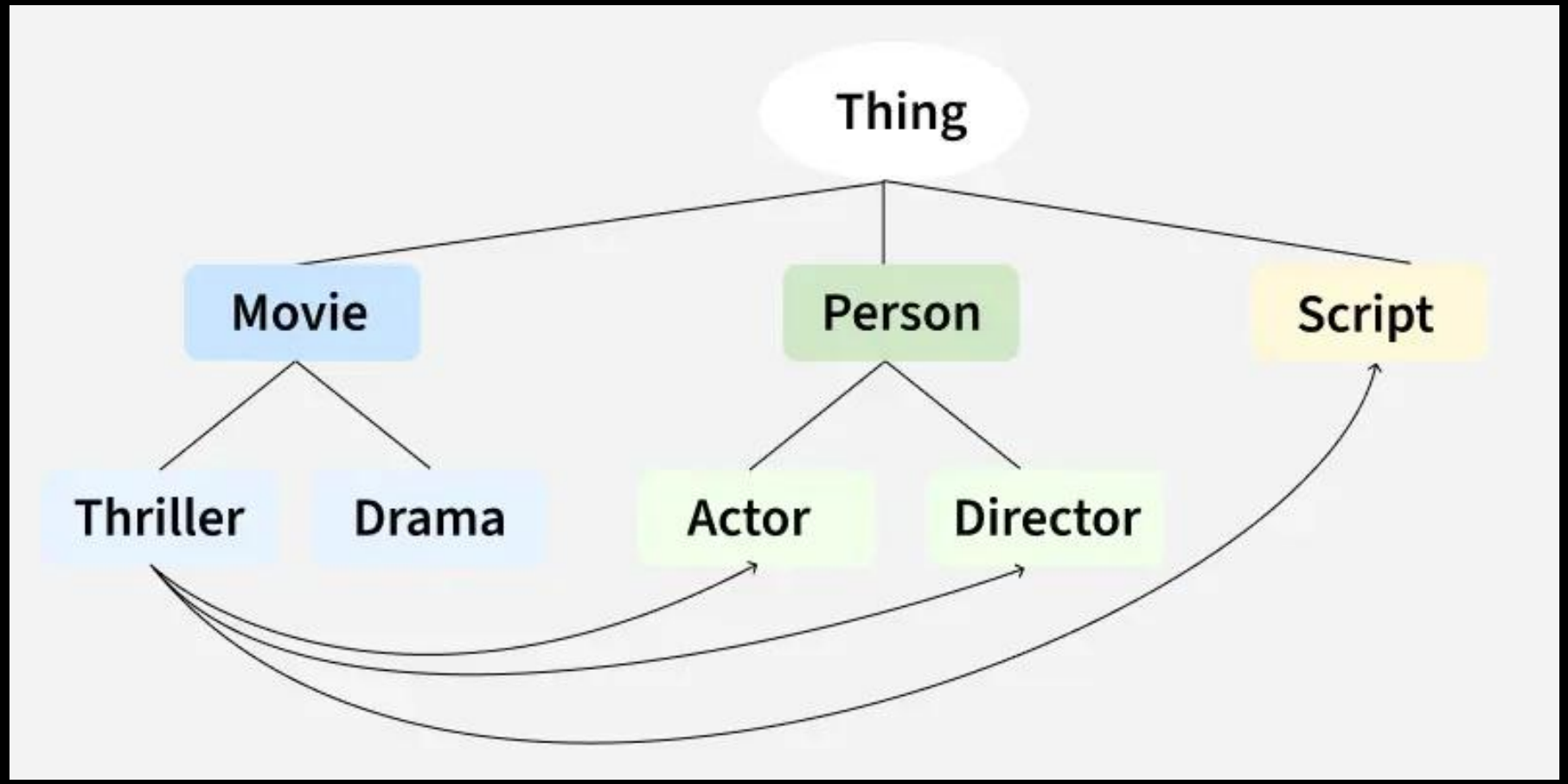
Let's Look at a Simple Example: Movies

Imagine you're building a database about movies. An ontology would help you logically organize all the movie information:



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🎬 Movie Ontology Example with *Raja Hindustani*

1. Individuals (Specific things – real examples)

These are actual items in the world:

- Movie: *Raja Hindustani*
- People: *Aamir Khan, Karisma Kapoor, Dharmesh Darshan (director), Nadeem-Shravan (music)*
- Studio: *Tips Industries Limited*

2. Classes (Categories / Groups)

Types of things:

- Movie type: *Romantic Drama*
- People type: *Actor, Actress, Director, Music Composer*
- Format: *Theater Release, DVD, TV Broadcast*

3. Properties (Attributes / Features)

Details that describe them:

Movie (*Raja Hindustani*) → *runtime: 176 minutes, release year: 1996, rating: blockbuster hit*

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Actor (*Aamir Khan*) → *age at that time: 31, nationality: Indian, awards: Filmfare Best Actor*
Studio (*Tips*) → *location: Mumbai, founded: 1975*

4. Relationships (Connections – how things link)

How they are connected:

- *Aamir Khan* → **acted in** → *Raja Hindustani*
- *Karisma Kapoor* → **acted in** → *Raja Hindustani*
- *Dharmesh Darshan* → **directed** → *Raja Hindustani*
- *Nadeem-Shravan* → **composed music for** → *Raja Hindustani*
- *Tips Industries* → **produced** → *Raja Hindustani*

🌟 **Why this is Ontology?**

Because we are:

✓ Identifying **things (individuals)** → Raja Hindustani, Aamir Khan
✓ Grouping into **categories (classes)** → Actor, Director, Movie type
✓ Giving **details (properties)** → runtime, age, rating
✓ Connecting them via **relationships** → acted in, directed, produced

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👉 Why it is needed?

- To give computers a **clear understanding** of knowledge.
- To make information **organized, reusable, and sharable**.
- To help in **AI, expert systems, knowledge-based applications**.

👉 Steps in Ontological Engineering:

1. **Decide the domain** (e.g., movies, healthcare, education).
2. **List important concepts** (e.g., Movie, Actor, Director).
3. **Define categories (classes)** (e.g., Action movie, Drama).
4. **Define properties** (e.g., movie has runtime, person has age).
5. **Define relationships** (e.g., Actor → acts in → Movie).
6. **Test and refine** the ontology.

📌 ***Example***: In Movie Ontology, we define → Actor, Director, Movie, Script and their connections.

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❖ Ontological Reasoning

Ontological reasoning means **using ontology to make logical conclusions**.

It is the process where the system **reasons (thinks)** using the structured knowledge in ontology.

👉 How it works?

The computer checks the relationships and rules in the ontology.

Then it **derives new facts** or **answers queries**.

👉 Example:

- Knowledge in ontology:
- Aamir Khan is an **Actor**.
- An **Actor** is a type of **Person**.
- **Reasoning result**: Aamir Khan is also a **Person**.

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Thank You....

Download Notes : https://rzp.io/rzp/VsXqbmB

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Unit – 4 Uncertainty & Learning Techniques

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IV	**Uncertainty & Learning Techniques:** Introduction to uncertainty and probabilistic reasoning, Bayes' Rule, Bayesian Networks, Fuzzy logic and handling imprecision, Neural Networks (basics only): Perceptron, Backpropagation (intro level), Fundamentals of Machine Learning in AI context, Introduction to supervised and unsupervised learning.

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❖ What is Uncertainty in AI?

Uncertainty means **not being 100% sure about something**.

Uncertainty in AI means **lack of complete confidence in decisions**.

In AI, uncertainty comes because:

- Data may be incomplete or wrong.
- Models may not be perfect.
- Real-world is complex and unpredictable.

Example: A self-driving car sees an object on the road. Is it a plastic bag or a rock? The AI is **uncertain**.

❖ Understanding Uncertain Domains

An uncertain domain refers to an environment where information is incomplete or unpredictable. Unlike deterministic systems where outcomes are precisely predictable, AI systems operating in uncertain domains must handle:

- **Incomplete Information:** Missing data needed for fully informed decisions
- **Ambiguity:** Inputs that support multiple interpretations

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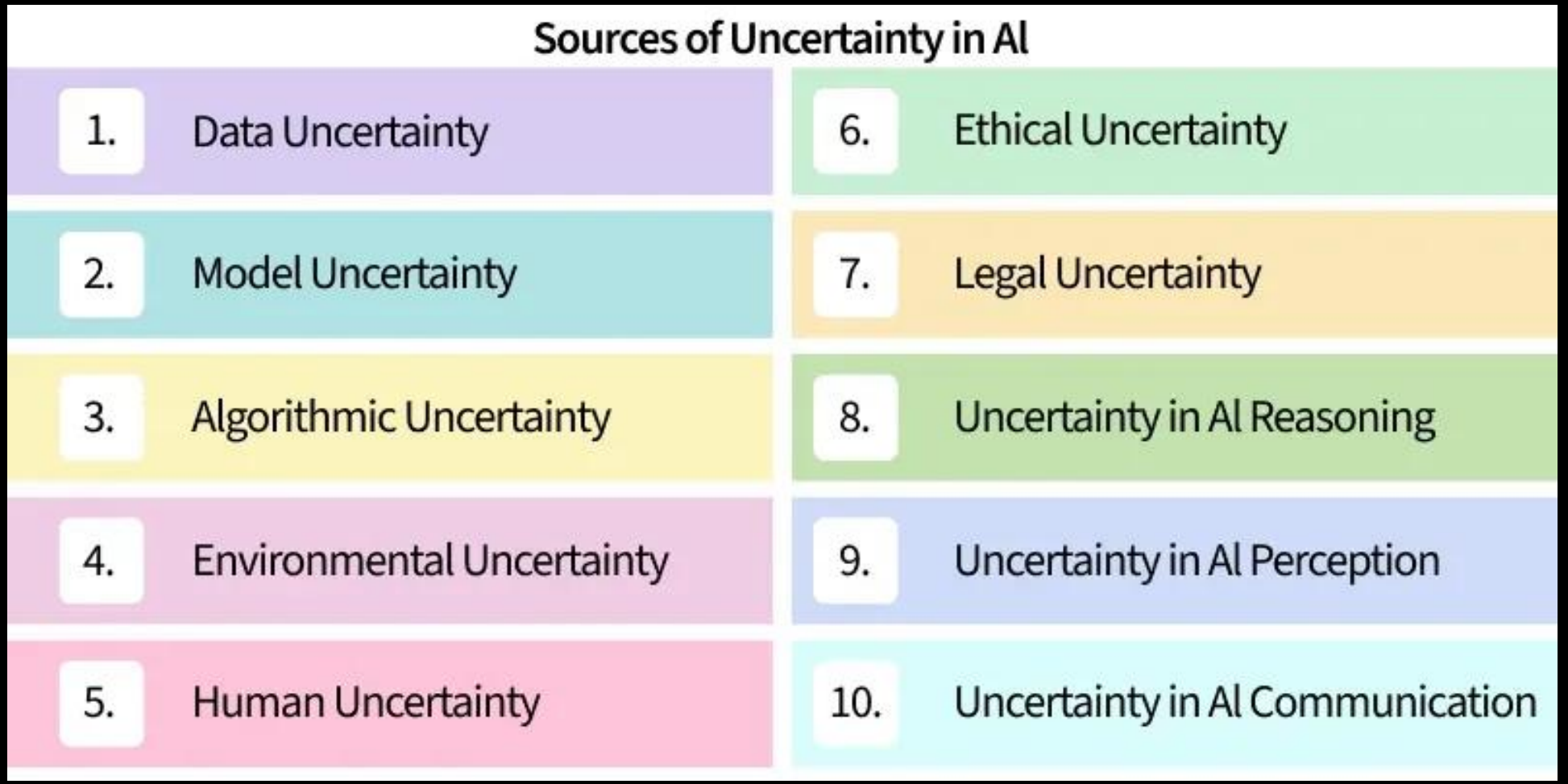
- **Noise**: Data corruption due to measurement errors or external factors
- **Stochastic Processes**: Events with probabilistic outcomes that can't be predicted deterministically



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❖ Sources of Uncertainty in AI

1. Data Uncertainty

Data may be incomplete, noisy, or biased.

Example: If medical data is missing symptoms, diagnosis may be uncertain.

2. Model Uncertainty

AI models are simplifications of reality → they cannot capture everything.

Example: A weather prediction model may not include all factors.

3. Algorithmic Uncertainty

Algorithms may not always give the right solution (approximation, randomness).

Example: A search algorithm may not find the best path always.

4. Environmental Uncertainty

Real-world environment changes unexpectedly.

Example: Self-driving car → sudden rain, fog, or unexpected obstacles.

5. Human Uncertainty

Humans provide wrong or unclear input.

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Example: Voice assistant mishears commands.

6. Ethical Uncertainty

Confusion about what is right or wrong for AI to do.

Example: Should a self-driving car save the passenger or pedestrian in an accident?

7. Legal Uncertainty

No clear laws or rules about AI in many situations.

Example: Who is responsible if AI causes an accident?

8. Uncertainty in AI Reasoning

AI might not always make the right logical conclusions.

Example: Medical expert system may misinterpret symptoms.

9. Uncertainty in AI Perception

AI sensors (camera, microphone) may not capture information correctly.

Example: Image recognition system mistakes a cat for a dog.

10. Uncertainty in AI Communication

Misunderstanding between AI and human during communication.

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Example: Chatbot misinterprets user's question.

🌩️What is Probabilistic Reasoning?

Probabilistic reasoning means making decisions or predictions when we are **not 100% sure** but have **some probability (chances)**.
Instead of saying "Yes/No," it says "How likely" something will happen.

Example:

- If it rains → Cricket match may get canceled.
- If it does not rain → Cricket match will happen.
- AI assigns probability: Rain (60%), Match cancel (60%).

So, probabilistic reasoning helps AI handle **uncertainty**.

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Key Concepts in Probabilistic Reasoning

1. Probability

A number between 0 and 1 that shows the chance of an event.

Example: Probability of rain tomorrow = 0.7 (70%).

2. Random Variables

Variables whose values are uncertain.

Example: Weather = {Rain, No Rain}.

3. Prior Probability

Probability based on previous knowledge.

Example: In rainy season, chance of rain = 80%.

4. Conditional Probability

Probability of an event happening, given that another event has already happened.

Example: P(Match Cancel | Rain) = Probability match cancels if it rains.

5. Bayes' Rule

A formula to update probabilities when new evidence comes.

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Example: If dark clouds appear, the probability of rain increases.

6. Joint Probability

Probability of two events happening together.

Example: Probability of (Rain AND Match Cancel).

7. Inference

Process of drawing conclusions from probabilities.

Example: If rain probability is high → infer match cancellation is likely.

? Why Do We Need Probabilistic Reasoning?

- Real world is uncertain – data is incomplete or noisy.
- Deterministic (yes/no) reasoning is not enough.
- Helps AI make better predictions with limited information.
- More realistic decision-making like humans (we also think in terms of chances).

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📌 Applications of Probabilistic Reasoning

1. Medical Diagnosis

Predicting diseases based on symptoms.

Example: Fever + cough → 70% flu, 30% normal cold.

2. Weather Forecasting

Predicting rain, storms, temperature with probabilities.

3. Self-Driving Cars

Predicting chances of pedestrian crossing, road accidents.

4. Spam Filtering in Email

Probability that an email is spam or not.

5. Speech & Image Recognition

AI calculates probability of which word/image is correct.

6. Games and Sports

Predicting chances of win/loss, match cancellation.

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❖ Bayes Theorem (Bayes Rule) :

- Bayes' Theorem is a formula that tells you **how to update the probability of a hypothesis** when you get new evidence.
- **Bayes' Theorem** is a mathematical formula used to determine the **conditional probability** of an event based on prior knowledge and new evidence.
- It adjusts probabilities when new information comes in and helps make better decisions in uncertain situations.

❖ Conditional Probability

Conditional probability is the probability of one event happening **given that** another event has already happened.

It tells us **"what is the chance of A if we already know B has happened?"**

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Iska matlab:

- $P(A|B)$ = A hone ki chance **jab B ho chuka hai.**
- $P(A \cap B)$ = A aur B dono hone ki chance.
- $P(B)$ = B hone ki chance.

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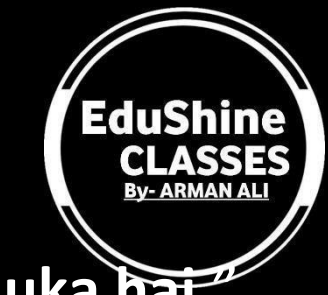

"Kisi event ke hone ki probability, **jab hume already pata hai ki koi aur event ho chuka hai.**"

❑ Example (Daily life)

- Suppose tumhare sheher me **rain hone ki probability 0.3 (30%)** hai.
- Agar rain hoti hai to **umbrella lene ki probability 0.9 (90%)** hai.

Ab sawal: **Tumhare umbrella lene ki probability kitni hai, jab already rain ho chuki hai?**

Given Data:

- P(B) = Probability of Rain = **0.3**
- P(A|B) = Probability of taking Umbrella if Rain happens = **0.9**

☞ Find probability of umbrella lene ki, **jab already rain ho chuki hai**.

Yaani P(A|B) ?

Step 1: Formula of Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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Step 2: Hume $P(A \cap B)$ nikalna hoga

$$P(A \cap B) = P(B) \times P(A|B)$$

$$P(A \cap B) = 0.3 \times 0.9 = 0.27$$

Step 3: Ab formula me value put karo

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = \frac{0.27}{0.3} = 0.9$$

✓ **Final Answer:**

Probability of taking Umbrella given Rain = **0.9 (90%)**

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❖ Bayes Theorem Formula

For any two events A and B, **Bayes's** formula for the Bayes theorem is given by:

$$P(A|B) = \frac{P(B|A)\ P(A)}{P(B)}$$

- **P(A) = Prior probability**
  मतलब पहले से जो मान्यता थी कि A होने की कितनी संभावना है।
  (पहले का विश्वास)
- **P(B|A) = Likelihood**
  अगर A सच है, तो B होने की कितनी संभावना है।
  (evidence की strength)
- **P(B) = Evidence (total probability of B)**
  B कितनी बार होता है, चाहे A हो या न हो।
  (कुल chance of B)
- **P(A|B) = Posterior probability**
  नया belief, जब हमें evidence B मिल चुका है।
  (update किया हुआ विश्वास)

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❑ Example of Bayes' Theorem 🐱🐶

Question:

In a pet shop,

- 60% animals are **cats** → P(Cat) = 0.6
- 40% animals are **dogs** → P(Dog) = 0.4

If an animal is a cat, the chance of meowing = 80% → P(Meow|Cat) = 0.8

If an animal is a dog, the chance of meowing = 10% → P(Meow|Dog) = 0.1

Now, if you hear a **Meow**, what is the probability that it is a **Cat**?

Step 1: Write the Formula

$$P(Cat|Meow) = \frac{P(Meow|Cat) \cdot P(Cat)}{P(Meow)}$$

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Step 2: Find Evidence P(Meow)

$$P(Meow) = P(Meow|Cat) \cdot P(Cat) + P(Meow|Dog) \cdot P(Dog)$$

$$P(Meow) = (0.8 \times 0.6) + (0.1 \times 0.4)$$

$$P(Meow) = 0.48 + 0.04 = 0.52$$

Step 3: Put Values in Formula

$$P(Cat|Meow) = \frac{0.8 \times 0.6}{0.52}$$

$$P(Cat|Meow) = \frac{0.48}{0.52} \approx 0.923$$

✅ **Final Answer:**

The probability that the animal is a **Cat** after hearing a **Meow** = **92.3%**.

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🌐 Bayesian Network (BN) :

A **Bayesian Network** (also called Bayesian Belief Network) is a **probabilistic graphical model** that shows how different variables (events) are related to each other using **probabilities**.
It is very useful to **reason under uncertainty** and to **predict outcomes**.
It is Probalistic graphical model that represent a set of variables and their conditional dependencies via a DAG (Directed acyclic graph).

✨ Main Points

1. Structure

- Represented as a **Directed Acyclic Graph (DAG)**.
- **Nodes** = Random variables (e.g., Rain, Traffic, Late).
- **Edges (arrows)** = Relationships between variables (cause → effect).

2. Conditional Probability Table (CPT)

- Each node has a CPT that defines the probability of the node **given its parent nodes**.

Example: If Rain happens, then probability of Traffic is higher.

3. Joint Probability

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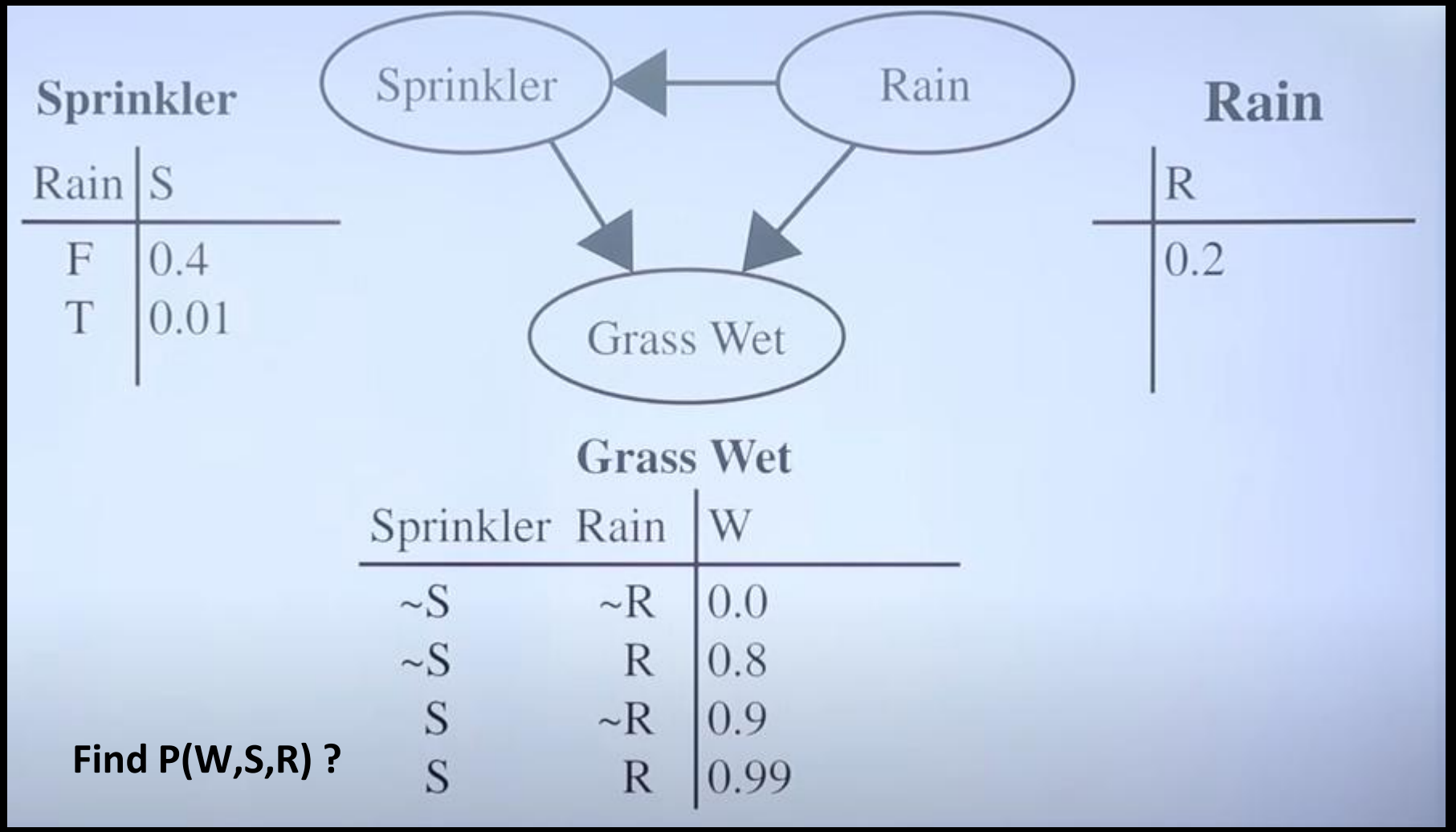
By multiplying all CPTs, we can calculate the probability of any event combination.

Example 1 :



Sprinkler

Rain	S
F	0.4
T	0.01

Rain

R
0.2

Grass Wet

Sprinkler	Rain	W
~S	~R	0.0
~S	R	0.8
S	~R	0.9
S	R	0.99

Find P(W,S,R) ?

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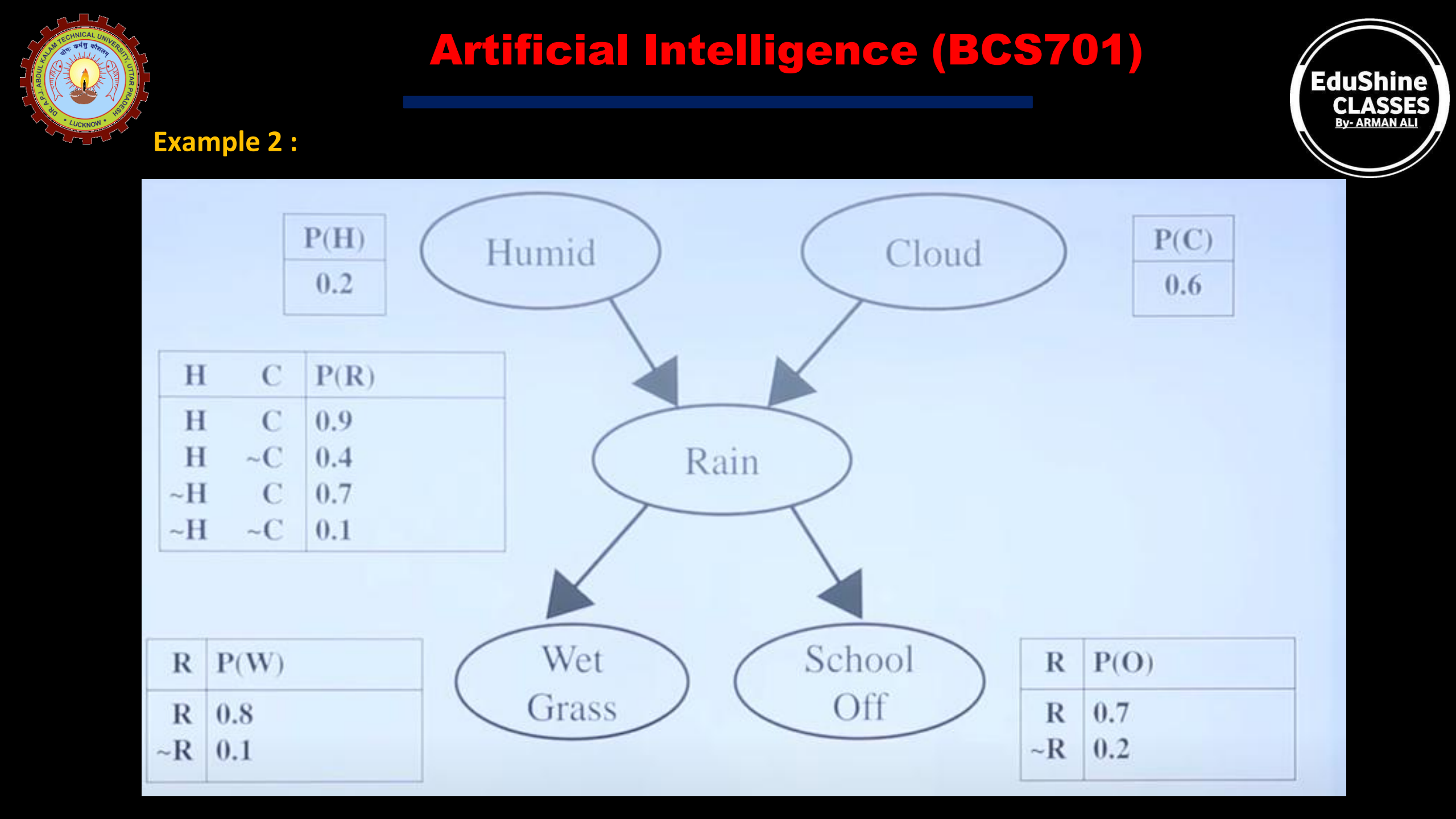

Example 2 :

P(H)
0.2

P(C)
0.6

H	C	P(R)
H	C	0.9
H	~C	0.4
~H	C	0.7
~H	~C	0.1

R	P(W)
R	0.8
~R	0.1

R	P(O)
R	0.7
~R	0.2

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🌟 Fundamentals of Machine Learning (ML) in Artificial Intelligence (AI)

❖ What is Machine Learning?

- **Machine Learning (ML)** is a branch of **AI** that allows computers to **learn from data** without being explicitly programmed.
- Instead of giving fixed instructions, we give the system data and let it **find patterns** and **make predictions/decisions**.

👉 Example:

If we want a computer to recognize cats in pictures:

- Traditional programming → Write rules: “If it has whiskers, 4 legs, tail = Cat.” (hard, because cats vary).
- ML → Show thousands of cat and non-cat images, the system **learns patterns** automatically.

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❖ Key Components of ML

1. **Data** – The information we use to train the model (images, numbers, text, etc.).
2. **Features** – Important characteristics taken from data (for image: color, shape).
3. **Model** – A mathematical representation that makes predictions.
4. **Training** – The process of teaching the model by giving it data + correct answers.
5. **Testing** – Checking how well the model performs on new, unseen data.

❖ Types of Machine Learning

I. **Supervised Learning** (Learn from labeled data)
II. **Unsupervised Learning** (Learn from unlabeled data)
III. **Reinforcement Learning** (Learn by trial and error)

❖ Applications of ML in AI

✓ **Healthcare** – Disease prediction, drug discovery.
✓ **Finance** – Fraud detection, stock prediction.
✓ **E-commerce** – Product recommendation.
✓ **Self-driving cars** – Detecting traffic, pedestrians.

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✓ **Natural Language Processing (NLP)** – ChatGPT, translation, voice assistants.

Why ML is Important in AI?

- AI needs **intelligence** → ML provides **learning ability**.
- Instead of manually coding rules, ML adapts itself with data.
- Helps AI systems improve **accuracy and decision making** over time.

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to how people think.

Neuron :-

* Biological Neuron are also called Nerve cell.
* Basic working unit of the brain.
* Specialized cells responsible for sending and receiving signals from the brain.
* Neurons send and receive messages allowing us to move, feel, think and react to the world around us.

Structure of a Neuron :-

① ~~Dendrites~~ -

① Cell body (Soma) - This is the main part of the neuron that contains the nucleus (which controls the cell) it helps keep the neuron alive & process information.

* Sum up all the incoming i/p signals it also consist of threshold unit.

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(2) Dendrites -
* They are branch-like structures that come out of the cell body.
* Dendrites receive signals from other neurons and bring that information to the cell body.

(3) Synapse -
The is the small gap between neurons. Signals cross the gap called synapse using chemical neurotransmitters to get from one neuron to the next.

---

* Structure of a Biological Neuron

(1) Dendrite - (I/P)
It receives signals from other neurons.

(2) Soma (Cell body) -
Sums up all the incoming input signals. It also consist of threshold unit

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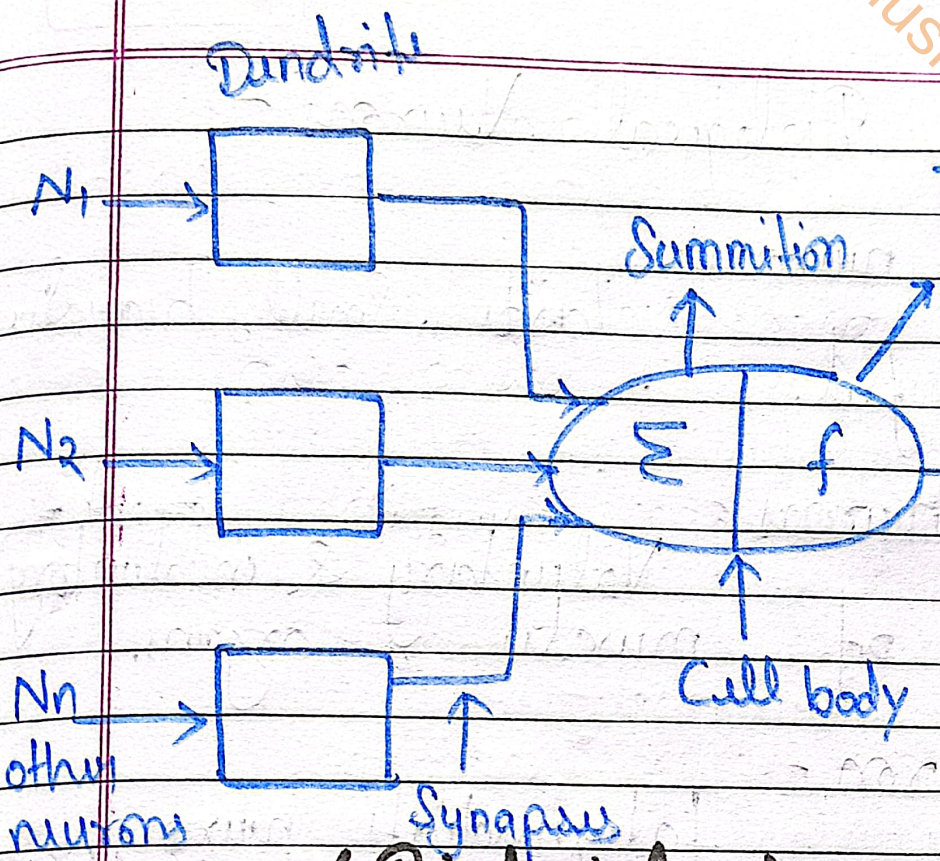

यदि Signal का Sum एक a threshold value से Cross कर देता है then o/p generate होता है

(Biological Nuron)

③ Synapse - (Weighted Connection) - The point of interconnection of one nuron with other nurons. The amount of Signal transmitted depends upon the strength (Synaptic weights) of the connection.

Connection can be inhibitory (decreasing strength) or excitatory (increasing strength) in natur

④ Axon (output) - When the Sum reaches a threshold value, Nuron fires and generates an output Signal which travel down to the other nurons.

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Types of Biological Neurons -

① Sensory neurons - Taste, Smell, Hear, See and Feel.

② Motor neurons - Voluntary & involuntary movements of muscles & organs.

③ Interneuron - Intermediate neurons found in Spinal Cord & brain they pass signals from Sensory neurons to motor neurons.

Artificial Neurons -

An artificial Neurons is a mathmatical function of biological neurons, based on the model of biological neurons.

Each artificial Neuron has the following main functions -

① Receives inputs from a no. of input signals.

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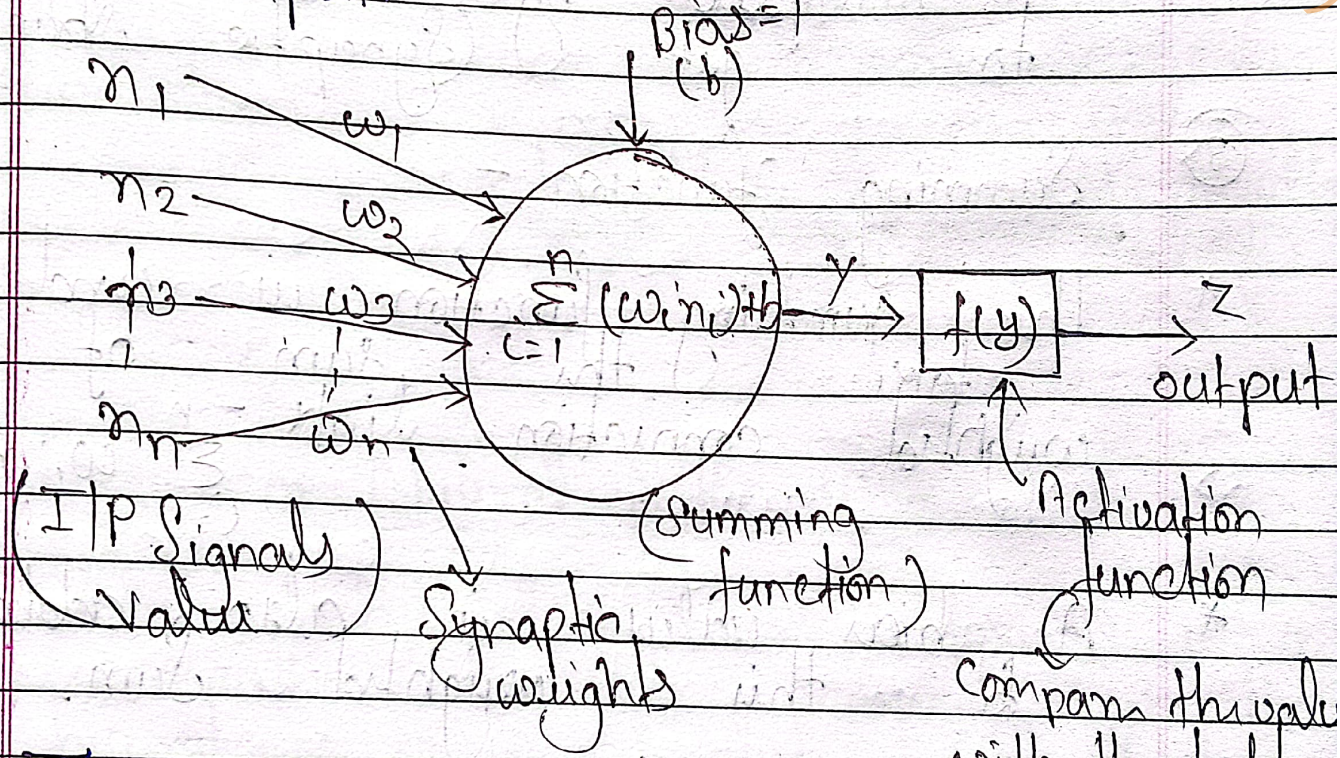
② Weight each input separately and sums them up.

③ Pass this sum through an activation function to produce output.



There are 5 Components of an artificial Neuron -

① Input values / One input layer -

* Input values are passed to a neuron using this layer.
* It is similar to dendrite in biological neurons.

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② Synapsis / Weighted Connection link -
Each connection links is characterized by a weight or strength (w_i). Each input n_i is multiplied by its weight (w_i) to get. (Synaptic strength)

③ Summing function -
* The summing function is used to take the sum of all weighted connection link - $\sum_{i=1}^{n} w_i \cdot n_i$
* A bias value is also added to this weighted sum.

④ Activation function -
* Activation function decides whether or not a neuron is fired based on the output value produced.

⑤ Output / output layer -
Output layer gives the final output of a neuron which can then be passed to

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other neurons in the network.

So,

An artificial neurons is a mathmatical model inspired by the neurons in our brain. just like biological neurons artificial neurons process numbers (data) and send results. They are the building blocks of an artificial neural network (ANN).

Artificial Neural Network -

* An artificial Neural network is a computer model inspired by the human brain. just like the brain uses neurons to process information, and ANN uses artificial neurons (also called processing elements) to solve problems like recognizing image, predicting outcomes or making decision.

* ANN is like a virtual brain that learns and makes decision by analyzing patterns in data.

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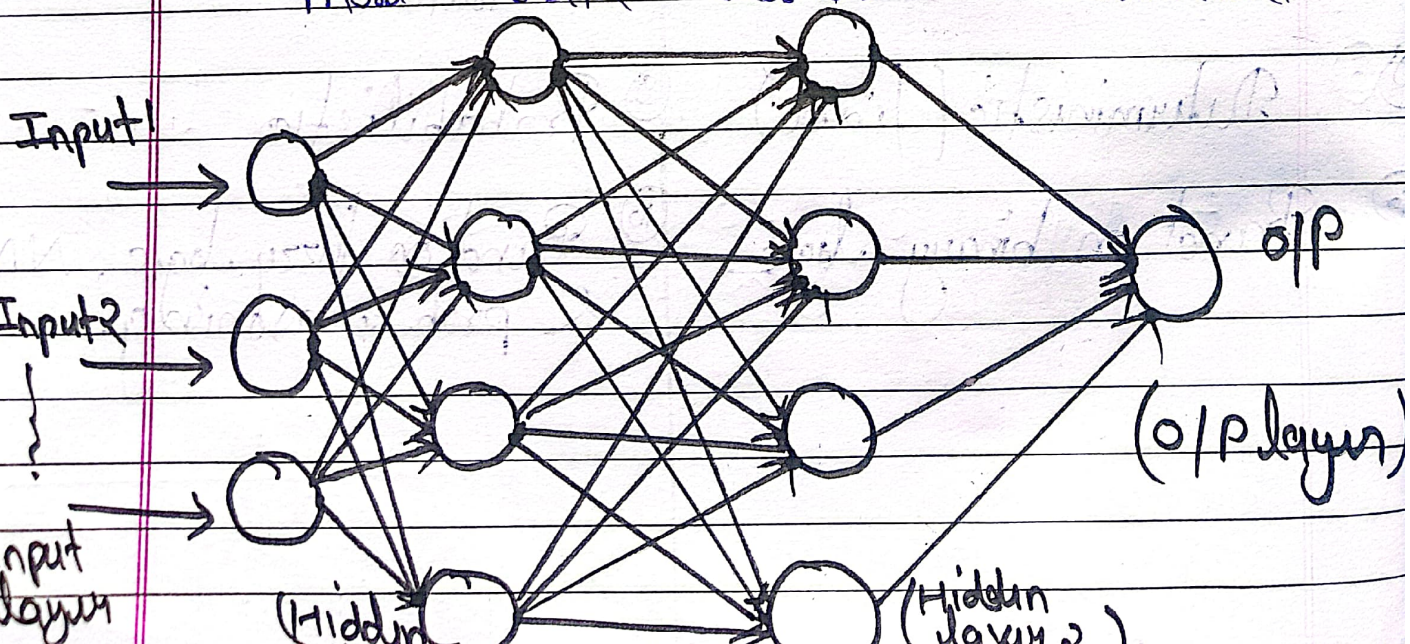
Artificial Neural Network -

* An artificial Neural network (ANN) is the piece of a computing system design to simulate the way the human brain analyze and process.

* It is also defined as a data processing system consisting of a large no. of interconnected processing elements or artificial neurons (Nodes)

* Then artificial Neurons are connected to each other through connection links.

* ANN have self-learning capabilities that enable them to produce better results as more data becomes available.



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Architecture of ANN-

① **Input layer** - * This is where the network starts.

* It takes the data you provide (I/P)
* Ex- if you're giving an image, the input layer takes pixel values of the image.

② **Hidden layer** - * They are the layers between the input and output.

* They process the data and find patterns or features.
* For a face recognition app, hidden layer identifies shapes like eyes or nose.
* Can be one or more hidden layers, depending on the complexity.

③ **Output layer** - * This is where the n/w ends and gives the final result

* For recognizing animals, the o/p layer might say 'cat' or 'dog'.

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There are 2 fundamentally different classes of ANN.

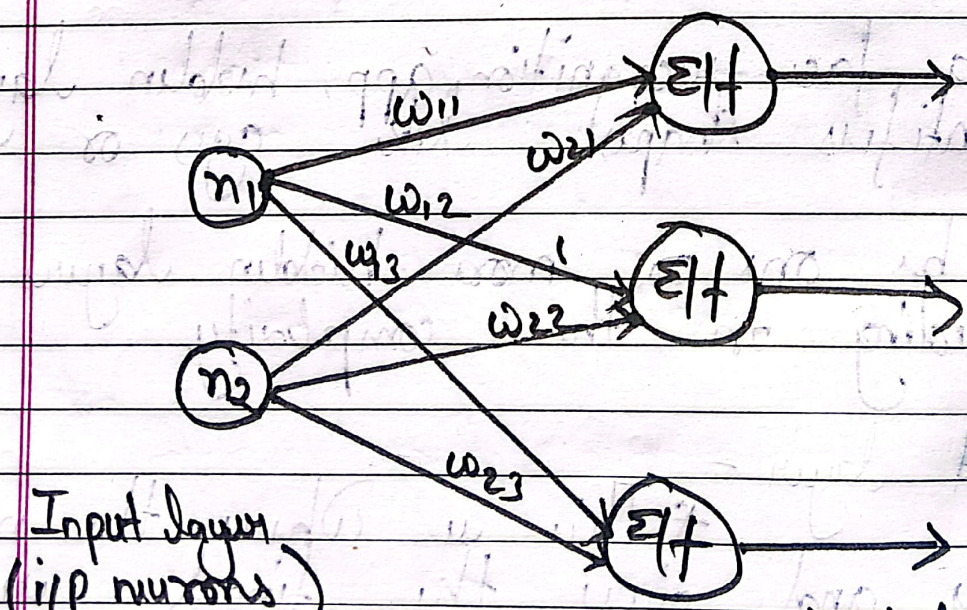
① **Single layer Feedforward Network** -

Single layer (or Computing layer) -

Single layer = This is only one computational layer (output layer) so it is a single layer ANN. However there are two layer (input layer & output layer)

Feed forward Network -

Input layer forward data to output layer not vice-versa.



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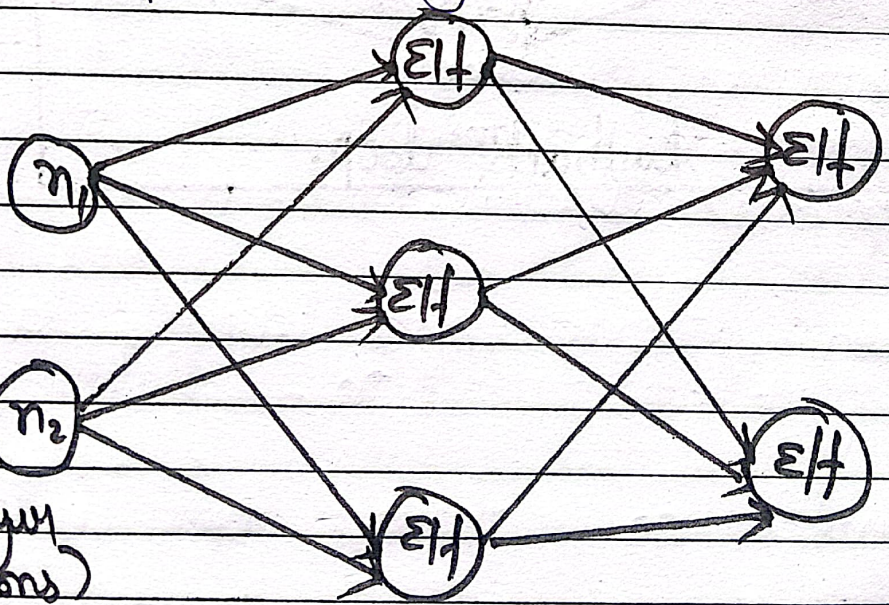
② Multilayer feedforward network -

Multilayer -

* In this architecture, there is an input layer, an output layer and one or more intermediate layers called **'Hidden layer'**. The computational units of the hidden layer is called 'Hidden neurons'.

* The hidden layer helps in performing intermediate computations before directing the input to the output layer.



इन दोनों के आर पर Result Produce होगा तो उसे further Connect नहीं कर सकते यहाँ Result नहीं होना चाहिए

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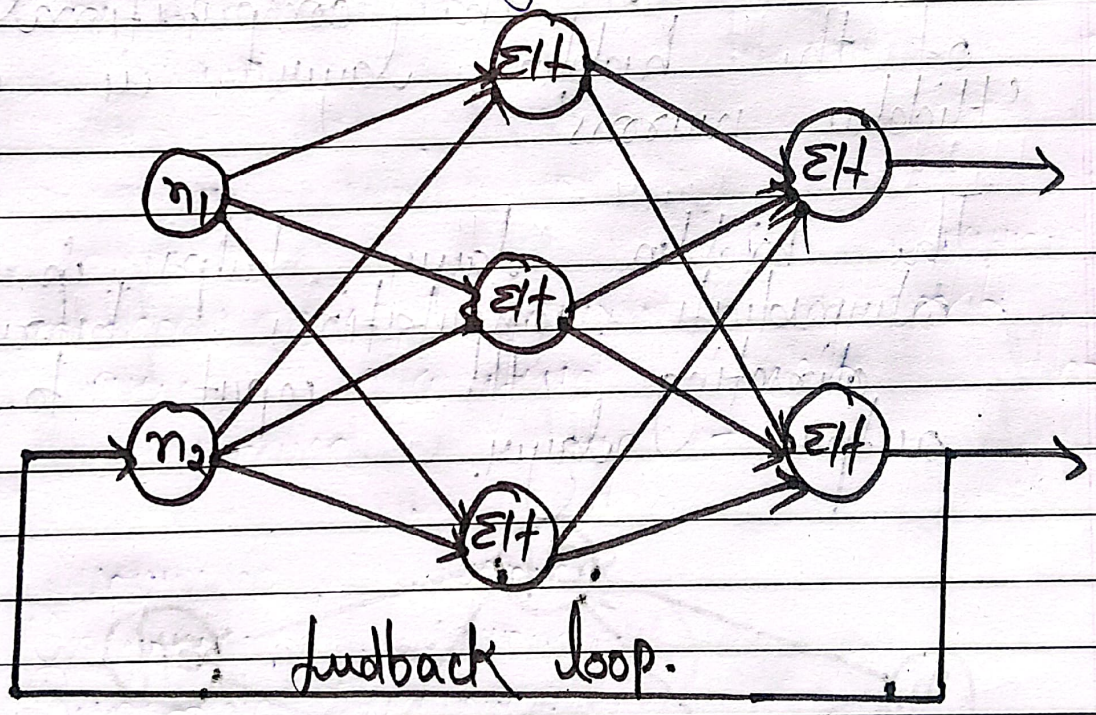
③ Recurrent Network -

* Thus networks differ from feed forward network architecture in the sense that there is atleast one feedback loop thy can be single layer or multilayer recurrent network.



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Learning in Artificial Neural network :-

* An artificial neural network (ANN) learning rule is a method, mathmatical logic or algorithm which improves the network's performance and or training time.
* This rule is applied repeatedly over the network (iterative process)
* It is done by updating the parameters (weights, bias) levels of a network when a network is stimulated in a specific data environment.
* A learning rule may accept initially the existing parameters (weights and bias) of the network and then will compare the expected result and actual result of the network to give new and improved values for weights & bias.
* The learning rule is one of the factors which decides how fast or how accurately the artificial network can be developed.

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* So, learning rule is procedure (learning algorithm) for training algorithm for modifying the weights and biases of a network.

* The purpose of the learning rule is training the network to perform some task.

Learning methods -

The learning methods in neural network are classified into three basic types -

(i) Supervised learning
(ii) Unsupervised learning
(iii) Reinforced learning

Then, these types are classified based on -

→ Presence or absence of teacher. (training data)

→ The information provided for the system to learn.

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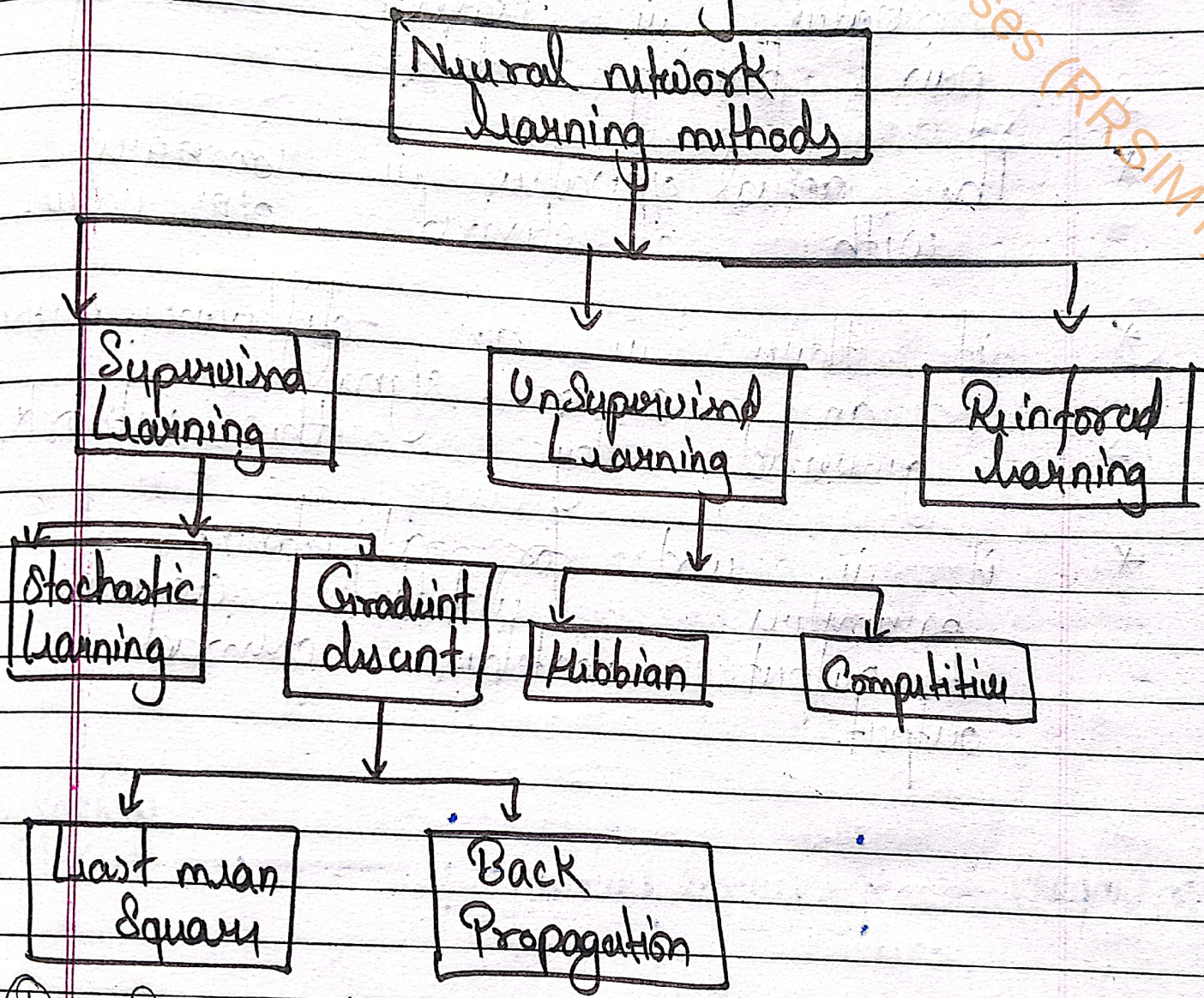
* Classification of Learning Methods based on learning rules -



① Supervised Learning -

* A teacher is present during learning process and presents expected output.

* In ANN, each input value requires a corresponding target value which supervises the desired output value.

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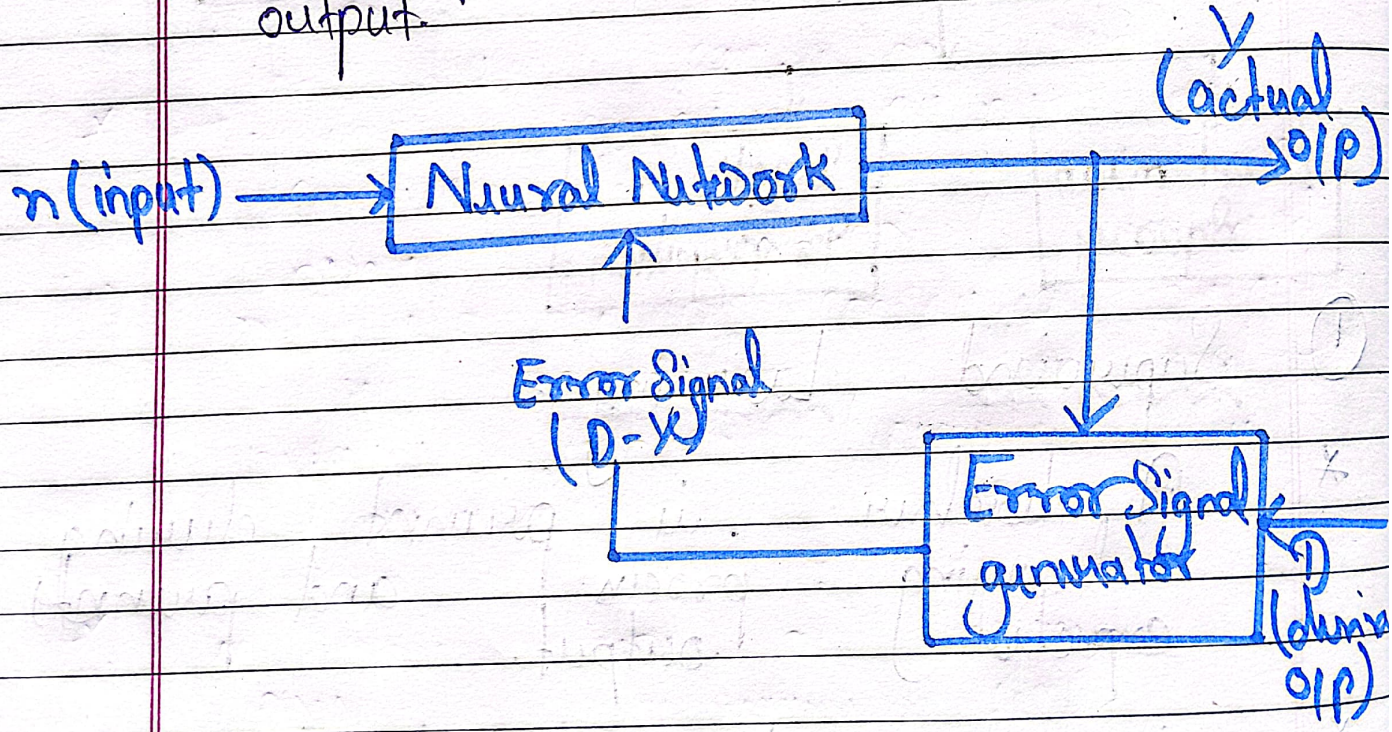
* The input value along with target value is called training pair / training data.
* The actual o/p value is compared with desired o/p value.
* if there is an difference, means an error, signal is generated by the network.
* it is used for adjustment of parameters unfill actual output matches desired output.



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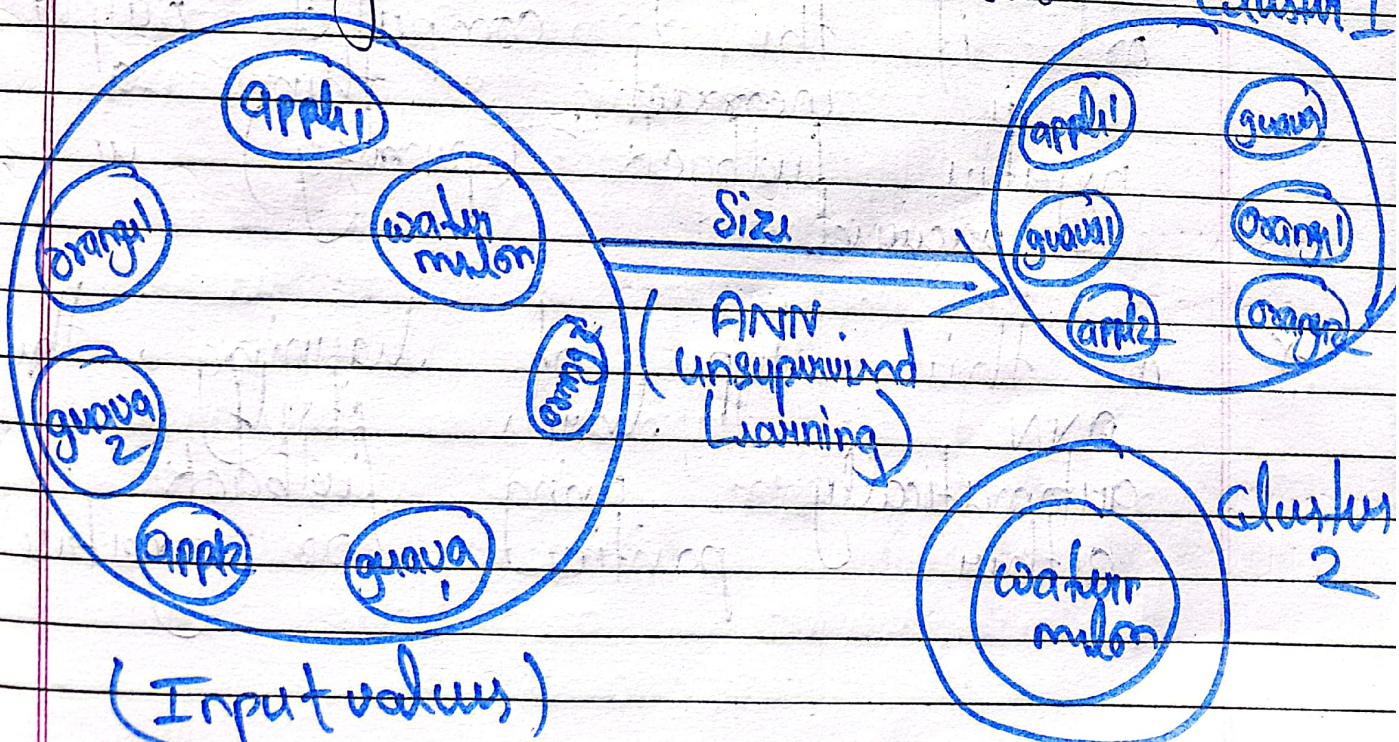
② Unsupervised learning - (I/P & O/P)

* Learning without a teacher.
* The expected (desired) output is not known. only inputs are known.
* No explicit error information can be used to improve network performance or accuracy.
* It is also called Self-organized learning (through the experience)
* In this type of learning the ANN will try to find some kind of pattern using the input data set without any external aids.



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* All Similar input patterns are grouped together as cluster.
* So clustering is a process of grouping similar input patterns.

③ § Reinforced Learning – Partial role of teacher

* A teacher is present but does not present the expected or desired output but only indicates if the computed output is correct or incorrect.
* if the computed output is correct then it gets a positive feedback (reward) or if the computed output is incorrect then a negative feedback (penalty) is received.
* in this type of learning, the ANN automatically learns using either positive feedback or negative

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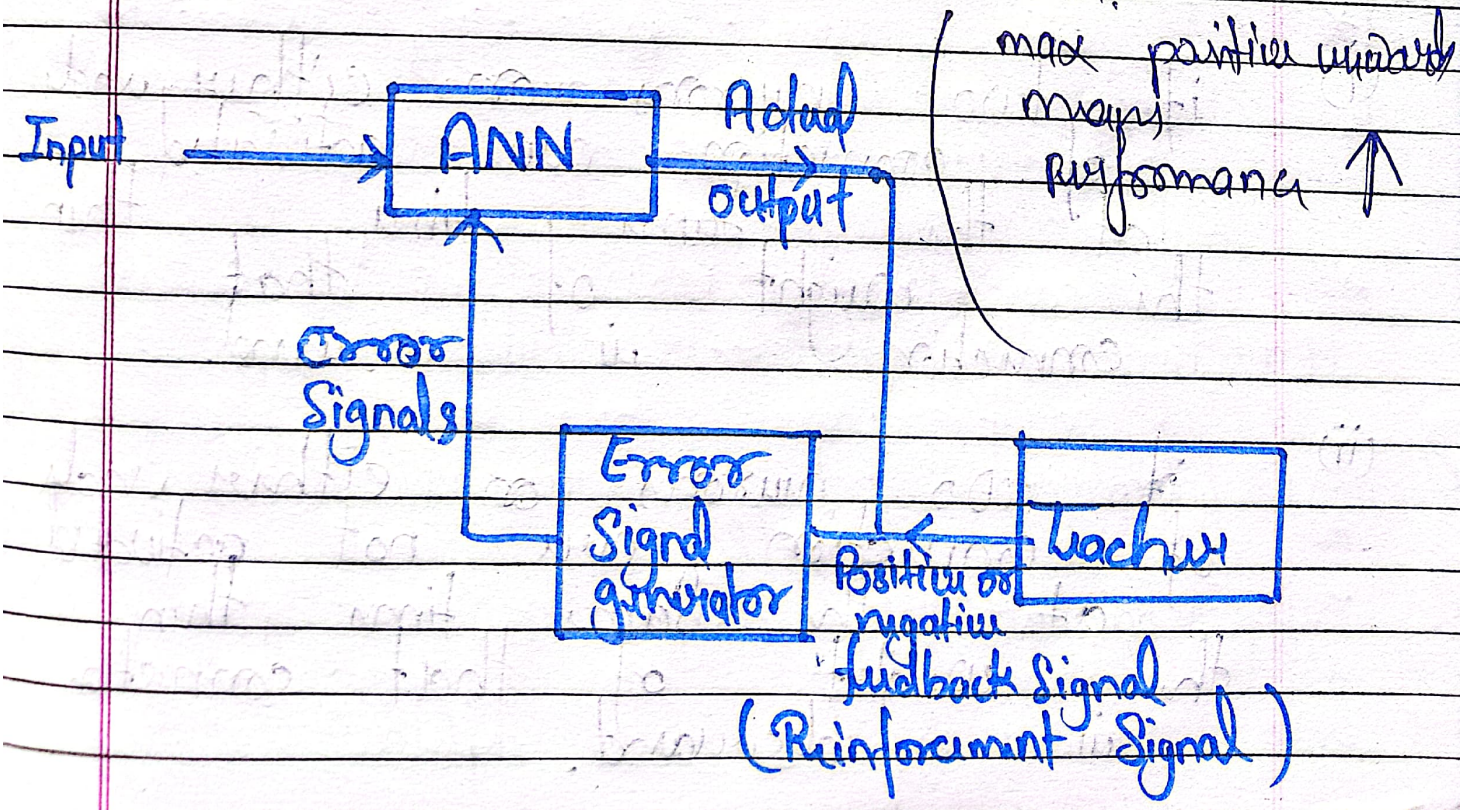
* Since, there is no expected output so the ANN is bound to learn by its experiment only.

Key Points -

* it is based on the hit and trial process.
* ANN take the next action and changes parameters (weights etc) according to the feedback of the previous action.
* In this there is no instruction about environment & what action needs to be learn.



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Perceptron architecture

* A perceptron is the simplest type of artificial neural network and foundation for understanding more complex neural networks
* It consist of a single neuron with adjustable synaptic weights and bias.
* Perceptron was introduced by Frank Rosenblatt in 1957.
* A perceptron has one or more inputs a process and only one output.
* It is used as an algorithm to facilitate supervised learning of binary classifier.
* A perceptron takes input values, process them through a mathematical formula and produces an output (either 0 or 1).
* It works like a decision-making tool based on the inputs it decides whether

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to 'activate' (output 1) or not (equal 0).

Architecture (Component) of a Perceptron:

① Inputs -

Features are taken as inputs in the perceptron algorithm. inputs are denoted as $x_1, x_2, x_3, x_4 \dots x_n$.

Ex- For a weather Prediction System, inputs could be temperature, humidity etc.

② Weights -

* Each input is assigned a weight, which determines how important that input is in the decision.

* Weights can be positive or negative.

③ Bias -

* A bias is a special input type.

* it allows the classifier to move the decision boundary around from its original position to the right, left, up and down.

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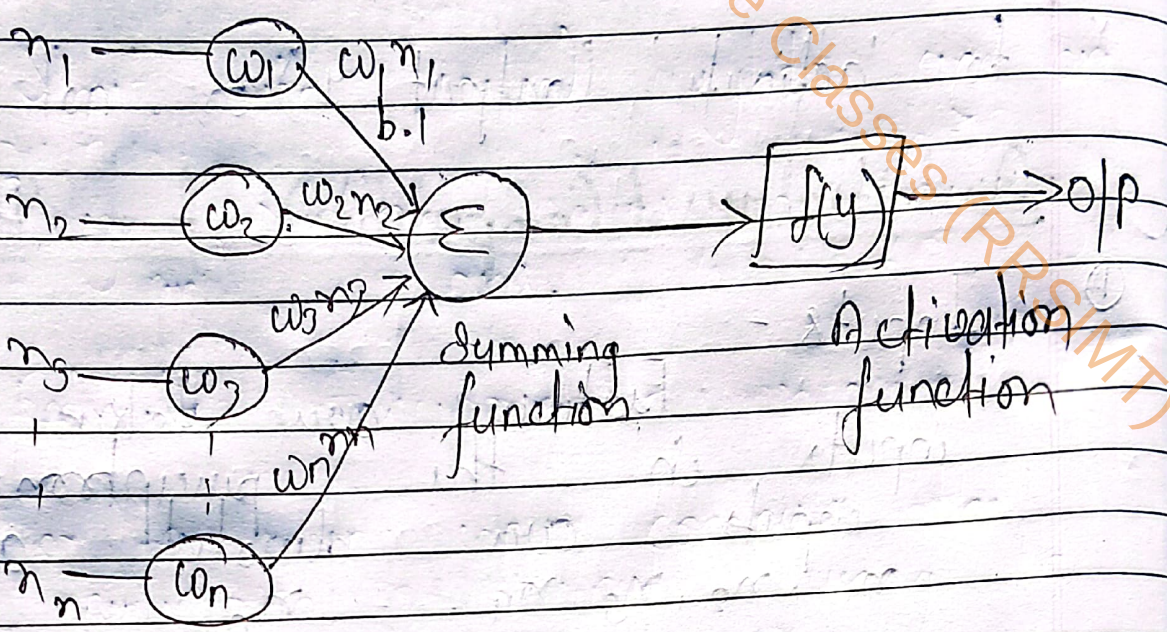

① Summing function -

* All the inputs are multiplied by their respective weights and then summed up.

* Mathmatically -

Weighted Sum = $w_1x_1 + w_2x_2 + \cdots w_nx_n$

② Activation function -

* The summation result is passed through an activation function, typically a step function.

* if the Sum is greater than or equal to a certain threshold, the perceptron outputs 1 (activated)

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* if it is less that the threshold, the output is 0 (not activated)

Perceptron Learning Rule -

* The perceptron learning rule is a simple method used to train a perceptron to correctly classify data.
* it adjust the weight and bias of the perceptron based on the error between the actual output and the target output.
* The goal of this rule is to make the perceptron learn from its mistakes by minimizing the difference between predicted & target outputs.

How the Perceptron Learning Rule works -

The learning rules updates the weights of the perceptron using the following steps -

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① Start with initial weights and bias.

Assign random values to the weights and bias.

② Calculate the output -

For a given input, compute the perceptron's output using the formula -

$$z = (\text{weighted sum of inputs}) + \text{bias}$$

Then pass z through the activation function to get the perceptron's output.

③ Compare Output with Target:

* check whether the perceptron's output matches the desired target output.
* if the output is correct, no changes are needed.
* if the output is incorrect, adjust weight & bias.

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④ Update the weights and bias -

On Perceptron Learning Rule formula -

$$w_{new} = w_{old} + \eta \cdot (t - o) \cdot x$$
$$b_{new} = b_{old} + \eta \cdot (t - o)$$

where,

w - weight

η - learning rate (a small value like 0.001 or 0.1 that controls speed of learning)

t - target output

o - perceptrons output

x - input value.

⑤ Repeat for all inputs -

* Continue the process for all training example until the perceptron makes no mistake.

output $(Y) = \begin{cases} 1, & \text{if } wx + b > 0 \\ 0, & \text{if } wx + b < 0 \end{cases}$

radio exp

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③ Repeat :- Keep updating the weights for all inputs untill the error becomes very small (or zero)

* Back Propagation algorithm :-

Back Propagation :-

* Back propagation is a standard method of training artificial neural network.

* "Back Propagation is a method of repeatedly adjusts the weights of the connections in the network so as to minimize the difference between actual output value and desired output value."

* This method/algorithm looks for a minimum value of the error function in weight space using either delta rule/gradient descents

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Steps in Back Propagation algorithm -

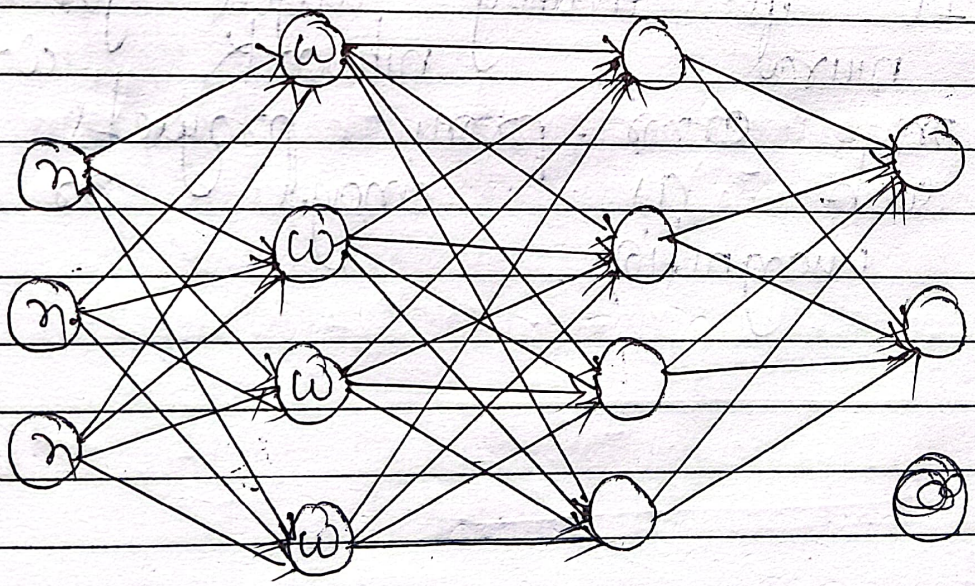
① Inputs x arrives through the ~~random~~ preconnected path.

② Inputs are modified using randomly assigned weight 'w'

③ Calculate the output for each neuron from the input layer, to the hidden layer, to the output layer.

④ Calculate the error in the output

$Error_B$ = Actual output − Desired output.



* Travel back from the output layer, to the hidden layer to adjust the

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weights such that error is decreased.
Repeat the process until the desired output is achieved.

Advantages of backpropagation -

1. Backpropagation is fast, simple & easy to program.
2. It is a flexible method as it does not require prior knowledge about the network.
3. It is mainly useful for deep neural network working on error-prone projects such as image or speech recognition.

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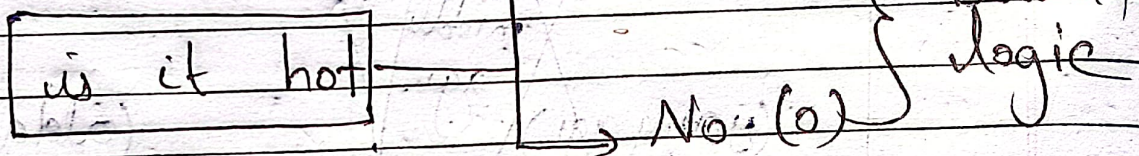
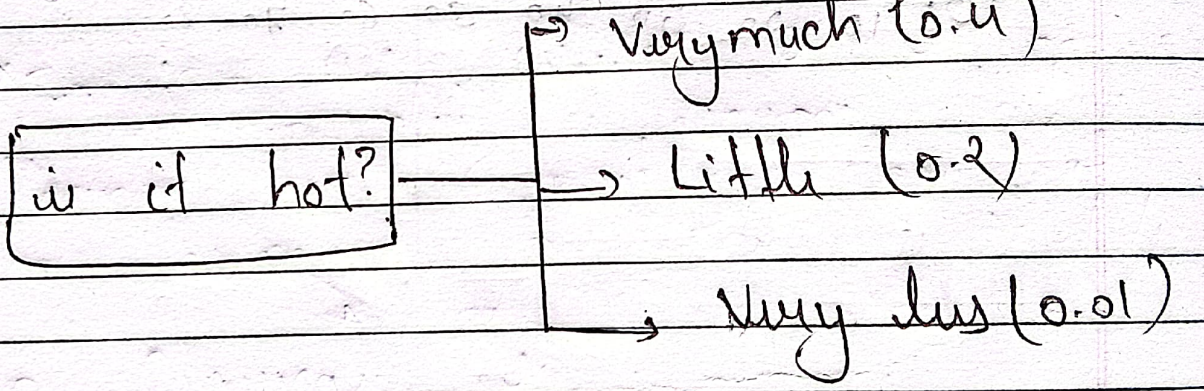
Unit-3
Fuzzy Logic-I (Introduction)

Fuzzy logic - * The term "Fuzzy" refers to things which aren't clear.

* In real world Sometimes we face a situation where we can't determine whether the state is true or false





* Fuzzy logic is an approach of computing based on "degrees of truth" rather than usual "true or false (1 or 0) boolean logic. on which the modern computer is based.

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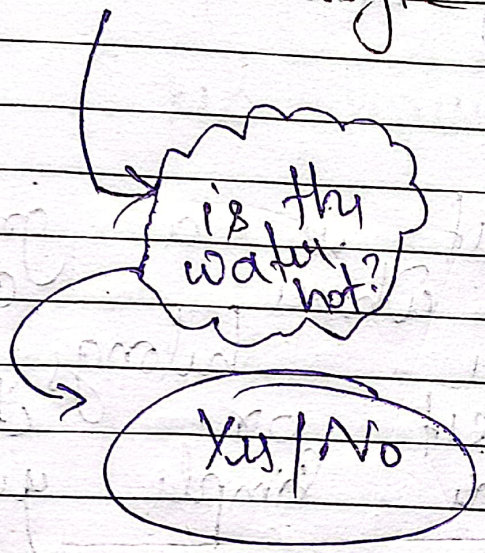
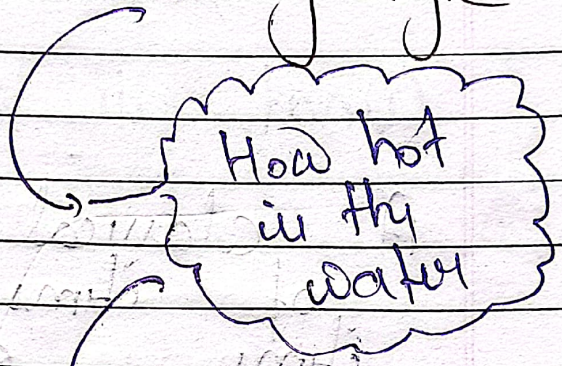
* The idea of fuzzy logic was given by Dr. Lotfi Zadeh of University of California in the 1965.

Traditional logic



Fuzzy logic



→ it could be 50%, 75%, 90% etc

* Importance of fuzzy logic in Daily life -

① Home appliance -

→ Air conditioning - Adjust Cooling based on how hot room is.

② Vehicles -

→ Automatic breaking - Gradually applies breaks based on the car's speed and distance from an object.

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③ Medical Systems -

→ Diagnosis Tools - Helps doctors make better decision by analysing symptoms that aren't clearly yes/No (eg- mild fever, High fever)

What is Classical Set?

A classical set is a group of items where each item either belongs to the set or it does not. its like simple yes/No.

Ex- if we define a set of "Tall people" anyone taller than 6 feet belongs to the set (Yes)

In Classical Set -

Membership is clear-cut (0 or 1)

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What is Fuzzy Set?

A Fuzzy set is a group where items can partially belong to the set. Instead of just 'yes' or 'No' it works with degree of membership between 0 and 1.

Ex- For a Fuzzy set of "tall people" Someone 5'8 might be 0.5 tall while someone 6'3 might be 0.9 tall.

In Fuzzy Set - membership is gradual ranging from 0 (not at all) to 1 (fully)

Importance of Fuzzy Set -

① Handle Uncertainity - Fuzzy sets are perfect for scenario where "yes" or "no" isn't enough like "yes" how hot is the weather or how fast is the car?

② Better decision making - In smart system like AI fuzzy sets allow more natural and flexible decision.

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Fuzzification –

* Fuzzification is the process of converting a crisp input (a precise, exact value) into a fuzzy value.
* This fuzzy value represents the degree to which the input belongs to different fuzzy set.

Why fuzzification –

* Real world data is often not black or white (eg. hot or cold) instead it lies somewhere in between (eg - warm)
* Fuzzification helps handle this uncertainity by allowing an input to partially belong to multiple categories (fuzzy sets).

How fuzzification Works?

I/P Crisp value → [Fuzzification] → Fuzzy value.

1) Input Crisp value –
Start with a precise, measurable input (like - temp. = 75°F)

2) Membership Functions -
Define membership functions for different fuzzy set (like - 'Cold' 'Warm' 'Hot')

This functions assign a membership value (between 0 and 1) to the input, indicating how well it fits into each fuzzy set.

For example - 75°F might have as -

0.2 membership in 'Cold'
0.7 membership in 'Warm'
0.1 membership in 'Hot'

3) Output fuzzy value -
The crisp input is now represented as fuzzy value across the defined sets.

Example -
Crisp Input - Temp = 75°F
Fuzzy sets - 'Cold' (membership) = 0.2
'Warm' (membership) = 0.7
'Hot' (membership) = 0.1

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*** * Defuzzification (Fuzzy to Crisp Conversion) -

* Defuzzification is the opposite of fuzzification.
* It converts fuzzy output value back into a crisp value that can be used in the real world.
* Defuzzification is a process of converting fuzzy quantity to crisp quantity
* Calculation of a crisp value from a fuzzy value is called "Defuzzification"

Fuzzy I/P Set → [Defuzzification] → Crisp O/P Set

Various methods of defuzzification process -

i) Lambda Cut - method -
ii) Maxima methods - } Complicity.
iii) Weighted Sum method -
iv) Centroid methods

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Thank You....

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Unit – 5 Applications of AI & Multi-Agent Systems

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V	**Applications of AI & Multi-Agent Systems:** Natural Language Processing, Machine Translation, Information Retrieval and Extraction, Robotics: Perception, Planning, and Motion, Speech Recognition, Software Agents: Architecture, Communication, Trust, Multi-agent Negotiation and Reputation. **Explainable AI (XAI)** – Importance of interpretability, techniques for explaining black-box models, trust in AI, case studies in NLP and vision.

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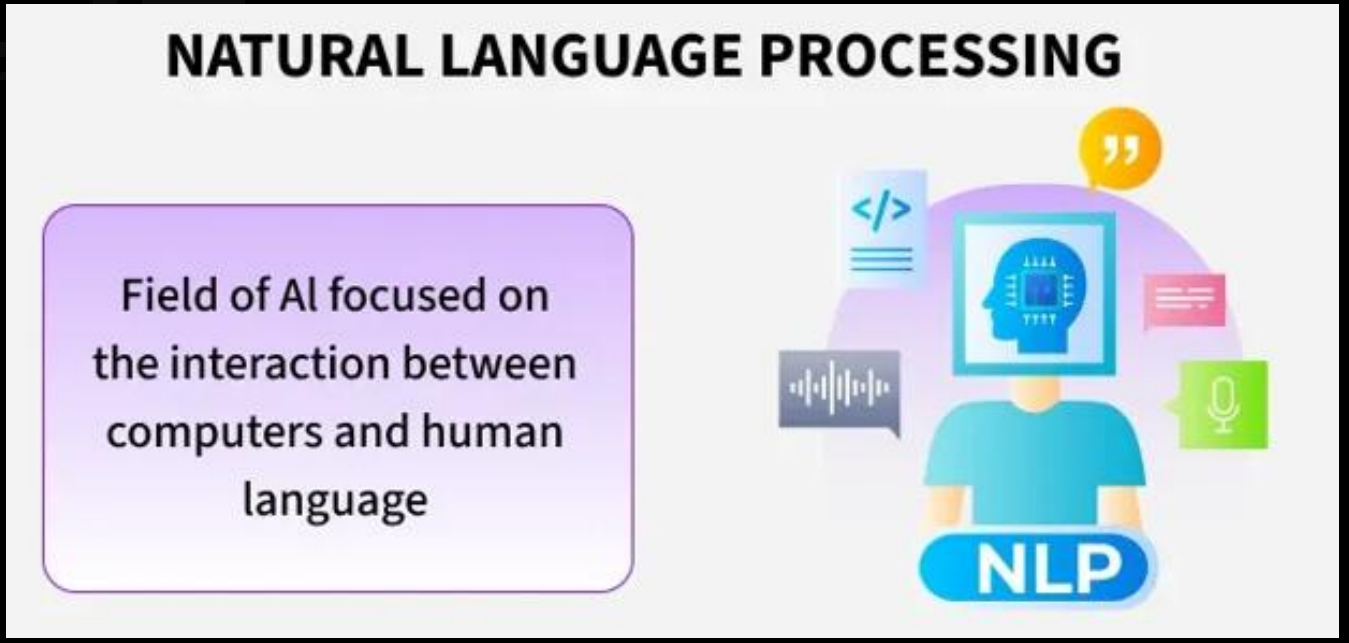
🌍 What is Natural Language Processing (NLP)?

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that helps computers **understand, interpret, and respond** to human language (like English, Hindi, etc.).

Example:

- Google Translate (understands one language → converts to another)
- ChatGPT (understands your question → gives an answer)
- Alexa / Siri (listens to voice → performs task)



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❖ How Natural Language Processing (NLP) Works :

1. Text Input and Data Collection

Collect data → text, documents, speech transcripts, chats.

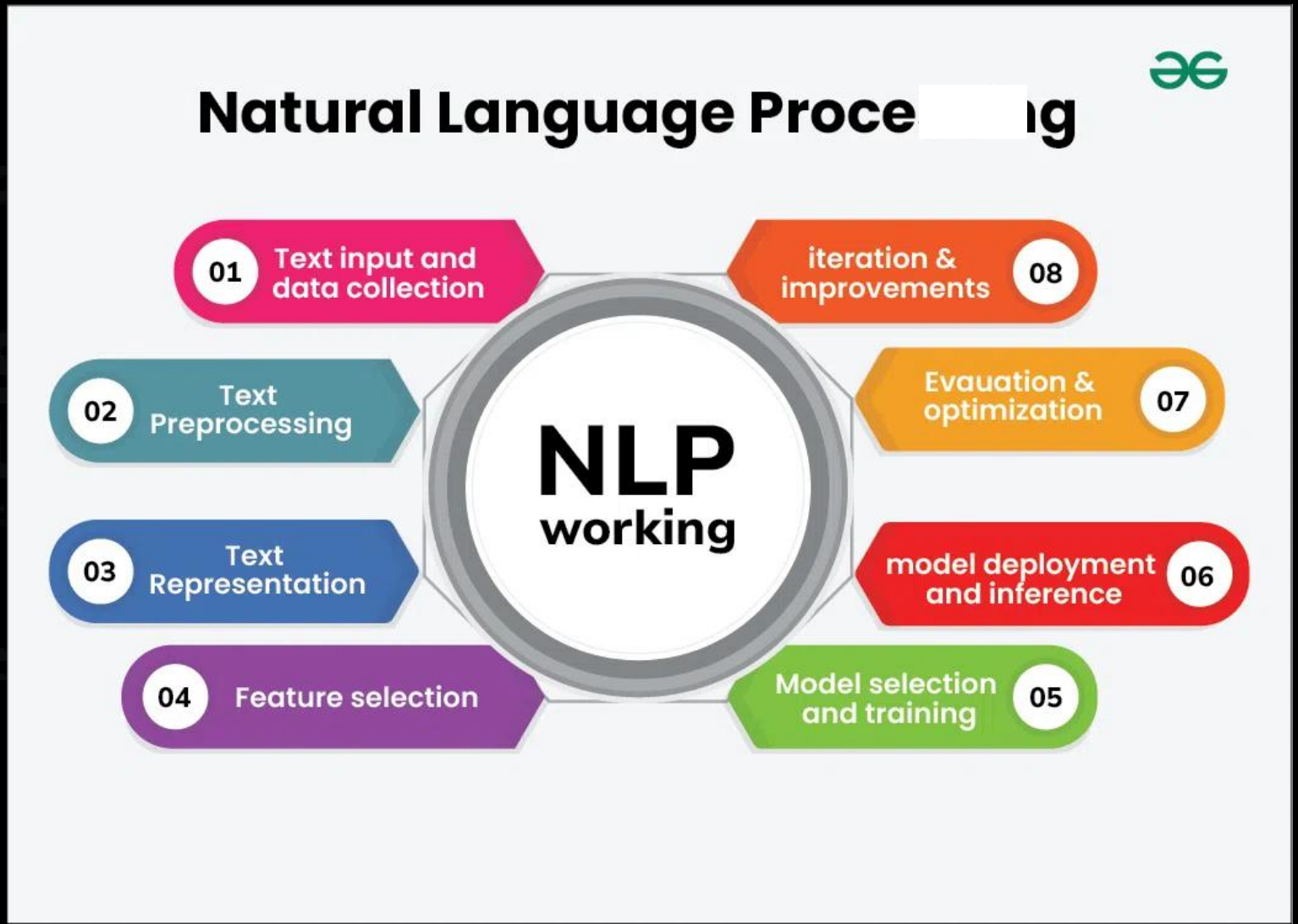
Example: Collecting tweets to analyze "public opinion."

2. Text Preprocessing

Clean the text so that machine can read it.

Includes:

- Removing punctuation, stop words ("is, the, a").
- Converting all words to lowercase.
- Breaking text into tokens (words).



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Example:

Input: *"ChatGPT is Amazing!"*

Preprocessed: *["chatgpt", "amazing"]*

3. Text Representation

Convert words into numbers (since machines understand numbers, not text).

Methods:

- Bag of Words
- TF-IDF
- Word Embeddings (Word2Vec, BERT, GPT)

Example:

"dog" → [0.21, 0.56, -0.11] (vector form).

4. Feature Selection

Select important features from the text.

Example: In spam detection, words like *"free, money, win"* are important features.

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5. Model Selection and Training

Choose an AI/ML model → Train it with data.

Example: Train a model to classify “spam” vs “not spam.”

6. Model Deployment and Inference

Deploy the trained model → now it can be used in real world.

Example: Gmail spam filter is deployed → checks all new emails.

7. Evaluation & Optimization

Test the model → How accurate is it?

Optimize → Improve accuracy.

Example: Accuracy = 95% → try tuning parameters to reach 98%.

8. Iteration & Improvements

NLP is **not perfect** → keep improving with more data and better models.

Example: ChatGPT gets updated regularly for better answers.

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✅ Summary

👉 NLP = Teaching computers to understand human language.

👉 **It works in steps:**

- Collect text →
- Clean it →
- Convert to numbers →
- Select useful features →
- Train AI model →
- Deploy it →
- Evaluate performance →
- Improve continuously.

💡 Example Flow:

If you say **“Play music”** to Alexa:

1.Collect speech →
2.Convert to text →
3.Process text →
4.Extract meaning (intent = play music) →
5.AI model understands →
6.Alexa plays the song. ♪

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🌍 What is Machine Translation (MT)?

Machine Translation is the process where a computer automatically translates text or speech from one language (source) to another language (target).

Example:

Input (English): **“Hello”**
Output (Hindi): **“नमस्ते”**
Output (Korean): **“안녕하세요”**

⚙ How does it work?

The **Machine Translation System** uses Artificial Intelligence (AI), Natural Language Processing (NLP), and statistical/learning models to perform translation.

Steps:

1. Input (Source Language)

The user provides a sentence in a source language (e.g., English).

2. Processing by MT System

The system understands the meaning and structure using:

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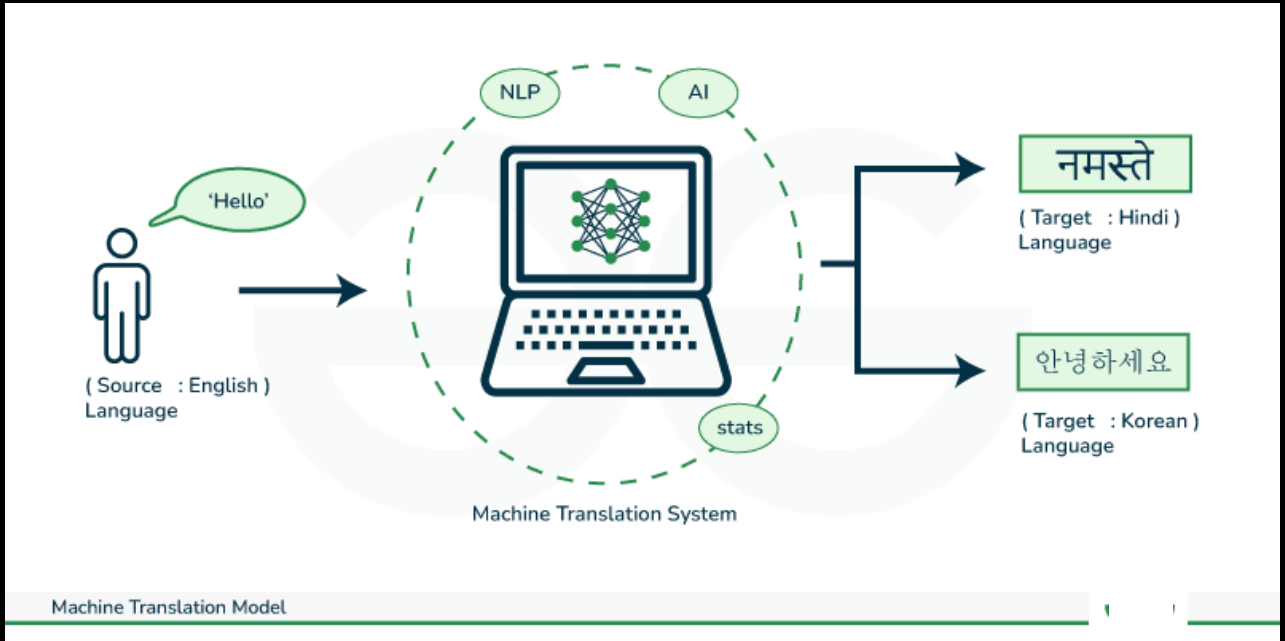
- **NLP (Natural Language Processing):** Breaks the sentence into words, grammar, and context.
- **AI & Neural Networks:** Learn from huge datasets of translated sentences.
- **Stats & Probability:** Decide which translation is most likely correct.

3. Output (Target Language)

- The system generates the sentence in the target language (e.g., Hindi, Korean).
- It tries to preserve both **meaning** and **grammar**.



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🌍 Key Approaches in Machine Translation

👉 Machine Translation works in **two steps**:

1. **Decode** the source language (understand meaning + structure).
2. **Encode** it into the target language (generate correct sentence).

Different approaches are used for this process:

1. Rule-Based Machine Translation (RBMT)

- Works like a **dictionary + grammar book**.
- Uses pre-defined **rules** for grammar and word meanings.
- Process: Parse sentence → convert into intermediate form → apply rules → output in target language.
- ✅ Good for **technical/legal content** where precision is needed.
- ❌ Not flexible for casual or complex sentences.

Example:

English "He is eating" → Grammar rules + dictionary → Hindi "वह खा रहा है"

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2. Statistical Machine Translation (SMT)

- Works like **probability & pattern matching**.
- Learns from **large collections of human-translated texts** (corpus).
- Finds the **most likely translation** based on statistics.
- ✓ Learns automatically, handles many languages.
- ✗ Can make **grammatical mistakes**.

Example:

If in many texts "Hello" → "नमस्ते", the system guesses that's the most probable translation.

3. Neural Machine Translation (NMT)

Works like the **human brain (neural networks)**.

Uses **deep learning models** to understand the whole sentence (not just word by word).

Considers **context** → gives more natural and fluent translation.

✓ Best accuracy, used in **Google Translate, DeepL**.

✗ Needs **huge data & computing power**.

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Example:

English "He kicked the bucket" → Instead of literal translation, it knows the phrase means **"He died"**, so it translates meaningfully.

4. Hybrid Machine Translation (HMT)

A **mix of two or more approaches** (usually RBMT + SMT).

Takes **best parts** of each system.

✓ More flexible and accurate.

✗ More complex to build.

Example:

Uses **rules for grammar** + **statistics for word choices** → final translation is better.

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Approach	How it works	Pros	Cons
Rule-Based (RBMT)	Grammar rules + dictionary	Precise for specific content	Not flexible, hard to scale
Statistical (SMT)	Probability & patterns from large text data	Learns automatically	May be grammatically wrong
Neural (NMT)	Deep learning neural networks	Most accurate & natural	Needs huge data & power
Hybrid (HMT)	Mix of RBMT + SMT (sometimes NMT)	Balanced, better accuracy	Complex to design

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🕮 What is Information Retrieval (IR)?

☞ **Information Retrieval (IR)** is the process of **finding useful information from a large collection of data** (like search engines, digital libraries, or research databases).

Simple Example:

- You search **"machine learning basics"** on Google.
- Google finds and shows you the most relevant documents/webpages from billions of stored documents.

🛠Components of Information Retrieval (IR Model)

The IR model works in **two sides**:

❖ User Side (Search Process)

This is what the **user does**.

1. Problem Identification – User realizes they need info.

Example: Student wants to know about *machine learning*.

2. Representation – Convert need → keywords.

Instead of asking: *"How do machines learn?"*

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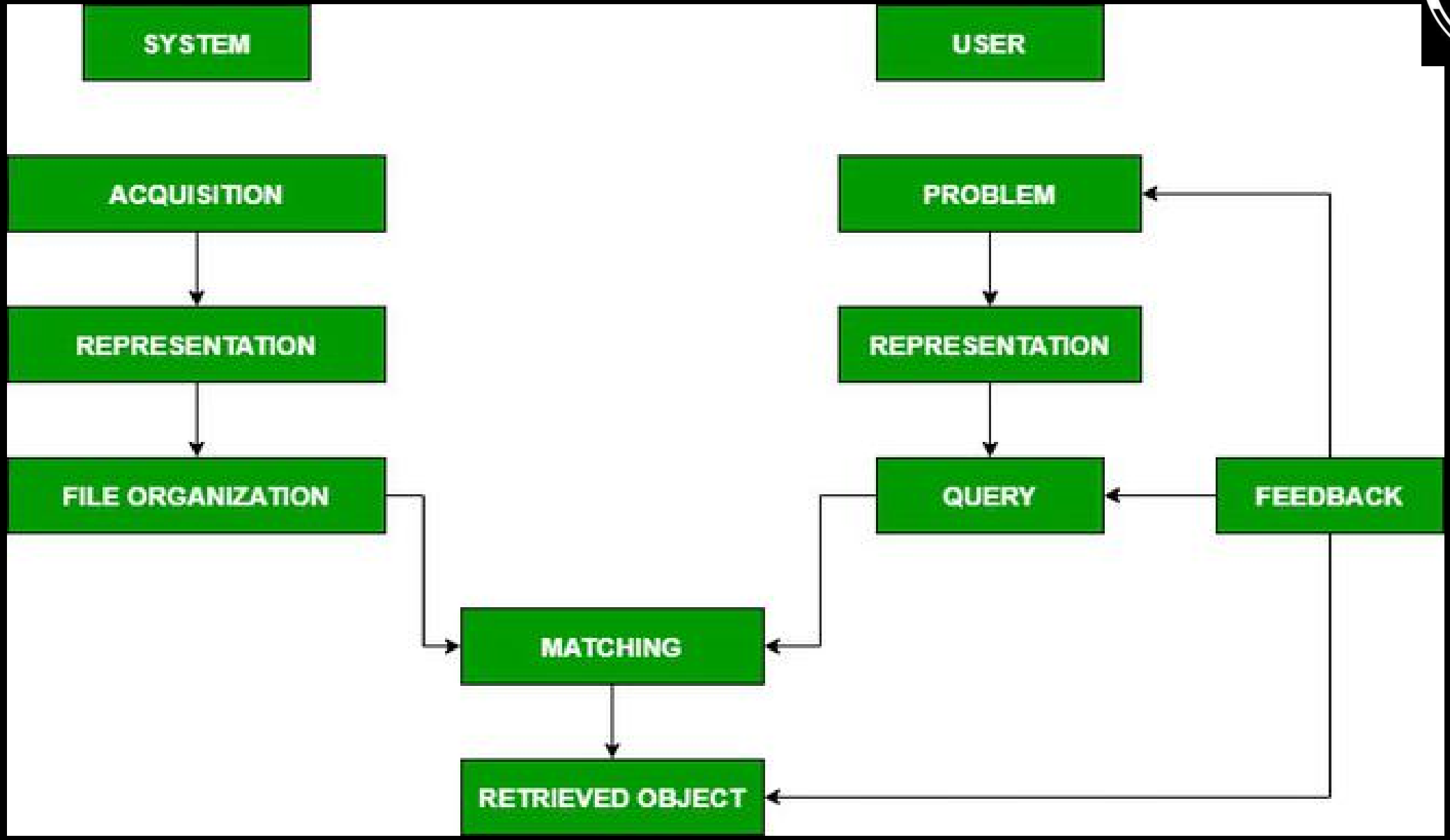

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User types: *"machine learning basics"*.

3. Query – The keywords are entered into the search box.

Example: Google search.

4. Feedback – User checks results. If not satisfied, they **refine the query**.

Example: Change *"machine learning basics"* to *"machine learning algorithms"*.

❖ **System Side (Retrieval Process)**

This is what the **system (Google/search engine)** does.

1. Acquisition (Collecting Data)

The system collects documents (web pages, books, articles) using **web crawlers**.

Stores them in a huge database.

2. Representation (Indexing)

Each document is analyzed and tagged with keywords (terms).

Example: A document on *machine learning* may get terms like **AI, deep learning, algorithms, models**.

Also includes metadata like title, author, abstract.

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3. File Organization

Documents are stored in such a way that they can be found quickly.

Two methods:

- ✓ **Sequential file** → stored one after another (slow).
- ✓ **Inverted file** → keywords point to documents (like an index in a book).

4. Matching

User query is compared with indexed documents.

System calculates **relevance score** (how well document matches query).

Example: If query = “machine learning basics”, documents with keywords **machine learning, introduction, AI** get higher scores.

5. Retrieved Object (Results)

The system shows the most relevant documents.

Results are **ranked** so best ones appear first.

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❖ Interaction Between User & System

User checks results → refines query → system updates search → better results.
This loop continues until the **user is satisfied**.

👉 Information Retrieval is like a librarian:

User asks: “Books on Machine Learning?”
Librarian (system) searches shelves (database), checks index cards (indexing), finds matching books, and gives the best ones first.

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🕮 What is Information Extraction (IE)?

☞ **Information Extraction (IE)** is the process of **automatically pulling out useful and structured information from unstructured text** (like news articles, research papers, social media posts, etc.).

Simple Example:

Text: *"Steve Jobs founded Apple in 1976 in California."*

IE System extracts:

- **Person** → Steve Jobs
- **Organization** → Apple
- **Year** → 1976
- **Place** → California

So instead of reading the whole text, IE gives structured facts.

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🛠Key Components of Information Extraction

Information Extraction usually works in steps:

1. Text Input & Preprocessing

- The system takes raw text (article, report, social media post).
- Cleans it → removes noise (extra symbols, punctuation, stopwords).
- Breaks into sentences and words (tokenization).

2. Named Entity Recognition (NER)

Identifies important entities like:

- **People** (e.g., “Steve Jobs”)
- **Organizations** (e.g., “Apple”)
- **Locations** (e.g., “California”)
- **Dates** (e.g., “1976”)

3. Relation Extraction

Finds relationships between entities.

Example: “Steve Jobs **founded** Apple.”

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- Relation = *founded*
- Entities = *Steve Jobs* → *Apple*

4. Event Extraction

Detects events mentioned in text.

Example: *"Apple released iPhone in 2007."*

- Event = *release*
- Actor = *Apple*
- Object = *iPhone*
- Time = *2007*

5. Coreference Resolution

Links words that refer to the same entity.

Example: *"Steve Jobs founded Apple. He was its CEO."*

"He" → Steve Jobs

"its" → Apple

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6. Template Filling (Structured Output)

Organizes extracted info into a structured database or template.

Example:

Person: Steve Jobs
Organization: Apple
Founded: 1976
Location: California

❑ **in One Line:**

☞ **Information Retrieval (IR)** finds *relevant documents*.

☞ **Information Extraction (IE)** digs inside those documents to pull out *important facts*.

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❖ What is Robotics?

☞ **Robotics** is the branch of science and engineering that deals with designing, building, and using robots.

A **robot** is a machine that can sense the environment, process information, and act to achieve tasks (like moving, picking objects, delivering, etc.).

👀 What is Perception in Robotics?

☞ **Perception** means how a robot **sees, hears, and understands** the world around it using sensors.

- Just like humans use **eyes, ears, and skin** to sense the environment, robots use **cameras, microphones, LiDAR, GPS, and touch sensors**.
- The data collected is then processed with **AI + Computer Vision + Machine Learning** to understand the surroundings.

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⚙ How Robotics Perception Works (Step by Step)

1. Sensing

The robot collects data from the environment.

Example:

- **Camera** → captures images
- **LiDAR** → scans distances using lasers
- **Microphone** → records sound
- **Touch Sensor** → detects pressure/force

2. Preprocessing

Raw data is messy (like blurry images or noisy sounds).

The robot cleans and organizes data for better understanding.

Example: removing background noise from audio.

3. Feature Extraction

Important details are picked from the data.

Example: From a camera image, features like **edges, shapes, colors** of objects are extracted.

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4. Interpretation (Understanding)

AI and ML algorithms analyze features to recognize objects and situations.

Example:

- Identifying a **human face** in a camera feed.
- Detecting an **obstacle** in front of a self-driving car.

5. Decision Making

Based on perception, the robot decides what to do next.

Example:

- If obstacle detected → Stop or change direction.
- If object identified → Pick it up.

6. Action (Execution)

The robot performs the action using motors, wheels, or robotic arms.

Example: A delivery robot avoiding a wall and moving forward.

👉 Robotics is about creating intelligent machines.

👉 Perception is how robots **sense → understand → act** in the real world.

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❖ What is Planning and Motion in Robotics?

Planning = Deciding **what path** or sequence of actions a robot should take to reach its goal.
Motion = Actually moving the robot along that path safely and smoothly.
☞ Together, **Planning + Motion** help a robot go from **start point → goal point** without crashing or wasting energy.

⚙ How Planning and Motion Work (Step by Step)

1. Goal Definition

The robot is given a goal or task.
Example: *"Go to the kitchen," "Pick up the box," "Deliver to person."*

2. Perception & Mapping

Robot uses sensors (camera, LiDAR, GPS) to build a **map of the environment**.
Example: Detect walls, tables, people, obstacles.

3. Path Planning (High-Level Planning)

Robot plans the **best path** from start to goal.

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Uses algorithms like:

- **Dijkstra / A*** → find shortest path.
- **PRM (Probabilistic Roadmaps)** → random sampling for complex spaces.
- **RRT (Rapidly Exploring Random Tree)** → for dynamic environments.

Example: Like Google Maps giving shortest driving route.

4. Motion Planning (Low-Level Planning)

Breaks the path into **small movements** that the robot can actually execute.

Considers:

- **Kinematics** (robot's joints, wheels, arms).
- **Dynamics** (forces, friction, speed, turning ability).

Example: Self-driving car plans turns, accelerations, and braking points.

5. Collision Avoidance

While moving, robot must avoid obstacles (static or moving).

Uses **real-time sensor input** → updates path if needed.

Example: If a human suddenly walks in front of robot, it stops or reroutes.

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6. Control & Execution

Finally, motion is executed by motors and actuators.

Robot continuously checks if it is **on the planned path** (feedback loop).

Example: Drone adjusting its flight to fight wind.

✓ In Short

A. **Planning** = Deciding *where to go* (like Google Maps route).

B. **Motion** = Actually *moving along that route* (driving carefully, avoiding bumps).

Together, they make robots intelligent and safe movers.

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🔊 Speech Recognition :

Speech recognition is the technology that allows a **computer or machine to listen to human speech, understand the words, and convert them into text or actions**.

Example:

When you say *"Hey Siri, set an alarm for 7 AM"*, Siri listens, understands, and creates the alarm.

⚙ How it Works (Step by Step)

1. Speech Input (Voice Capture) 🎤

You speak into a microphone.

The sound waves of your voice are captured as an audio signal.

2. Preprocessing (Cleaning the Audio)

The system removes background noise, echoes, and unwanted sounds.

It breaks down your speech into small units (called *frames*) for analysis.

3. Feature Extraction 🔍

The computer analyzes the audio signal to find patterns.

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It extracts features like pitch, tone, speed, and frequency.
Common technique: **MFCC (Mel-Frequency Cepstral Coefficients)** → helps convert speech into a machine-readable form.

4. Acoustic Model 🎶 → 📝

This model maps **sounds (phonemes)** to possible **letters/words**.
Example: The sound "k-æ-t" matches the word "cat".

5. Language Model 📚

This checks the context to figure out the most likely word sequence.
Example:
If you say *"recognize speech"*, it should not be understood as *"wreck a nice beach"*.
It uses grammar, probability, and dictionaries to make sense of words.

6. Decoder

Combines **acoustic model** + **language model** + **probabilities**.
Chooses the most accurate sentence or command from possible options.

7. Output 💻

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Converts recognized speech into **text** or **action**.

Example: Displays words on screen, executes a command, or interacts with an app.

❑ Key Components of Speech Recognition

1. **Microphone** – Captures your speech.
2. **Preprocessor** – Removes noise, prepares audio.
3. **Feature Extractor** – Converts speech into numeric features.
4. **Acoustic Model** – Matches sounds with phonemes/words.
5. **Language Model** – Understands grammar and context.
6. **Decoder** – Finds the best match for the spoken sentence.
7. **Output** – Shows text or performs action.

🎯 Applications of Speech Recognition

- **Virtual Assistants** (Alexa, Siri, Google Assistant)
- **Transcription tools** (converting lectures, meetings to text)
- **Voice Commands** (car navigation, smart home devices)
- **Accessibility** (helping people who cannot type)

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❖ What is a Software Agent?

A **software agent** is like a **computer program that works for you automatically**.
It can **sense the environment**, **make decisions**, and **take actions** to achieve a goal without you controlling every step.

👉 Example:

- Your **email spam filter** → automatically detects and moves spam emails.
- **Chatbots** → reply to users without human help.
- **Recommender systems** (like Netflix/YouTube suggestions).

❑ Architecture of a Software Agent

The **architecture** is basically the *internal design or structure* of how an agent works.
It decides:

✓ How the agent **perceives (senses)** the environment.
✓ How it **decides** what to do.
✓ How it **acts** in the environment.

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❖ Main Components of Agent Architecture

1. Sensors (Input) 👂

Collect information from the environment.

Example: In a chatbot, sensors are text input from the user.

2. Actuators (Output) 🗣️

Perform actions to affect the environment.

Example: Sending a reply message, moving a robot's arm.

3. Agent Program (Brain)

The decision-making part.

Takes input from sensors → decides → sends output to actuators.

❖ Types of Agent :

Already covered in Unit 1

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❖ Communication – How Agents Talk to Each Other

- In a **multi-agent system**, you don't have just one AI program, you have **many agents** (small AI units) working together.
- To solve a problem, they need to **share information**.
- This sharing of information is called **communication**.

❖ How do they communicate?

1. Message Passing → One agent sends a message to another (like chatting).

Example: In an online shopping site, the *recommendation agent* might tell the *inventory agent* which products are trending.

2. Common Language / Protocols → Agents need a common "language" (like humans use English or Hindi).

Example: The FIPA-ACL (Agent Communication Language) is often used.

3. Negotiation → Agents may discuss and come to an agreement.

Example: Two delivery drones deciding **who will deliver which package**.

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❖ Trust – Ensuring Agents are Reliable

- In multi-agent systems, not all agents can be trusted blindly.
- **Trust** means making sure that the information or actions of an agent are **correct, safe, and not misleading**.

❖ Why trust is important?

1. Correctness → If one agent gives wrong data, the whole system can fail.
Example: A *weather prediction agent* must provide correct data for a *farming decision agent*.
2. Security → Some agents might be faulty or even malicious.
Example: In online banking, a *payment agent* must be trusted to send money securely.
3. Reputation Systems → Agents can build trust over time.
Example: Like **ratings on Amazon**—if a seller (agent) has 5★ ratings, other agents (buyers) will trust it more.

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❖ What is Multi-Agent Negotiation?

Imagine you have **many intelligent agents** (like small AI programs or robots).
Sometimes, they **need the same resource** (money, time, energy, product, etc.) or have **different goals**.
To solve this conflict, agents **negotiate** → they **discuss, bargain, and agree** on a solution that works for everyone.

☞ Just like people negotiate in a market:

- Buyer: "I'll pay ₹900."
- Seller: "No, I want ₹1000."
- Finally, they agree at ₹950.

❖ Why do agents negotiate?

✓ To **share limited resources** (bandwidth, energy, money).
✓ To **coordinate tasks** (who does what).
✓ To **reach an agreement** without fighting or wasting resources.

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◆ Types of Multi-Agent Negotiation

1. Cooperative Negotiation

Agents work **together** to reach the best result for the group.

They **share information** openly and trust each other.

Example:

In a **smart factory**, different robots negotiate who will assemble which part so the whole car gets built faster.

2. Competitive Negotiation ⚔

Agents act in their **own self-interest**.

Each agent wants the **best deal for itself**, not for others.

Example:

In **online shopping auctions** (like eBay), every bidder agent wants to win the product at the lowest cost possible.

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3. Bargaining 🗣️

Two or more agents **exchange offers and counteroffers** until they agree.

Example:

In **energy trading**, one household's smart meter agent bargains with the electricity company's agent for cheaper night-time rates.

4. Auction-Based Negotiation

One agent (auctioneer) sells something, and others (bidders) compete by offering prices.

Different styles:

- **English Auction** → Price goes up (like eBay).
- **Dutch Auction** → Price starts high, then goes down until someone accepts.
- **Sealed-Bid Auction** → All agents secretly place bids; highest wins.

Example:

Cloud servers auction computing power, and agents from different companies bid for it.

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5. Contract Net Protocol (CNP)

One agent announces a **task** → other agents bid to do it → best agent gets the job.

Example:

In **delivery drones**, one agent announces "Who can deliver this package fastest?" → others bid with their delivery times.

→ **Negotiation** = discussion among agents to solve conflicts and agree.

❖ What is Reputation in Multi-Agent Systems?

- **Reputation** means the **record of past behavior** of an agent.
- It is how **trustworthy and reliable** an agent is **based on its history**.
- Other agents use this reputation to decide **whether to trust it or not**.

👉 Example in real life:

- On **Amazon**, sellers with 5 ⭐ ratings have a **good reputation**.
- On **Uber**, drivers with high ratings are more trusted by passengers.

Same idea applies to agents in AI systems.

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❖ Why is Reputation Important?

1. **Builds Trust** → If an agent has good past performance, others trust it.
2. **Avoids Cheating** → Bad agents can't easily trick the system because their reputation becomes low.
3. **Improves Cooperation** → Agents prefer to work with reliable partners.

❖ How is Reputation Calculated?

- Agents **observe behavior** or **receive feedback** from others.
- Reputation = **past performance + ratings + recommendations**.

👉 Example:

- If a food delivery agent always delivers on time, other agents give it **positive feedback**.
- Over time, its **reputation score increases**.

❑ Example in Multi-Agent System

Imagine a **smart city** with delivery drones:

Drone A always delivers on time → **high reputation**.

Drone B sometimes loses packages → **low reputation**.

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When a new delivery comes, the system chooses Drone A because of its better reputation.

✅ **In short:**

Reputation = history-based trust score of an agent.

Good reputation → more trusted.

Bad reputation → less trusted, fewer chances to get tasks.

Artificial Intelligence (BCS701)

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❖ Explainable AI (XAI) :

The Problem with Today's AI (Black Box)

In normal AI (top part of diagram):

1. **Training Data** → AI is trained on huge amounts of data.
2. **Machine Learning Process** → AI builds patterns inside.
3. **Learned Function (Model)** → AI learns to make predictions.
4. **Decision/Recommendation** → AI gives output to the user.

☞ **But the problem is: users don't know HOW the AI made the decision.**

Example: AI says *"This person is denied a loan."*

The user asks: *"Why?"* → No clear answer (Black Box).

❖ Explainable AI (XAI)

In XAI (bottom part of diagram):

- **Training Data** → Same as before.
- **New Machine Learning Process** → Builds models designed to be explainable.
- **Explainable Model** → The model is built in a way that can show reasons.

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- **Explanation Interface** → Gives **clear explanations** to the user about *why* the AI made that decision.

👉 Now the user not only gets the **decision**, but also the **reason behind it**.

Example: AI says *"Loan Denied."*

Then it adds: *"Because income is too low and credit score is 550."*

The user says: *"Ah, Got it."* ✅

❖ Why XAI is Important?

✓ **Transparency** → AI becomes less of a mystery.

✓ **Trust** → Users trust AI when they understand it.

✓ **Debugging** → Developers can find mistakes in AI.

✓ **Legal & Ethical** → In healthcare, finance, and law, explanations are required.

✅ In short:

Normal AI = "Answer only" (user asks *HOW?*).

XAI = "Answer + Reason" (user says *GOT IT*).

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❖ What is Interpretability?

Interpretability means **how easily a human can understand the reason behind an AI's decision**.

In simple words → *"Can I understand why the AI gave this answer?"*

☞ Example:

- If AI says: *"This email is spam."*
- **Low interpretability (black box)** → It just says *"spam"*, but doesn't explain.
- **High interpretability** → It explains: *"spam because it has suspicious links + repeated money words."*

❖ Importance of Interpretability :

1. Builds Trust

People will trust AI only if they understand it.

Example: In healthcare, a doctor will trust AI more if it explains *why it suggests a treatment*.

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2. Debugging AI 🛠

If AI makes a mistake, developers need to know **what went wrong**.
Without interpretability, fixing errors is very hard.

3. Ethical & Legal Compliance ⚖

In finance, law, and medicine, decisions must be explainable.
Example: A bank must explain *why someone's loan was rejected* (legally required in many countries).

4. Safety & Reliability 🚦

If AI controls cars, planes, or robots, humans must **understand its actions** to avoid accidents.
Example: A self-driving car should explain why it suddenly braked (e.g., "pedestrian crossing detected").

5. User Learning & Feedback 📚

Users can learn from AI explanations and give feedback to improve the system.
Example: Students using an AI tutor understand *why their answer was wrong*, not just that it's wrong.

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❖ Interpretability vs Accuracy (The Trade-off)

- Sometimes **simple models** (like Decision Trees, Linear Regression) are more interpretable but less accurate.
- **Complex models** (like Deep Neural Networks) are very accurate but hard to interpret.

☞ Example:

- A **linear model** may say: "Marks depend on study hours (70%) + attendance (30%)." → Easy to understand.
- A **deep neural network** might give very accurate predictions but without clear reasons.

That's why **XAI techniques** are used to make even complex models more interpretable.

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❖ What is a Black Box Model?

A **black box model** is an **AI or machine learning model** that makes predictions or decisions, but we **cannot easily see or understand how it works inside**.

We only see the **input** (data we give it) and the **output** (result it gives), but the **process in between is hidden or too complex**.

☞ Like a **magic box**:

You put something in → you get something out → but you don't know exactly *why or how*.

❖ Techniques for Explaining Black Box Models

1. LIME (Local Interpretable Model-agnostic Explanations)

Works by **approximating the black-box model locally** (for one prediction).

It builds a **simple, understandable model** (like linear regression) around that prediction.

☞ **Example:**

AI says: *"Loan Rejected."*

LIME explains: *"Because income (−0.6 weight) is low and credit score (−0.3 weight) is below 600, but savings (+0.1 weight) helped a little."*

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You see which factors mattered most.

2. SHAP (SHapley Additive exPlanations)

- Based on **game theory** 🎮.
- It calculates **how much each feature contributed** to the decision (positive or negative).
- Fair distribution of credit/blame among features.

👉 **Example:**

AI predicts: *"This customer will buy a product."*

SHAP explains:

- Age: +0.25
- Salary: +0.40
- Location: −0.10
- Past Purchases: +0.45

So Salary + Past Purchases influenced the decision most.

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3. Partial Dependence Plots (PDPs)

Shows how **changing one feature** affects predictions, while keeping others fixed.

☞ **Example:**

In a house price model:

PDP shows: *"If square footage increases, predicted price increases steadily."*

4. Feature Importance

Ranks features by how important they are for the model's prediction.

☞ **Example:**

For exam score prediction:

- Study Hours = 50% importance
- Attendance = 30%
- Sleep = 15%
- Internet Usage = 5%

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5. Counterfactual Explanations

Answers the question: *"What should be different for a different outcome?"*

👉 **Example:**

AI says: *"Loan Rejected."*

Counterfactual: *"If your income was ₹50,000 instead of ₹30,000, the loan would be approved."*

6. Surrogate Models

Train a **simpler model** (like a decision tree) to mimic the black-box model's behavior.

Easier to understand.

👉 **Example:**

A decision tree explains a complex neural network by showing rules like:

If income > 40K AND credit score > 700 → Loan Approved.

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❖ What is Trust in AI?

Trust in AI means how much a **user believes** that an AI system is **reliable, safe, fair, and correct** in its decisions.

In simple words → *"Can I depend on this AI without fear of mistakes, bias, or harm?"*

☞ Example:

If a doctor uses an AI system for diagnosing diseases:

- **Trusted AI** → Doctor feels confident in following its advice.
- **Untrusted AI** → Doctor ignores it because they doubt its accuracy.

❖ How Do We Build Trust in AI?

✓ **Accuracy** → AI must give correct and consistent results.
✓ **Explainability** → AI must explain its decisions (XAI).
✓ **Fairness** → No discrimination or hidden bias.
✓ **Robustness** → Works well even in unexpected situations.
✓ **Security** → Protects against hacking or manipulation.

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◆ 1. Case Study in NLP (Natural Language Processing)

👉 **Problem**: AI makes decisions on text, but people don't know *why*.

👉 **Solution**: Use **XAI** to explain the decision.

Example 1: Spam Email Detection

Black box AI: Just says → "This is spam."

With XAI: Explains →

- Contains "Free money" (+0.6)
- Has too many links (+0.3)
- Unknown sender (+0.1)

✓ User now knows *why* the email is marked spam.

Example 2: Sentiment Analysis (Positive/Negative review detection)

Black box AI: Says → "This review is negative."

With XAI: Explains →

- Words like "worst", "boring" gave negative weight.
- Words like "good" gave positive weight, but less impact.

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✅ User can see which words influenced the result.

2. Case Study in Computer Vision

👉 **Problem:** AI predicts images, but users don't know *what features it looked at*.

👉 **Solution:** Use **XAI (heatmaps, saliency maps, Grad-CAM, etc.)**.

Example 1: Medical Imaging (X-ray Diagnosis)

Black box AI: Says → "This X-ray shows pneumonia."

With XAI (heatmap): Highlights the lung region where pneumonia patterns were found.

✅ Doctors can verify if AI is looking at the correct place.

Example 2: Self-Driving Cars 🚗

Black box AI: Car suddenly brakes.

With XAI (object detection explanation): Explains → "Stopped because pedestrian detected in crosswalk."

✅ Driver understands and feels safer.

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Thank You....

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